



Samples



Also by the author

The Electric Book Works Knowledge Base

Designing for Digital: What every book designer needs to know

Killing Time



Samples

Sample content for Electric Book print, epub
and web output

Arthur Attwell





Samples

© Arthur Attwell

ISBN (Print): 978-1-928313-13-7

ISBN (Digital download): 978-1-928313-13-7

ISBN (Digital download): 9780000000000

ISBN (Digital online): 978-1-928313-13-7

This work is licensed under a Creative Commons Attribution 4.0 International License. This means you are free to share (copy and redistribute the material in any medium or format) and adapt it (remix, transform, and build upon the material) for any purpose, even commercially, as long as you give appropriate credit, with a link to your source, and indicate if changes were made. You may do so in any reasonable manner, but not in any way that suggests the licensor endorses you or your use.

Contents

Half title page	i
Previous publications	ii
Title page	iii
Copyright	iv
Contents	v
Dedication	vii
Epigraph	ix
Frontispiece	x
Other frontmatter	xi
Part 1: Text	
Plain text	3
I	3
II	11
III	20
Headings	25
Quotes	29
Blockquotes	29
Pullquotes	31
Quote sources	31
Lists	33
Simple lists	33
Chapter tables of contents	35
Bibliographies	36
Definitions and glossaries	37
Tables	39
Boxes	41
Notes	49
Dividers	59



CONTENTS

Poetry	63
Questions and quizzes	71
Part 2: Images	
Plain images	75
Figures	89
Part 3: Code and mathematics	
Code	99
Mathematics	101
Part 4: Interactivity	
Links and buttons	105
Video	107
Audio	109
iFrames	111
Slides	113
Part 5: Indexes	
Traditional index	117
Dynamic index	139



To Ms Sleep, who said I could be a writer.
You were right, though much of it's in `code`.





In a good bookroom you feel in some mysterious way that you are absorbing the wisdom contained in all the books through your skin, without even opening them.

Mark Twain





Look at this lovely picture from *The Count of Monte Cristo* for the frontispiece of our sample book.

Other frontmatter

In a real book, this might be a Preface or a Foreword or some other kind of prefatory section. But right now we just want to show you what it looks like when a book file is generic frontmatter. To keep you entertained as you stare at these pages, here is the first part of Hippolyte Adolphe Taine's Introduction to the *History of English Literature*.

History, within a hundred years in Germany, and within sixty years in France, has undergone a transformation owing to a study of literatures.

The discovery has been made that a literary work is not a mere play of the imagination, the isolated caprice of an excited brain, but a transcript of contemporary manners and customs and the sign of a particular state of intellect. The conclusion derived from this is that, through literary monuments, we can retrace the way in which men felt and thought many centuries ago. This method has been tried and found successful.

We have meditated over these ways of feeling and thinking and have accepted them as facts of prime significance. We have found that they were dependent on most important events, that they explain these, and that these explain them, and that henceforth it was necessary to give them their place in history, and one of the highest. This place has been assigned to them, and hence all is changed in history—the aim, the method, the instrumentalities, and the conceptions of laws and of causes. It is this change as now going on, and which must continue to go on, that is here attempted to be set forth.

On turning over the large stiff pages of a folio volume, or the yellow leaves of a manuscript, in short, a poem, a code of laws, a confession of faith, what is your first comment? You say to yourself that the work before you is not of its own creation. It is simply a mold like a fossil shell, an imprint similar to one of those forms embedded in a stone by an animal which once lived and perished. Beneath the shell was an animal and behind the document there was a man. Why do you study the shell unless to form some idea of the animal? In the same way do you study the document in order to comprehend the man; both shell and document are dead fragments and of value only as indications of the complete living being. The aim is to reach this being; this is what you strive to reconstruct. It is a mistake to study the document as if it existed alone by itself. That



is treating things merely as a pedant, and you subject yourself to the illusions of a book-worm. At bottom mythologies and languages are not existences; the only realities are human beings who have employed words and imagery adapted to their organs and to suit the original cast of their intellects. A creed is nothing in itself. Who made it? Look at this or that portrait of the sixteenth century, the stern, energetic features of an archbishop or of an English martyr. Nothing exists except through the individual; it is necessary to know the individual himself. Let the parentage of creeds be established, or the classification of poems, or the growth of constitutions, or the transformations of idioms, and we have only cleared the ground. True history begins when the historian has discerned beyond the mists of ages the living, active man, endowed with passions, furnished with habits, special in voice, feature, gesture and costume, distinctive and complete, like anybody that you have just encountered in the street. Let us strive then, as far as possible, to get rid of this great interval of time which prevents us from observing the man with our eyes, the eyes of our own head. What revelations do we find in the calendared leaves of a modern poem? A modern poet, a man like De Musset, Victor Hugo, Lamartine, or Heine, graduated from a college and traveled, wearing a dress-coat and gloves, favored by ladies, bowing fifty times and uttering a dozen witticisms in an evening, reading daily newspapers, generally occupying an apartment on the second story, not over-cheerful on account of his nerves, and especially because, in this dense democracy in which we stifle each other, the discredit of official rank exaggerates his pretensions by raising his importance, and, owing to the delicacy of his personal sensations, leading him to regard himself as a Deity. Such is what we detect behind modern meditations and sonnets.

Again, behind a tragedy of the seventeenth century there is a poet, one, for example, like Racine, refined, discreet, a courtier, a fine talker, with majestic perruque and ribboned shoes, a monarchist and zealous Christian, "God having given him the grace not to blush in any society on account of zeal for his king or for the Gospel," clever in interesting the monarch, translating into proper French "the gaulois of Amyot," deferential to the great, always knowing how to keep his place in their company, assiduous and respectful at Marly as at Versailles, amid the formal creations of a decorative landscape and the reverential bows, graces, intrigues, and fineness of the braided seigniors who get up early every morning to obtain the reversion of an office, together with the charming ladies who count on their fingers the pedigrees which entitle them to a seat on a footstool. On this point consult Saint-Simon and the engravings of Pérelle, the same as you have just consulted Balzac and the watercolor drawings of Eugène Lami.

In like manner, on reading a Greek tragedy, our first care is to figure to ourselves the Greeks, that is to say, men who lived halfnaked in the gymnasiums or on a public square under a brilliant sky, in full view of the noblest and most delicate landscape, busy in rendering their bodies strong and agile, in conversing together, in arguing, in voting, in carrying out patriotic piracies, and yet idle and temperate, the furniture of their houses consisting of three earthen jars and their food of two pots of anchovies preserved in oil, served by slaves who afford them the time to cultivate their minds and to exercise their limbs, with no other concern than that of having the most beautiful city, the most beautiful processions, the most beautiful ideas, and the most beautiful men. In this respect, a statue like the "Meleager" or the "Theseus" of the Parthenon, or again a

sight of the blue and lustrous Mediterranean, resembling a silken tunic out of which islands arise like marble bodies, together with a dozen choice phrases selected from the works of Plato and Aristophanes, teach us more than any number of dissertations and commentaries.

And so again, in order to understand an Indian Purana, one must begin by imagining the father of a family who, "having seen a son on his son's knees," follows the law and, with ax and pitcher, seeks solitude under a banyan tree, talks no more, multiplies his fastings, lives naked with four fires around him under the fifth fire, that terrible sun which endlessly devours and resuscitates all living things; who fixes his imagination in turn for weeks at a time on the foot of Brahma, then on his knee, on his thigh, on his navel, and so on, until, beneath the strain of this intense meditation, hallucinations appear, when all the forms of being, mingling together and transformed into each other, oscillate to and fro in this vertiginous brain until the motionless man, with suspended breath and fixed eyeballs, beholds the universe melting away like vapor over the vacant immensity of the Being in which he hopes for absorption. In this case the best of teachings would be a journey in India; but, for lack of a better one, take the narratives of travelers along with works in geography, botany, and ethnology. In any event, there must be the same research. A language, a law, a creed, is never other than an abstraction; the perfect thing is found in the active man, the visible corporeal figure which eats, walks, fights, and labors. Set aside the theories of constitutions and their results, of religions and their systems, and try to observe men in their workshops or offices, in their fields along with their own sky and soil, with their own homes, clothes, occupations and repasts, just as you see them when, on landing in England or in Italy, you remark their features and gestures, their roads and their inns, the citizen on his promenades and the workman taking a drink. Let us strive as much as possible to supply the place of the actual, personal, sensible observation that is no longer practicable, this being the only way in which we can really know the man; let us make the past present; to judge of an object it must be present; no experience can be had of what is absent. Undoubtedly, this sort of reconstruction is always imperfect; only an imperfect judgment can be based on it; but let us do the best we can; incomplete knowledge is better than none at all, or than knowledge which is erroneous, and there is no other way of obtaining knowledge approximatively of bygone times than by seeing approximatively the men of former times.

Such is the first step in history. This step was taken in Europe at the end of the last century when the imagination took fresh flight under the auspices of Lessing and Walter Scott, and a little later in France under Chateaubriand, Augustin Thierry, Michelet, and others. We now come to the second step.



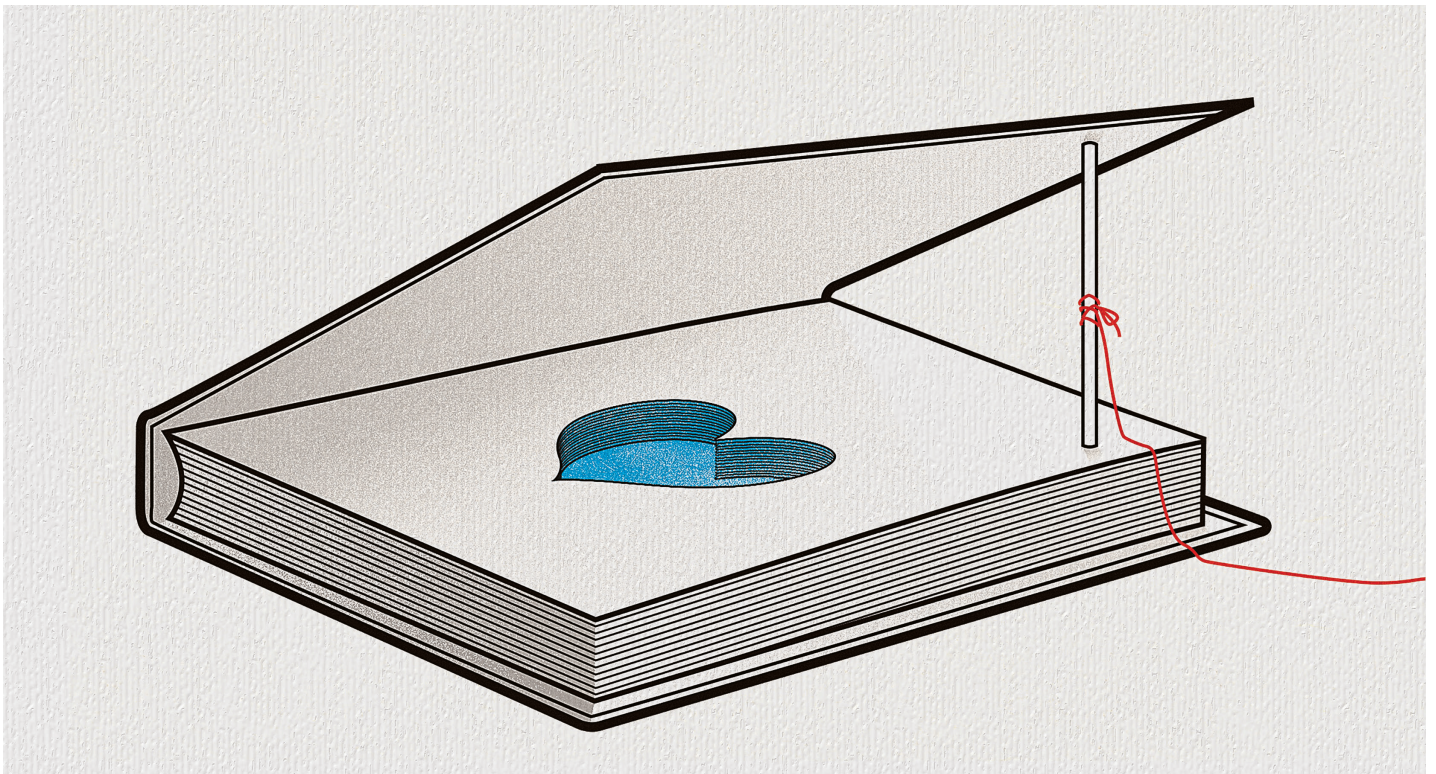


Part 1

Text







Plain text

Most books comprise pages and pages of running text. A good design makes them lovely to look at and easy to read. The heart of a design is how it handles both plain text and special forms like letters. There is a letter towards the end of this chapter (page 21).

The text for this chapter is taken from Sir Arthur Conan Doyle's *Adventures of Sherlock Holmes*.

A Scandal in Bohemia

I.

To Sherlock Holmes she is always THE woman. I have seldom heard him mention her under any other name. In his eyes she eclipses and predominates the whole of her sex. It was not that he felt any emotion akin to love for Irene Adler. All emotions, and that one particularly, were abhorrent to his cold, precise but admirably balanced mind. He was, I take it, the most perfect reasoning and observing machine that the world has seen, but as a lover he would have placed himself in a false position. He never spoke of the softer passions, save with a gibe and a sneer. They were admirable things for the observer—excellent for drawing the veil from men's motives and actions. But for the trained reasoner to admit such intrusions into his own delicate and finely adjusted temperament was to introduce a



Scan the code to see
this page online.

distracting factor which might throw a doubt upon all his mental results. Grit in a sensitive instrument, or a crack in one of his own high-power lenses, would not be more disturbing than a strong emotion in a nature such as his. And yet there was but one woman to him, and that woman was the late Irene Adler, of dubious and questionable memory.

I had seen little of Holmes lately. My marriage had drifted us away from each other. My own complete happiness, and the home-centred interests which rise up around the man who first finds himself master of his own establishment, were sufficient to absorb all my attention, while Holmes, who loathed every form of society with his whole Bohemian soul, remained in our lodgings in Baker Street, buried among his old books, and alternating from week to week between cocaine and ambition, the drowsiness of the drug, and the fierce energy of his own keen nature. He was still, as ever, deeply attracted by the study of crime, and occupied his immense faculties and extraordinary powers of observation in following out those clues, and clearing up those mysteries which had been abandoned as hopeless by the official police. From time to time I heard some vague account of his doings: of his summons to Odessa in the case of the Trepoff murder, of his clearing up of the singular tragedy of the Atkinson brothers at Trincomalee, and finally of the mission which he had accomplished so delicately and successfully for the reigning family of Holland. Beyond these signs of his activity, however, which I merely shared with all the readers of the daily press, I knew little of my former friend and companion.

One night—it was on the twentieth of March, 1888—I was returning from a journey to a patient (for I had now returned to civil practice), when my way led me through Baker Street. As I passed the well-remembered door, which must always be associated in my mind with my wooing, and with the dark incidents of the Study in Scarlet, I was seized with a keen desire to see Holmes again, and to know how he was employing his extraordinary powers. His rooms were brilliantly lit, and, even as I looked up, I saw his tall, spare figure pass twice in a dark silhouette against the blind. He was pacing the room swiftly, eagerly, with his head sunk upon his chest and his hands clasped behind him. To me, who knew his every mood and habit, his attitude and manner told their own story. He was at work again. He had risen out of his drug-created dreams and was hot upon the scent of some new problem. I rang the bell and was shown up to the chamber which had formerly been in part my own.

His manner was not effusive. It seldom was; but he was glad, I think, to see me. With hardly a word spoken, but with a kindly eye, he waved me to an armchair, threw across his case of cigars, and indicated a spirit case and

a gasogene in the corner. Then he stood before the fire and looked me over in his singular introspective fashion.

“Wedlock suits you,” he remarked. “I think, Watson, that you have put on seven and a half pounds since I saw you.”

“Seven!” I answered.

“Indeed, I should have thought a little more. Just a trifle more, I fancy, Watson. And in practice again, I observe. You did not tell me that you intended to go into harness.”

“Then, how do you know?”

“I see it, I deduce it. How do I know that you have been getting yourself very wet lately, and that you have a most clumsy and careless servant girl?”

“My dear Holmes,” said I, “this is too much. You would certainly have been burned, had you lived a few centuries ago. It is true that I had a country walk on Thursday and came home in a dreadful mess, but as I have changed my clothes I can’t imagine how you deduce it. As to Mary Jane, she is incorrigible, and my wife has given her notice, but there, again, I fail to see how you work it out.”

He chuckled to himself and rubbed his long, nervous hands together.

“It is simplicity itself,” said he; “my eyes tell me that on the inside of your left shoe, just where the firelight strikes it, the leather is scored by six almost parallel cuts. Obviously they have been caused by someone who has very carelessly scraped round the edges of the sole in order to remove crusted mud from it. Hence, you see, my double deduction that you had been out in vile weather, and that you had a particularly malignant boot-slitting specimen of the London slavey. As to your practice, if a gentleman walks into my rooms smelling of iodoform, with a black mark of nitrate of silver upon his right forefinger, and a bulge on the right side of his top-hat to show where he has secreted his stethoscope, I must be dull, indeed, if I do not pronounce him to be an active member of the medical profession.”

I could not help laughing at the ease with which he explained his process of deduction. “When I hear you give your reasons,” I remarked, “the thing always appears to me to be so ridiculously simple that I could easily do it myself, though at each successive instance of your reasoning I am baffled until you explain your process. And yet I believe that my eyes are as good as yours.”

“Quite so,” he answered, lighting a cigarette, and throwing himself down into an armchair. “You see, but you do not observe. The distinction is clear. For example, you have frequently seen the steps which lead up from the hall to this room.”

“Frequently.”



“How often?”

“Well, some hundreds of times.”

“Then how many are there?”

“How many? I don’t know.”

“Quite so! You have not observed. And yet you have seen. That is just my point. Now, I know that there are seventeen steps, because I have both seen and observed. By-the-way, since you are interested in these little problems, and since you are good enough to chronicle one or two of my trifling experiences, you may be interested in this.” He threw over a sheet of thick, pink-tinted note-paper which had been lying open upon the table. “It came by the last post,” said he. “Read it aloud.”

The note was undated, and without either signature or address.

“There will call upon you to-night, at a quarter to eight o’clock,” it said, “a gentleman who desires to consult you upon a matter of the very deepest moment. Your recent services to one of the royal houses of Europe have shown that you are one who may safely be trusted with matters which are of an importance which can hardly be exaggerated. This account of you we have from all quarters received. Be in your chamber then at that hour, and do not take it amiss if your visitor wear a mask.”

“This is indeed a mystery,” I remarked. “What do you imagine that it means?”

“I have no data yet. It is a capital mistake to theorize before one has data. Insensibly one begins to twist facts to suit theories, instead of theories to suit facts. But the note itself. What do you deduce from it?”

I carefully examined the writing, and the paper upon which it was written.

“The man who wrote it was presumably well to do,” I remarked, endeavouring to imitate my companion’s processes. “Such paper could not be bought under half a crown a packet. It is peculiarly strong and stiff.”

“Peculiar—that is the very word,” said Holmes. “It is not an English paper at all. Hold it up to the light.”

I did so, and saw a large “E” with a small “g,” a “P,” and a large “G” with a small “t” woven into the texture of the paper.

“What do you make of that?” asked Holmes.

“The name of the maker, no doubt; or his monogram, rather.”

“Not at all. The ‘G’ with the small ‘t’ stands for ‘Gesellschaft,’ which is the German for ‘Company.’ It is a customary contraction like our ‘Co.’ ‘P,’ of course, stands for ‘Papier.’ Now for the ‘Eg.’ Let us glance at our Continental Gazetteer.” He took down a heavy brown volume from his shelves. “Eglow, Eglonitz—here we are, Egria. It is in a German-speaking

country—in Bohemia, not far from Carlsbad. ‘Remarkable as being the scene of the death of Wallenstein, and for its numerous glass-factories and paper-mills.’ Ha, ha, my boy, what do you make of that?” His eyes sparkled, and he sent up a great blue triumphant cloud from his cigarette.

“The paper was made in Bohemia,” I said.

“Precisely. And the man who wrote the note is a German. Do you note the peculiar construction of the sentence—‘This account of you we have from all quarters received.’ A Frenchman or Russian could not have written that. It is the German who is so uncourteous to his verbs. It only remains, therefore, to discover what is wanted by this German who writes upon Bohemian paper and prefers wearing a mask to showing his face. And here he comes, if I am not mistaken, to resolve all our doubts.”

As he spoke there was the sharp sound of horses’ hoofs and grating wheels against the curb, followed by a sharp pull at the bell. Holmes whistled.

“A pair, by the sound,” said he. “Yes,” he continued, glancing out of the window. “A nice little brougham and a pair of beauties. A hundred and fifty guineas apiece. There’s money in this case, Watson, if there is nothing else.”

“I think that I had better go, Holmes.”

“Not a bit, Doctor. Stay where you are. I am lost without my Boswell. And this promises to be interesting. It would be a pity to miss it.”

“But your client—”

“Never mind him. I may want your help, and so may he. Here he comes. Sit down in that armchair, Doctor, and give us your best attention.”

A slow and heavy step, which had been heard upon the stairs and in the passage, paused immediately outside the door. Then there was a loud and authoritative tap.

“Come in!” said Holmes.

A man entered who could hardly have been less than six feet six inches in height, with the chest and limbs of a Hercules. His dress was rich with a richness which would, in England, be looked upon as akin to bad taste. Heavy bands of astrakhan were slashed across the sleeves and fronts of his double-breasted coat, while the deep blue cloak which was thrown over his shoulders was lined with flame-coloured silk and secured at the neck with a brooch which consisted of a single flaming beryl. Boots which extended halfway up his calves, and which were trimmed at the tops with rich brown fur, completed the impression of barbaric opulence which was suggested by his whole appearance. He carried a broad-brimmed hat in his hand, while he wore across the upper part of his face, extending down past the cheekbones, a black vizard mask, which he had apparently adjusted that



very moment, for his hand was still raised to it as he entered. From the lower part of the face he appeared to be a man of strong character, with a thick, hanging lip, and a long, straight chin suggestive of resolution pushed to the length of obstinacy.

“You had my note?” he asked with a deep harsh voice and a strongly marked German accent. “I told you that I would call.” He looked from one to the other of us, as if uncertain which to address.

“Pray take a seat,” said Holmes. “This is my friend and colleague, Dr. Watson, who is occasionally good enough to help me in my cases. Whom have I the honour to address?”

“You may address me as the Count Von Kramm, a Bohemian nobleman. I understand that this gentleman, your friend, is a man of honour and discretion, whom I may trust with a matter of the most extreme importance. If not, I should much prefer to communicate with you alone.”

I rose to go, but Holmes caught me by the wrist and pushed me back into my chair. “It is both, or none,” said he. “You may say before this gentleman anything which you may say to me.”

The Count shrugged his broad shoulders. “Then I must begin,” said he, “by binding you both to absolute secrecy for two years; at the end of that time the matter will be of no importance. At present it is not too much to say that it is of such weight it may have an influence upon European history.”

“I promise,” said Holmes.

“And I.”

“You will excuse this mask,” continued our strange visitor. “The august person who employs me wishes his agent to be unknown to you, and I may confess at once that the title by which I have just called myself is not exactly my own.”

“I was aware of it,” said Holmes dryly.

“The circumstances are of great delicacy, and every precaution has to be taken to quench what might grow to be an immense scandal and seriously compromise one of the reigning families of Europe. To speak plainly, the matter implicates the great House of Ormstein, hereditary kings of Bohemia.”

“I was also aware of that,” murmured Holmes, settling himself down in his armchair and closing his eyes.

Our visitor glanced with some apparent surprise at the languid, lounging figure of the man who had been no doubt depicted to him as the most incisive reasoner and most energetic agent in Europe. Holmes slowly reopened his eyes and looked impatiently at his gigantic client.

“If your Majesty would condescend to state your case,” he remarked, “I

should be better able to advise you.”

The man sprang from his chair and paced up and down the room in uncontrollable agitation. Then, with a gesture of desperation, he tore the mask from his face and hurled it upon the ground. “You are right,” he cried; “I am the King. Why should I attempt to conceal it?”

“Why, indeed?” murmured Holmes. “Your Majesty had not spoken before I was aware that I was addressing Wilhelm Gottsreich Sigismond von Ormstein, Grand Duke of Cassel-Felstein, and hereditary King of Bohemia.”

“But you can understand,” said our strange visitor, sitting down once more and passing his hand over his high white forehead, “you can understand that I am not accustomed to doing such business in my own person. Yet the matter was so delicate that I could not confide it to an agent without putting myself in his power. I have come incognito from Prague for the purpose of consulting you.”

“Then, pray consult,” said Holmes, shutting his eyes once more.

“The facts are briefly these: Some five years ago, during a lengthy visit to Warsaw, I made the acquaintance of the well-known adventuress, Irene Adler. The name is no doubt familiar to you.”

“Kindly look her up in my index, Doctor,” murmured Holmes without opening his eyes. For many years he had adopted a system of docketing all paragraphs concerning men and things, so that it was difficult to name a subject or a person on which he could not at once furnish information. In this case I found her biography sandwiched in between that of a Hebrew rabbi and that of a staff-commander who had written a monograph upon the deep-sea fishes.

“Let me see!” said Holmes. “Hum! Born in New Jersey in the year 1858. Contralto—hum! La Scala, hum! Prima donna Imperial Opera of Warsaw—yes! Retired from operatic stage—ha! Living in London—quite so! Your Majesty, as I understand, became entangled with this young person, wrote her some compromising letters, and is now desirous of getting those letters back.”

“Precisely so. But how—”

“Was there a secret marriage?”

“None.”

“No legal papers or certificates?”

“None.”

“Then I fail to follow your Majesty. If this young person should produce her letters for blackmailing or other purposes, how is she to prove their authenticity?”

“There is the writing.”

PLAIN TEXT

“Pooh, pooh! Forgery.”

“My private note-paper.”

“Stolen.”

“My own seal.”

“Imitated.”

“My photograph.”

“Bought.”

“We were both in the photograph.”

“Oh, dear! That is very bad! Your Majesty has indeed committed an indiscretion.”

“I was mad—insane.”

“You have compromised yourself seriously.”

“I was only Crown Prince then. I was young. I am but thirty now.”

“It must be recovered.”

“We have tried and failed.”

“Your Majesty must pay. It must be bought.”

“She will not sell.”

“Stolen, then.”

“Five attempts have been made. Twice burglars in my pay ransacked her house. Once we diverted her luggage when she travelled. Twice she has been waylaid. There has been no result.”

“No sign of it?”

“Absolutely none.”

Holmes laughed. “It is quite a pretty little problem,” said he.

“But a very serious one to me,” returned the King reproachfully.

“Very, indeed. And what does she propose to do with the photograph?”

“To ruin me.”

“But how?”

“I am about to be married.”

“So I have heard.”

“To Clotilde Lothman von Saxe-Meningen, second daughter of the King of Scandinavia. You may know the strict principles of her family. She is herself the very soul of delicacy. A shadow of a doubt as to my conduct would bring the matter to an end.”

“And Irene Adler?”

“Threatens to send them the photograph. And she will do it. I know that she will do it. You do not know her, but she has a soul of steel. She has the face of the most beautiful of women, and the mind of the most resolute of men. Rather than I should marry another woman, there are no lengths to which she would not go—none.”

“You are sure that she has not sent it yet?”

“I am sure.”

“And why?”

“Because she has said that she would send it on the day when the betrothal was publicly proclaimed. That will be next Monday.”

“Oh, then we have three days yet,” said Holmes with a yawn. “That is very fortunate, as I have one or two matters of importance to look into just at present. Your Majesty will, of course, stay in London for the present?”

“Certainly. You will find me at the Langham under the name of the Count Von Kramm.”

“Then I shall drop you a line to let you know how we progress.”

“Pray do so. I shall be all anxiety.”

“Then, as to money?”

“You have carte blanche.”

“Absolutely?”

“I tell you that I would give one of the provinces of my kingdom to have that photograph.”

“And for present expenses?”

The King took a heavy chamois leather bag from under his cloak and laid it on the table.

“There are three hundred pounds in gold and seven hundred in notes,” he said.

Holmes scribbled a receipt upon a sheet of his note-book and handed it to him.

“And Mademoiselle’s address?” he asked.

“Is Briony Lodge, Serpentine Avenue, St. John’s Wood.”

Holmes took a note of it. “One other question,” said he. “Was the photograph a cabinet?”

“It was.”

“Then, good-night, your Majesty, and I trust that we shall soon have some good news for you. And good-night, Watson,” he added, as the wheels of the royal brougham rolled down the street. “If you will be good enough to call to-morrow afternoon at three o’clock I should like to chat this little matter over with you.”

//.

At three o’clock precisely I was at Baker Street, but Holmes had not yet returned. The landlady informed me that he had left the house shortly after eight o’clock in the morning. I sat down beside the fire, however, with the intention of awaiting him, however long he might be. I was already deeply

interested in his inquiry, for, though it was surrounded by none of the grim and strange features which were associated with the two crimes which I have already recorded, still, the nature of the case and the exalted station of his client gave it a character of its own. Indeed, apart from the nature of the investigation which my friend had on hand, there was something in his masterly grasp of a situation, and his keen, incisive reasoning, which made it a pleasure to me to study his system of work, and to follow the quick, subtle methods by which he disentangled the most inextricable mysteries. So accustomed was I to his invariable success that the very possibility of his failing had ceased to enter into my head.

It was close upon four before the door opened, and a drunken-looking groom, ill-kempt and side-whiskered, with an inflamed face and disreputable clothes, walked into the room. Accustomed as I was to my friend's amazing powers in the use of disguises, I had to look three times before I was certain that it was indeed he. With a nod he vanished into the bedroom, whence he emerged in five minutes tweed-suited and respectable, as of old. Putting his hands into his pockets, he stretched out his legs in front of the fire and laughed heartily for some minutes.

"Well, really!" he cried, and then he choked and laughed again until he was obliged to lie back, limp and helpless, in the chair.

"What is it?"

"It's quite too funny. I am sure you could never guess how I employed my morning, or what I ended by doing."

"I can't imagine. I suppose that you have been watching the habits, and perhaps the house, of Miss Irene Adler."

"Quite so; but the sequel was rather unusual. I will tell you, however. I left the house a little after eight o'clock this morning in the character of a groom out of work. There is a wonderful sympathy and freemasonry among horsey men. Be one of them, and you will know all that there is to know. I soon found Briony Lodge. It is a bijou villa, with a garden at the back, but built out in front right up to the road, two stories. Chubb lock to the door. Large sitting-room on the right side, well furnished, with long windows almost to the floor, and those preposterous English window fasteners which a child could open. Behind there was nothing remarkable, save that the passage window could be reached from the top of the coach-house. I walked round it and examined it closely from every point of view, but without noting anything else of interest.

"I then lounged down the street and found, as I expected, that there was a mews in a lane which runs down by one wall of the garden. I lent the ostlers a hand in rubbing down their horses, and received in exchange

twopence, a glass of half and half, two fills of shag tobacco, and as much information as I could desire about Miss Adler, to say nothing of half a dozen other people in the neighbourhood in whom I was not in the least interested, but whose biographies I was compelled to listen to.”

“And what of Irene Adler?” I asked.

“Oh, she has turned all the men’s heads down in that part. She is the daintiest thing under a bonnet on this planet. So say the Serpentine-mews, to a man. She lives quietly, sings at concerts, drives out at five every day, and returns at seven sharp for dinner. Seldom goes out at other times, except when she sings. Has only one male visitor, but a good deal of him. He is dark, handsome, and dashing, never calls less than once a day, and often twice. He is a Mr. Godfrey Norton, of the Inner Temple. See the advantages of a cabman as a confidant. They had driven him home a dozen times from Serpentine-mews, and knew all about him. When I had listened to all they had to tell, I began to walk up and down near Briony Lodge once more, and to think over my plan of campaign.

“This Godfrey Norton was evidently an important factor in the matter. He was a lawyer. That sounded ominous. What was the relation between them, and what the object of his repeated visits? Was she his client, his friend, or his mistress? If the former, she had probably transferred the photograph to his keeping. If the latter, it was less likely. On the issue of this question depended whether I should continue my work at Briony Lodge, or turn my attention to the gentleman’s chambers in the Temple. It was a delicate point, and it widened the field of my inquiry. I fear that I bore you with these details, but I have to let you see my little difficulties, if you are to understand the situation.”

“I am following you closely,” I answered.

“I was still balancing the matter in my mind when a hansom cab drove up to Briony Lodge, and a gentleman sprang out. He was a remarkably handsome man, dark, aquiline, and moustached—evidently the man of whom I had heard. He appeared to be in a great hurry, shouted to the cabman to wait, and brushed past the maid who opened the door with the air of a man who was thoroughly at home.

“He was in the house about half an hour, and I could catch glimpses of him in the windows of the sitting-room, pacing up and down, talking excitedly, and waving his arms. Of her I could see nothing. Presently he emerged, looking even more flurried than before. As he stepped up to the cab, he pulled a gold watch from his pocket and looked at it earnestly, ‘Drive like the devil,’ he shouted, ‘first to Gross & Hankey’s in Regent Street, and then to the Church of St. Monica in the Edgeware Road. Half a guinea if

you do it in twenty minutes!’

“Away they went, and I was just wondering whether I should not do well to follow them when up the lane came a neat little landau, the coachman with his coat only half-buttoned, and his tie under his ear, while all the tags of his harness were sticking out of the buckles. It hadn’t pulled up before she shot out of the hall door and into it. I only caught a glimpse of her at the moment, but she was a lovely woman, with a face that a man might die for.

“‘The Church of St. Monica, John,’ she cried, ‘and half a sovereign if you reach it in twenty minutes.’

“This was quite too good to lose, Watson. I was just balancing whether I should run for it, or whether I should perch behind her landau when a cab came through the street. The driver looked twice at such a shabby fare, but I jumped in before he could object. ‘The Church of St. Monica,’ said I, ‘and half a sovereign if you reach it in twenty minutes.’ It was twenty-five minutes to twelve, and of course it was clear enough what was in the wind.

“My cabby drove fast. I don’t think I ever drove faster, but the others were there before us. The cab and the landau with their steaming horses were in front of the door when I arrived. I paid the man and hurried into the church. There was not a soul there save the two whom I had followed and a surpliced clergyman, who seemed to be expostulating with them. They were all three standing in a knot in front of the altar. I lounged up the side aisle like any other idler who has dropped into a church. Suddenly, to my surprise, the three at the altar faced round to me, and Godfrey Norton came running as hard as he could towards me.

“‘Thank God,’ he cried. ‘You’ll do. Come! Come!’

“‘What then?’ I asked.

“‘Come, man, come, only three minutes, or it won’t be legal.’

“I was half-dragged up to the altar, and before I knew where I was I found myself mumbling responses which were whispered in my ear, and vouching for things of which I knew nothing, and generally assisting in the secure tying up of Irene Adler, spinster, to Godfrey Norton, bachelor. It was all done in an instant, and there was the gentleman thanking me on the one side and the lady on the other, while the clergyman beamed on me in front. It was the most preposterous position in which I ever found myself in my life, and it was the thought of it that started me laughing just now. It seems that there had been some informality about their license, that the clergyman absolutely refused to marry them without a witness of some sort, and that my lucky appearance saved the bridegroom from having to sally out into the streets in search of a best man. The bride gave me a

sovereign, and I mean to wear it on my watch-chain in memory of the occasion."

"This is a very unexpected turn of affairs," said I; "and what then?"

"Well, I found my plans very seriously menaced. It looked as if the pair might take an immediate departure, and so necessitate very prompt and energetic measures on my part. At the church door, however, they separated, he driving back to the Temple, and she to her own house. 'I shall drive out in the park at five as usual,' she said as she left him. I heard no more. They drove away in different directions, and I went off to make my own arrangements."

"Which are?"

"Some cold beef and a glass of beer," he answered, ringing the bell. "I have been too busy to think of food, and I am likely to be busier still this evening. By the way, Doctor, I shall want your co-operation."

"I shall be delighted."

"You don't mind breaking the law?"

"Not in the least."

"Nor running a chance of arrest?"

"Not in a good cause."

"Oh, the cause is excellent!"

"Then I am your man."

"I was sure that I might rely on you."

"But what is it you wish?"

"When Mrs. Turner has brought in the tray I will make it clear to you. Now," he said as he turned hungrily on the simple fare that our landlady had provided, "I must discuss it while I eat, for I have not much time. It is nearly five now. In two hours we must be on the scene of action. Miss Irene, or Madame, rather, returns from her drive at seven. We must be at Briony Lodge to meet her."

"And what then?"

"You must leave that to me. I have already arranged what is to occur. There is only one point on which I must insist. You must not interfere, come what may. You understand?"

"I am to be neutral?"

"To do nothing whatever. There will probably be some small unpleasantness. Do not join in it. It will end in my being conveyed into the house. Four or five minutes afterwards the sitting-room window will open. You are to station yourself close to that open window."

"Yes."

"You are to watch me, for I will be visible to you."

“Yes.”

“And when I raise my hand—so—you will throw into the room what I give you to throw, and will, at the same time, raise the cry of fire. You quite follow me?”

“Entirely.”

“It is nothing very formidable,” he said, taking a long cigar-shaped roll from his pocket. “It is an ordinary plumber’s smoke-rocket, fitted with a cap at either end to make it self-lighting. Your task is confined to that. When you raise your cry of fire, it will be taken up by quite a number of people. You may then walk to the end of the street, and I will rejoin you in ten minutes. I hope that I have made myself clear?”

“I am to remain neutral, to get near the window, to watch you, and at the signal to throw in this object, then to raise the cry of fire, and to wait you at the corner of the street.”

“Precisely.”

“Then you may entirely rely on me.”

“That is excellent. I think, perhaps, it is almost time that I prepare for the new role I have to play.”

He disappeared into his bedroom and returned in a few minutes in the character of an amiable and simple-minded Nonconformist clergyman. His broad black hat, his baggy trousers, his white tie, his sympathetic smile, and general look of peering and benevolent curiosity were such as Mr. John Hare alone could have equalled. It was not merely that Holmes changed his costume. His expression, his manner, his very soul seemed to vary with every fresh part that he assumed. The stage lost a fine actor, even as science lost an acute reasoner, when he became a specialist in crime.

It was a quarter past six when we left Baker Street, and it still wanted ten minutes to the hour when we found ourselves in Serpentine Avenue. It was already dusk, and the lamps were just being lighted as we paced up and down in front of Briony Lodge, waiting for the coming of its occupant. The house was just such as I had pictured it from Sherlock Holmes’ succinct description, but the locality appeared to be less private than I expected. On the contrary, for a small street in a quiet neighbourhood, it was remarkably animated. There was a group of shabbily dressed men smoking and laughing in a corner, a scissors-grinder with his wheel, two guardsmen who were flirting with a nurse-girl, and several well-dressed young men who were lounging up and down with cigars in their mouths.

“You see,” remarked Holmes, as we paced to and fro in front of the house, “this marriage rather simplifies matters. The photograph becomes a double-edged weapon now. The chances are that she would be as averse

to its being seen by Mr. Godfrey Norton, as our client is to its coming to the eyes of his princess. Now the question is, Where are we to find the photograph?”

“Where, indeed?”

“It is most unlikely that she carries it about with her. It is cabinet size. Too large for easy concealment about a woman’s dress. She knows that the King is capable of having her waylaid and searched. Two attempts of the sort have already been made. We may take it, then, that she does not carry it about with her.”

“Where, then?”

“Her banker or her lawyer. There is that double possibility. But I am inclined to think neither. Women are naturally secretive, and they like to do their own secreting. Why should she hand it over to anyone else? She could trust her own guardianship, but she could not tell what indirect or political influence might be brought to bear upon a business man. Besides, remember that she had resolved to use it within a few days. It must be where she can lay her hands upon it. It must be in her own house.”

“But it has twice been burgled.”

“Pshaw! They did not know how to look.”

“But how will you look?”

“I will not look.”

“What then?”

“I will get her to show me.”

“But she will refuse.”

“She will not be able to. But I hear the rumble of wheels. It is her carriage. Now carry out my orders to the letter.”

As he spoke the gleam of the side-lights of a carriage came round the curve of the avenue. It was a smart little landau which rattled up to the door of Briony Lodge. As it pulled up, one of the loafing men at the corner dashed forward to open the door in the hope of earning a copper, but was elbowed away by another loafer, who had rushed up with the same intention. A fierce quarrel broke out, which was increased by the two guardsmen, who took sides with one of the loungers, and by the scissors-grinder, who was equally hot upon the other side. A blow was struck, and in an instant the lady, who had stepped from her carriage, was the centre of a little knot of flushed and struggling men, who struck savagely at each other with their fists and sticks. Holmes dashed into the crowd to protect the lady; but just as he reached her he gave a cry and dropped to the ground, with the blood running freely down his face. At his fall the guardsmen took to their heels in one direction and the loungers in the other, while a number of



better-dressed people, who had watched the scuffle without taking part in it, crowded in to help the lady and to attend to the injured man. Irene Adler, as I will still call her, had hurried up the steps; but she stood at the top with her superb figure outlined against the lights of the hall, looking back into the street.

“Is the poor gentleman much hurt?” she asked.

“He is dead,” cried several voices.

“No, no, there’s life in him!” shouted another. “But he’ll be gone before you can get him to hospital.”

“He’s a brave fellow,” said a woman. “They would have had the lady’s purse and watch if it hadn’t been for him. They were a gang, and a rough one, too. Ah, he’s breathing now.”

“He can’t lie in the street. May we bring him in, marm?”

“Surely. Bring him into the sitting-room. There is a comfortable sofa. This way, please!”

Slowly and solemnly he was borne into Briony Lodge and laid out in the principal room, while I still observed the proceedings from my post by the window. The lamps had been lit, but the blinds had not been drawn, so that I could see Holmes as he lay upon the couch. I do not know whether he was seized with compunction at that moment for the part he was playing, but I know that I never felt more heartily ashamed of myself in my life than when I saw the beautiful creature against whom I was conspiring, or the grace and kindness with which she waited upon the injured man. And yet it would be the blackest treachery to Holmes to draw back now from the part which he had intrusted to me. I hardened my heart, and took the smoke-rocket from under my ulster. After all, I thought, we are not injuring her. We are but preventing her from injuring another.

Holmes had sat up upon the couch, and I saw him motion like a man who is in need of air. A maid rushed across and threw open the window. At the same instant I saw him raise his hand and at the signal I tossed my rocket into the room with a cry of “Fire!” The word was no sooner out of my mouth than the whole crowd of spectators, well dressed and ill-gentlemen, ostlers, and servant-maids—joined in a general shriek of “Fire!” Thick clouds of smoke curled through the room and out at the open window. I caught a glimpse of rushing figures, and a moment later the voice of Holmes from within assuring them that it was a false alarm. Slipping through the shouting crowd I made my way to the corner of the street, and in ten minutes was rejoiced to find my friend’s arm in mine, and to get away from the scene of uproar. He walked swiftly and in silence for some few minutes until we had turned down one of the quiet streets which lead towards the

Edgeware Road.

"You did it very nicely, Doctor," he remarked. "Nothing could have been better. It is all right."

"You have the photograph?"

"I know where it is."

"And how did you find out?"

"She showed me, as I told you she would."

"I am still in the dark."

"I do not wish to make a mystery," said he, laughing. "The matter was perfectly simple. You, of course, saw that everyone in the street was an accomplice. They were all engaged for the evening."

"I guessed as much."

"Then, when the row broke out, I had a little moist red paint in the palm of my hand. I rushed forward, fell down, clapped my hand to my face, and became a piteous spectacle. It is an old trick."

"That also I could fathom."

"Then they carried me in. She was bound to have me in. What else could she do? And into her sitting-room, which was the very room which I suspected. It lay between that and her bedroom, and I was determined to see which. They laid me on a couch, I motioned for air, they were compelled to open the window, and you had your chance."

"How did that help you?"

"It was all-important. When a woman thinks that her house is on fire, her instinct is at once to rush to the thing which she values most. It is a perfectly overpowering impulse, and I have more than once taken advantage of it. In the case of the Darlington substitution scandal it was of use to me, and also in the Arnsworth Castle business. A married woman grabs at her baby; an unmarried one reaches for her jewel-box. Now it was clear to me that our lady of to-day had nothing in the house more precious to her than what we are in quest of. She would rush to secure it. The alarm of fire was admirably done. The smoke and shouting were enough to shake nerves of steel. She responded beautifully. The photograph is in a recess behind a sliding panel just above the right bell-pull. She was there in an instant, and I caught a glimpse of it as she half-drew it out. When I cried out that it was a false alarm, she replaced it, glanced at the rocket, rushed from the room, and I have not seen her since. I rose, and, making my excuses, escaped from the house. I hesitated whether to attempt to secure the photograph at once; but the coachman had come in, and as he was watching me narrowly it seemed safer to wait. A little over-precipitance may ruin all."

"And now?" I asked.

“Our quest is practically finished. I shall call with the King to-morrow, and with you, if you care to come with us. We will be shown into the sitting-room to wait for the lady, but it is probable that when she comes she may find neither us nor the photograph. It might be a satisfaction to his Majesty to regain it with his own hands.”

“And when will you call?”

“At eight in the morning. She will not be up, so that we shall have a clear field. Besides, we must be prompt, for this marriage may mean a complete change in her life and habits. I must wire to the King without delay.”

We had reached Baker Street and had stopped at the door. He was searching his pockets for the key when someone passing said:

“Good-night, Mister Sherlock Holmes.”

There were several people on the pavement at the time, but the greeting appeared to come from a slim youth in an ulster who had hurried by.

“I’ve heard that voice before,” said Holmes, staring down the dimly lit street. “Now, I wonder who the deuce that could have been.”

///.

I slept at Baker Street that night, and we were engaged upon our toast and coffee in the morning when the King of Bohemia rushed into the room.

“You have really got it!” he cried, grasping Sherlock Holmes by either shoulder and looking eagerly into his face.

“Not yet.”

“But you have hopes?”

“I have hopes.”

“Then, come. I am all impatience to be gone.”

“We must have a cab.”

“No, my brougham is waiting.”

“Then that will simplify matters.” We descended and started off once more for Briony Lodge.

“Irene Adler is married,” remarked Holmes.

“Married! When?”

“Yesterday.”

“But to whom?”

“To an English lawyer named Norton.”

“But she could not love him.”

“I am in hopes that she does.”

“And why in hopes?”

“Because it would spare your Majesty all fear of future annoyance. If the lady loves her husband, she does not love your Majesty. If she does not

love your Majesty, there is no reason why she should interfere with your Majesty's plan."

"It is true. And yet—Well! I wish she had been of my own station! What a queen she would have made!" He relapsed into a moody silence, which was not broken until we drew up in Serpentine Avenue.

The door of Briony Lodge was open, and an elderly woman stood upon the steps. She watched us with a sardonic eye as we stepped from the brougham.

"Mr. Sherlock Holmes, I believe?" said she.

"I am Mr. Holmes," answered my companion, looking at her with a questioning and rather startled gaze.

"Indeed! My mistress told me that you were likely to call. She left this morning with her husband by the 5:15 train from Charing Cross for the Continent."

"What!" Sherlock Holmes staggered back, white with chagrin and surprise. "Do you mean that she has left England?"

"Never to return."

"And the papers?" asked the King hoarsely. "All is lost."

"We shall see." He pushed past the servant and rushed into the drawing-room, followed by the King and myself. The furniture was scattered about in every direction, with dismantled shelves and open drawers, as if the lady had hurriedly ransacked them before her flight. Holmes rushed at the bell-pull, tore back a small sliding shutter, and, plunging in his hand, pulled out a photograph and a letter. The photograph was of Irene Adler herself in evening dress, the letter was superscribed to "Sherlock Holmes, Esq. To be left till called for." My friend tore it open and we all three read it together. It was dated at midnight of the preceding night and ran in this way:

"MY DEAR MR. SHERLOCK HOLMES,—You really did it very well. You took me in completely. Until after the alarm of fire, I had not a suspicion. But then, when I found how I had betrayed myself, I began to think. I had been warned against you months ago. I had been told that if the King employed an agent it would certainly be you. And your address had been given me. Yet, with all this, you made me reveal what you wanted to know. Even after I became suspicious, I found it hard to think evil of such a dear, kind old clergyman. But, you know, *I have been trained* as an actress myself. Male costume is nothing new to me. I often take advantage of the freedom which it gives. I sent John, the coachman, to watch you, ran up stairs, got into my walking-clothes, as I call them, and came down just as you departed.

PLAIN TEXT

“Well, I followed you to your door, and so made sure that I was really an object of interest to the celebrated Mr. Sherlock Holmes. Then I, rather imprudently, wished you good-night, and started for the Temple to see my husband.

“We both thought the best resource was flight, when pursued by so formidable an antagonist; so you will find the nest empty when you call to-morrow. As to the photograph, your client may rest in peace. I love and am loved by a better man than he. The King may do what he will without hindrance from one whom he has cruelly wronged. I keep it only to safeguard myself, and to preserve a weapon which will always secure me from any steps which he might take in the future. I leave a photograph which he might care to possess; and I remain, dear Mr. Sherlock Holmes,

“Very truly yours,
“IRENE NORTON, née ADLER.”

“What a woman—oh, what a woman!” cried the King of Bohemia, when we had all three read this epistle. “Did I not tell you how quick and resolute she was? Would she not have made an admirable queen? Is it not a pity that she was not on my level?”

“From what I have seen of the lady she seems indeed to be on a very different level to your Majesty,” said Holmes coldly. “I am sorry that I have not been able to bring your Majesty’s business to a more successful conclusion.”

“On the contrary, my dear sir,” cried the King; “nothing could be more successful. I know that her word is inviolate. The photograph is now as safe as if it were in the fire.”

“I am glad to hear your Majesty say so.”

“I am immensely indebted to you. Pray tell me in what way I can reward you. This ring—” He slipped an emerald snake ring from his finger and held it out upon the palm of his hand.

“Your Majesty has something which I should value even more highly,” said Holmes.

“You have but to name it.”

“This photograph!”

The King stared at him in amazement.

“Irene’s photograph!” he cried. “Certainly, if you wish it.”

“I thank your Majesty. Then there is no more to be done in the matter. I

have the honour to wish you a very good-morning." He bowed, and, turning away without observing the hand which the King had stretched out to him, he set off in my company for his chambers.

And that was how a great scandal threatened to affect the kingdom of Bohemia, and how the best plans of Mr. Sherlock Holmes were beaten by a woman's wit. He used to make merry over the cleverness of women, but I have not heard him do it of late. And when he speaks of Irene Adler, or when he refers to her photograph, it is always under the honourable title of the woman.





Headings

This chapter shows a variety of headings, from third- to sixth-level headings. For first-level headings, see the main part headings in this book, e.g. ‘Text’ (page 1). For second-level headings, see the secondary chapter headings like ‘Headings’ above this paragraph.

In the next few pages, we’ll include:

- short headings with no text between them
- long headings with no text between them
- short headings with text between them
- long headings with text between them.

An h3

An h4

An h5

An h6

I am a third-level heading, hear me roar

May the fourth-level heading be with you

Who ever uses fifth-level headings anyway?

Ah ye olde sixth-level heading, for lawyers and other section nesters

The h3

And now, to see how short and long headings interact with text, we’ll use the introduction to Concepción Arenal’s *La Igualdad Social y Política*.

Basta considerar la frecuencia con que se habla de igualdad, el calor con que se discute, la multitud de personas que toman parte en la discusión ó se interesan en ella, la vehemencia con que se ataca y se defiende, la pertinacia con que se afirma ó se niega, la confianza con que se invoca como un medio de salvación, el horror con que se rechaza como una causa de ruina; basta observar estos contrastes, no sólo[6] reproducidos, sino crecientes, para sospechar que la igualdad no es una de esas ideas fugaces que pasan con las circunstancias que las han producido, sino que tiene raíces profundas



en la naturaleza del hombre, y es, por lo tanto, un elemento poderoso y permanente de las sociedades humanas.

Esta sospecha se confirma, pasando á convencimiento, al ver en la historia la igualdad luchando con el privilegio; vencida, no exterminada, rebelarse cuando se la creía para siempre bajo el yugo; existir, si no en realidad, en idea y esperanza, y, derecho ó aspiración, aparecer en todo pueblo que tiene poderosos gérmenes de vida.

An h4

Aspiración generosa, instinto depravado, impulso ciego, deseo razonable, sueño loco; bajo todas estas formas se presenta la igualdad, ya matrona venerable, con balanza equitativa como la justicia, ya furia, que agita en sus manos rapaces tea incendiaria.

La igualdad en la abyección; la igualdad en el derecho; un populacho vil que quiere pasar, sobre todos, el nivel de su ignominia; un pueblo digno que se opone á que la justicia sea privilegio: el pensador, buen amigo de las mul[7]titudes, que procura ilustrarlas; el fanático ó el ambicioso, que las extravía, todos hablan de igualdad, aunque cada uno la comprenda de distinta manera.

An h5

Esta diferencia en el modo de concebir una misma cosa se observa en otras muchas; pero tal vez en ninguna es más perceptible que en la igualdad, porque no hay quizás aspiración que tan fácilmente pase de razonable á absurda, cuyos verdaderos límites sean tan fáciles de traspasar, que se ramifique y extienda tanto á todas las esferas de la vida, ni que haga tan estrecha alianza con una pasión implacable y vil: la envidia. La envidia enciende sus rencores y destila su veneno en los individuos y en las multitudes que convierten la igualdad en bandera de exterminio, y por eso son á veces tan sordas á la voz de la razón y á las súplicas de la misericordia.

Estudiando la igualdad en el pasado, no se la ve seguir un curso más ó menos rápido, más ó menos regular; su brillo no crece con las luces de la inteligencia; su marcha no es paralela á la del progreso humano: tiene resplandores de relámpago, movimientos vertiginosos, y á veces cada paso asemeja á una erupción.[8] Esto no es decir que carezca de ley, no; el huracán y la tempestad tienen la suya; pero es considerar cuán difícil ha de ser la observación de un fenómeno relacionado con tantos otros, y que no puede conocerse bien sino conociéndolos todos.

An h6

Mas por dificultoso que sea el estudio, parece necesario; la igualdad no se invoca ya por unos pocos, sino por el mayor número; no se limita á una ú otra esfera de la vida, pretende invadirlas todas, y sin saber lo que es, ni los obstáculos que halla, ni el modo de vencerlos, se pretende suprimir el tiempo necesario, el trabajo indispensable, y supliendo la fuerza con la violencia, lograr instantáneamente lo que sólo se realizará en el porvenir, ó lo que no podrá realizarse nunca. Estas aspiraciones las tiene el que padece, con la impaciencia de quien sufre, con la cólera del que halla un remedio ó un alivio que supone negado por la injusticia y el egoísmo. Y no son cientos ni miles, sino millones de cóleras impacientes y doloridas, que piden á la igualdad un recurso para su penuria y una satisfacción para su amor propio. Y estos millones de impacientes iracundos comunican entre sí; es decir, que mul[9]tiplican su impaciencia y su ira, que, contenida á intervalos, y á intervalos desenfrenada, es amenazadora siempre.

Enfrente de los que esperan en la igualdad están los que la temen, los que ven en ella una cosa monstruosa, imposible, absurda, injusta; un sueño de la fiebre popular, un producto de las malas pasiones de la plebe, ó un medio de explotarlas. Para éstos, la igualdad es sinónimo de anarquía, de caos, de degradación, hasta el punto que igualarse viene á ser rebajarse, y persona distinguida equivale á persona digna.

I am a third-level heading, hear me roar

El antagonismo no puede ser más evidente: lo que para éstos es un atentado, para aquéllos es un derecho; aberración para unos, dogma para otros.

Los dogmas se creen; los partidarios de la igualdad, las multitudes al menos, creen en ella, la afirman con la seguridad del que no ha pensado, con la vehemencia del que espera, y, como todo ignorante que sea apasionado, están dispuestos á imponer la creencia.

May the fourth-level heading be with you

El dogmatismo que suele aplicarse á las cosas espirituales aquí interviene en las materiales, y no tiene un reducido número de oráculos en el aula ó en el templo, sino que abre cátedra donde[10] quiera, en calles y plazas, en caminos y en veredas. El dogmatismo filosófico y religioso tiene máximas y preceptos que son promesas, reglas que enfrenan las pasiones, y aunque influya en las cosas materiales, no se dirige tan inmediata y directamente á ellas como el dogma de la igualdad. No se trata ya sólo de ser todos igualmente hijos de Dios, que no hará más distinción que entre justos y pecadores; de ser juzgados por la misma ley penal, y de suprimir todo

HEADINGS

privilegio en la política, sino de promulgar la económica de modo que desaparezcan las diferencias en las cosas que importan más, porque no se da tanto valor á tener voto en los comicios como pan y comodidades en casa. La insurrección económica, la huelga, es la más frecuente, casi la única, y manifiesta adónde se quiere aplicar el nivel con más empeño.

Who ever uses fifth-level headings anyway?

Mientras otros dogmas pierden prestigio, el de la igualdad aumenta el número de sus prosélitos, y extiende su acción en cada individuo; no hay fenómeno social en que no aparezca su influencia, difícil determinar hasta dónde llegará, al menos como aspiración. ¿Quién pone límites á la fe y á la esperanza?

Ah ye olde sixth-level heading, for lawyers and other section nesters

Y, no obstante, se comprende la necesidad[11] de ponerlos cuando la esperanza y la fe no se alimentan de espirituales promesas para otra vida, sino que quieren realizarse en ésta con la posesión inmediata de ventajas positivas y materiales bienes. Agréguese que éstos no se buscan siempre por la persuasión, sino recurriendo á la fuerza, y es de temer que la apelación á ella se repita más y más si no se contiene la fermentación de las impacencias. Uno de los medios de contenerlas es discutir las; citar ante el tribunal de la razón á los contendientes; oírlos con imparcialidad; no negar el derecho porque sea nuevo ni porque sea viejo, sino atendiendo á la justicia; precaverse contra la pasión, que no siempre es vocinglera; contra el egoísmo, que puede ser cínico ó hipócrita; determinar bien los puntos esenciales que se discuten para quitar al asunto mucho de lo vago que hoy tiene; señalar las contradicciones que hayan podido pasar desapercibidas, pero que en los hechos dan lugar á choques y conflictos, y de este modo contribuir á que, respecto á la igualdad, se tenga opinión que se discute, un elemento social que se analiza, y no un dogma que se impone ó un arma con que se amenaza.

Quotes

Everyone loves a good quotation. Good designs should support blockquotes, and might support pullquotes as a class of blockquote. Ideally, the source of a quotation should have a style, too.

Blockquotes

Generally, blockquotes are set off from body text with line spaces above and below, and an indent left and right. What does this design do with a blockquote? Thanks to The Quote Garden (<https://www.quote garden.com/quotations.html>) for these quotes about quotes.

Next to the originator of a good sentence is the first quoter of it. Many will read the book before one thinks of quoting a passage. As soon as he has done this, that line will be quoted east and west.

That's from Ralph Waldo Emerson's "Quotation and Originality" in *Letters and Social Aims*. How about blockquotes with two or three paragraphs?

While reading writers of great formulatary power — Henry James, Santayana, Proust — I find I can scarcely get through a page without having to stop to record some lapidary sentence.

Reading Henry James, for example, I have muttered to myself, "C'mon, Henry, turn down the brilliance a notch, so I can get some reading done." I may be one of a very small number of people who have developed writer's cramp while reading.

That's from Joseph Epstein's "Quotatious" in *A Line Out for a Walk*.

Whatever is felicitously expressed risks being worse expressed: it is a wretched taste to be gratified with mediocrity when the excellent lies before us.

We quote, to save proving what has been demonstrated, referring to where the proofs may be found.

We quote, to screen ourselves from the odium of doubtful opinions, which the world would not willingly accept from ourselves; and we may quote from the curiosity which only a quotation itself can give, when in our own words it would be divested of that tint of ancient phrase, that detail of narrative, and that naïveté, which we have for ever lost, and which we like to recollect once had an existence.



QUOTES

That's from Isaac D'Israeli's 'Quotation' in *A Second Series of Curiosities of Literature, Volume I*.

How about two separate but consecutive blockquotes?

I love quotations because it is a joy to find thoughts one might have, beautifully expressed with much authority by someone recognized wiser than oneself.

The quoting of an aphorism, like the angry barking of a dog or the smell of overcooked broccoli, rarely indicates that something helpful is about to happen.

Those are, respectively, Marlene Dietrich and Lemony Snicket (*The Vile Village*).

And just to be thorough, what about two separate, consecutive blockquotes each with more than one paragraph?

My readers, who may at first be apt to consider Quotation as downright pedantry, will be surprised when I assure them, that next to the simple imitation of sounds and gestures, Quotation is the most natural and most frequent habitude of human nature.

For, Quotation must not be confined to passages adduced out of authors. He who cites the opinion, or remark, or saying of another, whether it has been written or spoken, is certainly one who quotes; and this we shall find to be universally practiced.

What is all wisdom save a collection of platitudes? Take fifty of our current proverbial sayings—they are so trite, so threadbare, that we can hardly bring our lips to utter them. None the less they embody the concentrated experience of the race, and the man who orders his life according to their teaching cannot go far wrong.

How easy that seems! Has any one ever done so?

Never. Has any man ever attained to inner harmony by pondering the experience of others? Not since the world began! He must pass through the fire.

The first there is James Boswell in 'The Hypochondriack' (*The London Magazine: Or, Gentleman's Monthly Intelligencer*, June 1779). The second, Norman Douglas in *South Wind* (1921).

Pullquotes

Pull quotes are generally very short, punchy quotes from the surrounding text. They give readers a teasing hint at what's inside. There isn't an HTML element for pullquote, so we use a blockquote with a `pullquote` class.

Short, punchy ... a teasing hint at what's inside.

Some publications use pullquotes as part of the flowing text, rather than phrases extracted from the body. That may or may not be a good idea.

Quote sources

Sometimes the source of a quote is included with the quote, and needs to be styled specially. For that, we use a `source` class. That might be applied to an inline or a block element. Here as a paragraph class (block element):

Collecting quotations is an insidious, even embarrassing habit, like ragpicking or hoarding rocks or trying on other people's laundry. I got into it originally while trying to break an addiction to candy. I kicked candy and now seem to be stuck with quotations, which are attacking my brain instead of my teeth.

Robert Byrne, 'Sources, References, and Notes' in *The Other 637 Best Things Anybody Ever Said*, 1984

And here as an inline `<em>` element:

The multiplicity of facts and writings is become so great that every thing must soon be reduced to extracts and dictionaries.
Voltaire

And after these two quotations, just to see how consecutive quotations with block-element sources behave:

I enjoy collecting quotations. When I find a choice one I pounce on it like a lepidopterist. My day is made. When I lose one because I did not copy it out at once I feel bereft.

R.I. Fitzhenry, preface to *The David & Charles Book of Quotations*, September 1981

Books are the beehives of thought; laconics, the honey taken from

QUOTES

them.

James Ellis, quoted in *Edge-Tools of Speech* by Maturin M. Ballou, 1899



Lists

In HTML, there are three main kinds of lists: ordered lists (usually numbered), unordered lists (i.e. bullet lists) and definition lists (like a dictionary of entry words and their explanations).

In this chapter, an assortment of lists alone and in combination with each other and with surrounding text.

Simple lists

Here are the opening paragraphs of Charles Dickens's *A Tale of Two Cities*, as a list.

- It was the best of times,
- it was the worst of times,
- it was the age of wisdom,
- it was the age of foolishness,
- it was the epoch of belief,
- it was the epoch of incredulity,
- it was the season of Light,
- it was the season of Darkness,
- it was the spring of hope,
- it was the winter of despair,
- we had everything before us,
- we had nothing before us,
- we were all going direct to Heaven,
- we were all going direct the other way—
- in short, the period was so far like the present period, that some of
- its noisiest authorities insisted on its being received, for good or for
- evil, in the superlative degree of comparison only.

And the following paragraph as a numbered list:

1. There were a king with a large jaw and a queen with a plain face, on the throne of England.
2. There were a king with a large jaw and a queen with a fair face, on the throne of France.
3. In both countries it was clearer than crystal to the lords of the State preserves of loaves and fishes, that things in general were settled for ever.

Let's ruin some more of that fine book to show nested lists:

- It was the year of Our Lord one thousand seven hundred and seventy-five.
- Spiritual revelations were conceded to England at that favoured period,

LISTS

as at this.

We might have a paragraph here, followed by a nested list:

- Mrs. Southcott had recently attained her five-and-twentieth blessed birthday, of whom a prophetic private in the Life Guards had heralded the sublime appearance by announcing that arrangements were made for the swallowing up of London and Westminster.
- Even the Cock-lane ghost had been laid only a round dozen of years, after rapping out its messages, as the spirits of this very year last past (supernaturally deficient in originality) rapped out theirs.
- Mere messages in the earthly order of events had lately come to the English Crown and People, from a congress of British subjects in America: which, strange to relate, have proved more important to the human race than any communications yet received through any of the chickens of the Cock-lane brood.
- France, less favoured on the whole as to matters spiritual than her sister of the shield and trident, rolled with exceeding smoothness down hill, making paper money and spending it.
- Under the guidance of her Christian pastors, she entertained herself, besides, with such humane achievements as sentencing a youth to have his hands cut off, his tongue torn out with pincers, and his body burned alive, because he had not kneeled down in the rain to do honour to a dirty procession of monks which passed within his view, at a distance of some fifty or sixty yards.
- It is likely enough that, rooted in the woods of France and Norway, there were growing trees, when that sufferer was put to death, already marked by the Woodman, Fate, to come down and be sawn into boards, to make a certain movable framework with a sack and a knife in it, terrible in history.

We might have a paragraph here, followed by another list item:

- It is likely enough that in the rough outhouses of some tillers of the heavy lands adjacent to Paris, there were sheltered from the weather that very day, rude carts, bespattered with rustic mire, snuffed about by pigs, and roosted in by poultry, which the Farmer, Death, had already set apart to be his tumbrils of the Revolution.
- But that Woodman and that Farmer, though they work unceasingly, work silently, and no one heard them as they went about with muffled tread: the rather, forasmuch as to entertain any suspicion that they were awake, was to be atheistical and traitorous.

1. In England, there was scarcely an amount of order and protection to

justify much national boasting.

- a. Daring burglaries by armed men, and highway robberies, took place in the capital itself every night;
 - b. families were publicly cautioned not to go out of town without removing their furniture to upholsterers' warehouses for security;
(We might have a paragraph here, followed by a nested list.)
 - i. the highwayman in the dark was a City tradesman in the light, and, being recognised and challenged by his fellow-tradesman whom he stopped in his character of "the Captain," gallantly shot him through the head and rode away;
 - ii. the mail was waylaid by seven robbers, and the guard shot three dead, and then got shot dead himself by the other four, "in consequence of the failure of his ammunition:" after which the mail was robbed in peace;
 - iii. that magnificent potentate, the Lord Mayor of London, was made to stand and deliver on Turnham Green, by one highwayman, who despoiled the illustrious creature in sight of all his retinue;
 - c. prisoners in London gaols fought battles with their turnkeys, and the majesty of the law fired blunderbusses in among them, loaded with rounds of shot and ball;
 - d. thieves snipped off diamond crosses from the necks of noble lords at Court drawing-rooms;
 - e. musketeers went into St. Giles's, to search for contraband goods, and the mob fired on the musketeers, and the musketeers fired on the mob, and nobody thought any of these occurrences much out of the common way.
2. In the midst of them, the hangman, ever busy and ever worse than useless, was in constant requisition; now, stringing up long rows of miscellaneous criminals; now, hanging a housebreaker on Saturday who had been taken on Tuesday; now, burning people in the hand at Newgate by the dozen, and now burning pamphlets at the door of Westminster Hall; to-day, taking the life of an atrocious murderer, and to-morrow of a wretched pilferer who had robbed a farmer's boy of sixpence.

Chapter contents

In kramdown, you can get a list of all the headings in a file by adding the tag `{ :toc }` after a single list item. They automatically get an id of `markdown-toc`. How does this design handle those in-chapter tables of contents?

- Simple lists
- Chapter contents

LISTS

- Bibliographies
- Definitions and glossaries

You can exclude a heading from that in-chapter TOC by giving it a `no_toc` class, which we've done to the first 'Lists' heading in this chapter.

Bibliographies

In book-publishing convention, lists of references – known as bibliographies – usually have a particular layout. Most often, they have no marker (bullet or number) and a hanging indent, making it easy to scan their initial words, which are usually the surnames of the authors. We mark up bibliographies by giving a list the class `bibliography`. How does this design handle a bibliography? Here are Maria Popova's ten essential books on typography.

Ruder, Emil: *Typographie: A Manual of Design*

Rjeily, Rana Abou: *Cultural Connectives*

Bringhurst, Robert: *The Elements of Typographic Style*

Lupton, Ellen: *Thinking with Type: A Critical Guide for Designers, Writers, Editors, & Students*

Bantjes, Marian: *I Wonder*

Garfield, Simon: *Just My Type: A Book About Fonts*

Gill, Eric: *An Essay on Typography*

Fili, Louise: *Scripts: Elegant Lettering from Design's Golden Age*

Tholenaar, Jan: *Type: A Visual History of Typefaces and Graphic Styles, Vol. 1*

FL@33: *The 3D Type Book*

What about bibliographies with really long, complicated titles? Since we're using Jekyll, here are some things others have written while working with things called Jekyll – MLA style.

Hustis, Harriet. "Hyding Nietzsche in Robert Louis Stevenson's Gothic of Philosophy." *Studies in English Literature, 1500-1900* 49.4 (2009): 993-1007. Web.

Van Valkenburgh, Michael R. "The Flower Gardens of Gertrude Jekyll and Their Twentieth-Century Transformations." *Design Quarterly* 137 (1987): 1-32. Web.

Radchuk, Volodymyr, Borisjuk Ljudmilla, Radchuk Ruslana, Steinbiss Hans-Henning, Rolletschek Hardy, Broeders Sylvia, and Wobus Ulrich. "Jekyll Encodes a Novel Protein Involved in the Sexual Reproduction of Barley." *The Plant Cell* 18.7 (2006): 1652-666. Web.

Walsh, Emily C., and Didier Y. R. Stainier. "UDP-Glucose Dehydrogenase Required for Cardiac Valve Formation in Zebrafish." *Science* 293.5535 (2001): 1670-673. Web.

Wilkins, Thomas Hart. "Sir Joseph Jekyll and His Impact on Oglethorpe's Georgia." *The Georgia Historical Quarterly* 91.2 (2007): 119-34. Web.

Definitions and glossaries

In HTML definition lists are useful semantic structures, especially for things like glossaries. Here is a basic definition list, using definitions of new words from OxfordDictionaries.com in 2014.

acqui hire, n.

buying out a company primarily for the skills and expertise of its staff...

adorbs, adj.

(informal) arousing great delight; cute or adorable

air punch, n.

thrusting one's clenched fist up into the air, typically as a gesture of triumph...

amaze balls, adj.

(informal) very impressive, enjoyable, or attractive

anti-vax, adj.

(US informal) opposed to vaccination

Bank of Mum and Dad, phr.

(Brit. informal) a person's parents regarded as source of financial assistance

bare, adv.

(Brit. informal) very; really (used as an intensifier)

bedroom tax, n.

(in the UK) informal name for a measure introduced in the Welfare Reform Act 2012...

binge-watch, v.

(informal) watch multiple episodes of (a television programme) in rapid

LISTS

succession...

bro hug, n.

(US informal) friendly embrace between two men

clickbait, n.

(informal) (on the Internet) content...whose main purpose is to attract attention and draw visitors to a particular web page

cord cutting, n.

(informal) practice of cancelling a pay television subscription or landline phone connection in favour of an alternative Internet-based or wireless service

cotch, v.

(Brit. informal) spend time relaxing; stay or sleep somewhere on a temporary basis

cray, adj. (also cray cray)

(US informal) crazy



Tables

Here is a simple table with a caption. In markdown/HTML, the caption is a paragraph with a class of `.table-caption`.

Table 5. Subjective rankings of the Wilcoxon tests while food-serving with pincers- and scissors-pinching for various experimental chopsticks

Rank	Pincers-pinching	Scissors-pinching	Overall
1	3 (2.5)	3 (2.2)	3 (2.3)
2	1 (2.7)	2 (3.3)	2 (3.2)
3	2 (3.1)	6 (3.3)	1 (3.2)
4	6 (3.7)	1 (3.7)	6 (3.5)
5	5 (4.4)	5 (4.3)	5 (4.3)
6	4 (4.9)	4 (4.6)	4 (4.8)

That is data from an actual scientific study into the effectiveness of different kinds of chopsticks and ways to hold them.

Here’s a bigger table containing some lists. This table shows the skeletal muscles in your forehead and eyelid.



TABLES

Muscle	Origin	Insertion	Nerve	Action
occipitofrontalis	2 occipital bellies 2 frontal bellies	galea aponeurotica	facial nerve	raises the eyebrows
occipitalis	- superior nuchal line of the occipital bone - mastoid part of the temporal bone	galea aponeurotica	posterior auricular nerve (facial nerve)	
frontalis	galea aponeurotica	skin of frontal region	facial nerve	wrinkles eyebrow
orbicularis oculi	1. frontal bone 2. medial palpebral ligament 3. lacrimal bone	lateral palpebral raphe	zygomatic branch of facial	closes eyelids
corrugator supercilii	superciliary arches	- forehead skin - near eyebrow	facial nerve	wrinkles forehead
depressor supercilii	medial orbital rim	medial aspect of bony orbit	facial nerve	Depresses the eyebrow

Boxes

You should be able to throw a box around any block-level element (paragraphs, lists, blockquotes and more) by giving it a `box` class. Using some sample text from Gertrude Chandler Warner's *The Box-Car Children*, here is a selection of elements with `box` classes, interspersed with text and other elements to see how they interact.

The Flight

About seven o'clock one hot summer evening a strange family moved into the little village of Middlesex. Nobody knew where they came from, or who they were. But the neighbors soon made up their minds what they thought of the strangers, for the father was very drunk. He could hardly walk up the rickety front steps of the old tumble-down house, and his thirteen-year-old son had to help him. Toward eight o'clock a pretty, capable-looking girl of twelve came out of the house and bought a loaf of bread at the baker's. And that was all the villagers learned about the newcomers that night.

"There are four children," said the bakeshop woman to her husband the next day, "and their mother is dead. They must have some money, for the girl paid for the bread with a dollar bill."

The default `.box` has a border and the same background colour as the page.

"Make them pay for everything they get," growled the baker, who was a hard man. "The father is nearly dead with drink now, and soon they will be only beggars."

This happened sooner than he thought. The next day the oldest boy and girl came to ask the bakeshop woman to come over. Their father was dead.

The shaded `.box.box-shaded` has a light-coloured border and background.

She went over willingly enough, for someone had to go. But it was clear that she did not expect to be bothered with four strange children, with the bakery on her hands and two children of her own.

The colour-reversed `.box.box-reversed` has an accent-coloured border and background with light text.

BOXES

Boxes can also be sidenotes, with the appropriate classes, such as this one, which has .sidenote.box.box reversed.

“Haven’t you any other folks?” she asked the children.

“We have a grandfather in Greenfield,” spoke up the youngest child before his sister could clap her hand over his mouth.

- “Hush, Benny,” she said anxiously.
- This made the bakeshop woman suspicious. “What’s the matter with your grandfather?” she asked.

- “He doesn’t like us,” replied the oldest boy reluctantly. “He didn’t want my father to marry my mother, and if he found us he would treat us cruelly.”

- “Did you ever see him?”
- “Jess has. Once she saw him.”

“Well, did he treat you cruelly?” asked the woman, turning upon Jess.

“Oh, he didn’t see me,” replied Jess. “He was just passing through our—where we used to live—and my father pointed him out to me.”

- “Where did you use to live?” went on the questioner. But none of the children could be made to tell.

- “We will get along all right alone, won’t we, Henry?” declared Jess.

- “Indeed we will!” said Henry.

- “I will stay in the house with you tonight,” said the woman at last, “and tomorrow we will see what can be done.”

The four children went to bed in the kitchen, and gave the visitor the only other bed in the house. They knew that she did not at once go to bed, but sat by the window in the dark. Suddenly they heard her talking to her husband through the open window.

- “They must go to their grandfather, that’s certain,” Jess heard her say.
- “Of course,” agreed her husband. “Tomorrow we will make them tell us what his name is.”
- Soon after that Jess and Henry heard her snoring heavily. They sat up in the dark.
- “Mustn’t we surely run away?” whispered Jess in Henry’s ear.

“Yes!” whispered Henry. “Take only what we need most. We must be far off before morning, or they will catch us.”

Jess sat still for a moment, thinking, for every motion she made must count.

This is a blockquote in a box, by the way. The next three boxes are also blockquotes in boxes with multiple paragraphs. First singly separated with body text, then consecutively.—Ed.

“I will take both loaves of bread,” she thought, “and Violet’s little workbag. Henry has his knife. And all Father’s money is in my pocket.” She drew it out and counted it in the dark, squinting her eyes in the faint light of the moon. It amounted to nearly four dollars.

“You’ll have to carry Benny until he gets waked up,” whispered Jess.
 “If we wake him up here, he might cry.”
 She touched Violet as she spoke.
 “Sh! Violet! Come! We’re going to run away,” she whispered.

The little girl made no sound. She sat up obediently and tried to make out the dim shadow of her sister.

“What shall I do?” she said, light as a breath.

“Carry this,” said Jess, handing her the workbag and a box of matches.

BOXES

Jess tiptoed over to the tin box on the table, drew out the two loaves of bread, and slipped them into the laundry bag. She peered around the room for the last time, and then dropped two small clean towels and a cake of soap into the bag.

“All right. Pick him up,” she said to Henry.

Henry bent over the sleeping child and lifted him carefully. Jess took the laundry bag, turned the doorknob ever so softly, opened the door ever so slowly, and they tiptoed out in a ghostly procession.

Jess shut the door with as much care as she had opened it, listened to the bakeshop woman’s heavy snoring for a moment, and then they turned and picked their way without a sound to the country road.

“She may wake up before morning, you know,” whispered Henry. “We must do our fastest walking before then. If we can only get to another town before they find out we’re gone, they won’t know which way to go.”

Jess agreed, and they all walked briskly along in the faint moonlight.

1. “How far can you carry Benny?” asked Violet.
2. “Oh, at least a mile,” said Henry confidently, although his arms were beginning to ache. Benny was five years old, and he was a fat, healthy boy as well.
 - a. “I think we could all walk faster if we woke him up,” said Jess decidedly. “We could each take his hand and almost carry him along.”
 - b. Henry knelt by the roadside and set the little fellow against his knee.
3. “Come, Benny, you must wake up now and walk!” said Jess coaxingly.
4. “Go away!” Benny mumbled with his eyes shut, trying to lie down again.
 - a. “Let me try,” Violet offered softly.
 - i. “Say, Benny, you know little Cinnamon Bear ran away to find a nice warm bed for the winter? Now, you play you’re Cinnamon, and Henry and Jess will help you along, and we’ll find a bed.”

- ii. Violet’s little plan worked. Benny was never too cross to

listen to the wonderful stories his sister Violet could tell about Cinnamon Bear. He stood up bravely and marched along, yawning, while his big brother and sister almost swung him between them.

iii. Not a soul passed them on the country road. All the houses they saw were dark and still. And when the first faint streaks of morning light showed in the sky, all four children were almost staggering with sleep.

5. "I must go to sleep, Henry," murmured Jess at last. Little Benny was asleep already, and Henry was carrying him again.
6. "The first place we come to, then," panted Henry.

Violet said nothing, but she kept her eyes open.

Finally she caught Henry's sleeve. "Couldn't we make that haystack do?" she asked, pointing across a newly mown field.

1. "Indeed we could," said Henry thankfully. "What a big, enormous one it is! I was too sleepy to see it, I guess."
2. "And see how far away from the farmhouse and barn it is, too!" echoed Jess.
 - a. The sight gave them new courage. They climbed over two stone walls, got across a brook somehow with the heavy child, and arrived at the haystack.
 - b. Henry laid his brother down and stretched his aching arms, while Jess began to burrow into the haystack. Violet, after a moment of watching her, did the same.
 - i. "Here's his nest," said Jess sleepily, taking her head out of the deep round hole she had made. Henry lifted the child into the opening and was pleased to see that he curled up instantly, smiling in his sleep.
 - ii. Jess pulled wisps of hay over the opening so that it was absolutely invisible, and then proceeded to dig out a similar burrow for herself.
 - c. "We can stay here just—as long—as we like, can't we, Henry?" she murmured, digging with her eyes shut.
3. "We sure can," replied Henry. "You're an old brick, Jess. Get in,



BOXES

and I'll pull the hay over the hole."

4. Violet was already curled up in her nest, which was hidden so completely that Henry spoke to her to see if she were there. Then he wriggled himself backward into the haycock without stopping to hollow it out, pulled a handful of hay over his head, and laid his head on his arm.

Just as he did so he heard a heavy voice say, "Now, then, lass, git along!" Then he heard the rumble of a milk wagon coming down a near-by lane, and he realized thankfully that they had hidden themselves just before the first farmer in the neighborhood had set off toward Middlesex with his milk cans.

"He will say he didn't meet us coming this way," thought Henry, "so they will hunt for us the other way. And that will give us time to cover a lot more ground."

He dropped asleep just as the roosters all over the valley began to answer each other.

Expandable boxes

Expandable boxes can be toggled open and closed. They allow you to group more content together on a page without overwhelming readers. They can contain an optional descriptive preview.

Expandable box A box that can expand

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Aliquam pulvinar condimentum dui a ullamcorper. Integer eu purus in dui gravida luctus. Nulla imperdiet ante odio, vitae feugiat quam rhoncus sit amet. Phasellus porta leo at massa semper, et placerat erat posuere. Etiam suscipit eu justo a tempus.

Suspendisse potenti. Vivamus dapibus eu lorem ut posuere. Vivamus at volutpat justo, eget efficitur sapien. Nunc ac bibendum lacus. Nunc egestas auctor ex, id aliquam erat convallis a. Etiam ac dui sed erat dictum vehicula vitae sit amet augue. Maecenas malesuada orci nisi, non aliquet elit fringilla at. Aliquam vitae dui at nisl elementum ullamcorper at ac ligula. Duis tristique sollicitudin ex, id mollis lacus pretium sed.

Ut odio nulla, tempor ac felis vitae, mollis luctus enim. Nam in elit ut ipsum hendrerit tincidunt. Maecenas condimentum sit amet lacus ut lacinia. Nulla facilisi. Vivamus ut malesuada dui, vitae blandit sem. Pellentesque venenatis malesuada orci non convallis. Mauris id commodo felis. Curabitur urna massa, volutpat quis elementum at, imperdiet vitae lorem. Aliquam blandit finibus nunc, at bibendum urna elementum a. Cras ut sapien vitae purus semper scelerisque non sit amet enim. Nam pharetra magna risus, id interdum sapien posuere sit amet. Pellentesque id dui erat. Aliquam ac condimentum metus. Proin sit amet consequat velit. Nunc mauris est, accumsan nec volutpat egestas, eleifend ac nulla.





Notes

In Electric Book designs, we aim to support four kinds of notes: endnotes, footnotes, sidenotes, and (what we call) footer notes.

- **Endnotes** appear at the end of a document (web page or book chapter). In technical markdown terms, these are called footnotes, because on the web they appear at the bottom of a web page. To create them in markdown, follow the kramdown syntax for footnotes:
 - put a `[^1]` where the footnote reference should appear (the 1 there can be any numbers or letters, and should be different for each footnote in a document);
 - anywhere in the document (we recommend after the paragraph containing the footnote reference), put `[^1]: Your footnote text here..`
- **Footnotes** appear at the bottom of the page. To distinguish these from what markdown calls ‘footnotes’, we sometimes call these ‘on-page footnotes’. To create these, use the same syntax as for endnotes (above), but add a `move-to-footnote` class to the footnote text. This section of the samples book demonstrates endnotes with a few on-page footnotes. We only support these in PDF output.
- **Sidenotes** appear in a box to the right of the text. On wide screens, they might float far right of the text. On narrower screens, the text might wrap around them. In print, the text might wrap around them, too, or they might appear in an otherwise empty sidebar space. To create a sidenote, give the paragraph a `{:.sidenote}` tag. If the sidenote includes more than one paragraph, put them in a `<div class="sidenote" markdown="1"> ... </div>`.
- **Footer notes**, which in print are sidenotes at the bottom of the page. By adding `.bottom` to the `{:.sidenote}` tag, your sidenote should sit at the bottom of the page rather than on the right with text wrap, replicating a traditional footnote. So the markdown looks like this:

```
This is a sidenote at the bottom of the page in
print.
{:.sidenote .bottom}
```

This only affects PDF output. In other outputs, these remain regular sidenotes.

- **Definitions** appear in the sidebar in print and as popup definitions on the web at the words in bold. The definition text is stored in the book’s



NOTES

`glossary.yml` file, and added to the page here with the `definition` `include`.

- The glossary at the end of the chapter draws its definitions from `glossary.yml`, and is included on the page with the `glossary include`.
- The bibliography at the end of the chapter draws its sources from the book's `bibliography.yml` file.

Principles of Network Architecture Emerging from Comparisons of the Cerebral Cortex in Large and Small Brains

To show notes in action, we need a lot of text with a bunch of notes. For that, we'll use 'Principles of Network Architecture Emerging from Comparisons of the Cerebral Cortex in Large and Small Brains' by Barbara L. Finlay, which is on PLOS, licenced CC BY.

isocortex The largest part of the cerebral cortex: a layered sheet of neurons. Sometimes just called the cortex, it is disproportionately large in primates and grows mainly in surface area as brain volume increases.

How do you scale up a computing device? The mammalian brain has been presented a massive challenge: to retain its basic divisions and connectivity, despite brain volumes ranging from less than a gram in the smallest shrew to over 8,000 grams in the largest whale. In the cerebral cortex alone (**isocortex** or cortex), the problem is amplified, as the isocortex scales in volume disproportionately with respect to the rest of the brain, especially in primates.¹ Understanding how a computing device might scale would obviously be easier if its computational function were known. The computations the cortex performs are not definitively known, however, though there is no shortage of theories on this subject. Because of the bootstrapping nature of research on evolution, a description of how the brain adapts its connectivity to massive increases in numbers of neurons, however, may shed light on its essential computational role. The organizational features the cortex has defended over several hundred million years and massive size differences are important clues to its essential computations.

The isocortex in mammals is a layered sheet of neurons that, according to one view, offers considerable uniformity. Within any particular cortex, and across species, the same essential organization of layers can be seen, with each layer of neurons characterized by particular axonal inputs and outputs and identifiable **neuron** types. For example, the outermost two cell layers of the cortex, layers 2 and 3, are composed principally of asymmetric neurons called pyramidal cells, whose axons make long-distance interarea and short-distance local recurrent connections within the cortex itself. The larger pyramidal cells of layer 5 get axonal input from the cortex but send projections outside the cortex to diverse regions such as the midbrain, the

Copyright: © 2016 Barbara L. Finlay. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

basal ganglia of the forebrain, and the spinal cord. Computational theories that focus on overall cortex uniformity have offered detailed conjectures on the “canonical computation” enacted by the cortex over its whole surface.* **

On the other hand, the cortex also presents a striking mosaic of variability: the entire surface is tiled into a number of “areas,” with each area containing sensory, motor, cognitive, or other computed dimensions laid out in an orderly topographic manner (such as the primary motor or visual cortex, common to all species) as well as the sensory and motor specializations particular to each, ranging from electroreception in the platypus to trunks in elephants.² Systematic regional differences related to sensory versus motor processing requirements³ or neuroembryological gradients that progressively reorganize brains of different sizes in a predictable manner have also been proposed.

How a computational device scales also depends on what it is made up of. The cell (or in the case of the brain, neurons and glia) is the building block of the organism. In general, larger organisms (in this case, mammals) are made of more cells, not larger cells. In addition, fundamental cellular processes are largely scale invariant—in the case of the neuron, relevant examples of general scale invariance are the duration of the **action potential** and the fundamental biophysics of the propagation of action potentials down axons. The neuron represents a special problem for scaling at the cell level: while the cell bodies of neurons have a minimal relationship to the whole-organism mass in mammals, the axons may need to traverse distances varying directly with brain or body size.⁴ For example, comparable single neurons relaying information about the body midline across the cortex must travel millimeters in mice, but centimeters in macaque monkeys; to relay a motor command from the cortex to the spinal cord requires millimeters of travel in mice, but meters in giraffes. Though the speed

neuron A nerve cell, the basic signalling unit of the brain and nervous system. Neurons receive, process, and pass on information through electrical and chemical signals, and connect to one another by way of axons.

action potential A brief electrical impulse by which a neuron sends a signal along its axon. Its duration and underlying biophysics are largely the same regardless of the size of the brain or body.

axon The long, thread-like projection of a neuron that carries electrical signals away from the cell body towards other neurons. Because axons must span the whole brain, their length has to increase as brains and bodies get larger.

myelination The process of wrapping an axon in a fatty insulating sheath (myelin), which speeds up the electrical signals travelling along it. Faster transmission comes at the cost of adding mass to the nervous system. *See also axon.*

* Miller KD. Canonical computations of cerebral cortex. *Current Opinion in Neurobiology*, 2016; 37: 75–84. doi: 10.1016/j.conb.2016.01.008. pmid:26868041

** Bastos AM, Usrey WM, Adams RA, Mangun GR, Fries P, Friston KJ. Canonical microcircuits for predictive coding. *Neuron*, 2012; 76(4): 695–711. doi: 10.1016/j.neuron.2012.10.038. pmid:23177956

NOTES

of axonal transmission can be substantially increased by **axon myelination**, and is a solution to increasing brain volume, myelination adds mass to the nervous system, eventually exacerbating the problem it attempts to solve. So, with the affordances and constraints of these building blocks, how has evolution scaled up the cortex over five orders of magnitude?

Kennedy, along with Toroczka, Ercsey-Ravasz, and their colleagues,^{5 6} have provided an exceptionally interesting piece of information to both the puzzle of cortex scaling and cortex function. To this end, in a new study published in *PLOS Biology*, they have compared two relatively new comprehensive data sources from two species, the rhesus macaque and the laboratory mouse. The first is a comprehensive analysis of the connectivity of the cortex of the rhesus macaque, as determined by neuroanatomical tracing studies (as contrasted with functional or structural connectivity determined by various imaging methods), done by the same group,⁷ and the second is a similar study done in mice.^{8 9} Tracer injections made systematically across the cortical surface and in each cortical area determined the number of areas each cortical area projects to and receives projections from, with these tracer techniques exploiting the normal intracellular transport mechanisms of neurons. New to this study, the absolute axon length and cortical location each projection represents was added to the analysis. Over what range and with what pattern does a comparable location in cortex in mice versus monkeys project? Does the range reflect the surface area of the cortex, the growth constraints of neurons, or the idiosyncratic specializations of each species?

* Reviewed in Sz
Horvát, Gămănuț R,
Ercsey-Ravasz M,
Magrou L, Gămănuț B,
Van Essen DC, et al.
Spatial Embedding
and Wiring Cost
Constrain the
Functional Layout of
the Cortical Network of
Rodents and Primates.
PLoS Biol 2016 14(7):
e1002512. doi:
10.1371/
journal.pbio.1002512.
pmid:27441598)

Before the answer, a few basic facts about what is known about the scaling of the cortex should be put in play (Fig 1). As the cortex increases in volume, it principally increases in surface area, producing the characteristic folding of a large cortex, but the depth of the cortex increases as well; in the case of the mouse versus monkey, the surface area of the mouse is very roughly 100 mm² and the monkey 20,000 mm², while the mean depth of the cortex in the mouse is about 1–2 mm and the monkey 2–3 mm. Cortical area and depth reflect variable contributions of neuron number and connection volumes, both related to the order of neurogenesis and developmental duration.¹⁰ All mammals possess primary visual, auditory, and sensorimotor cortices, no matter how small the brain, and after that, the number of cortical areas scales regularly with overall cortical surface area.¹¹ Therefore, the greatest increase in number of cortical areas comes from the “association” regions of parietal, temporal, and frontal areas. Connections within the cortex are predominantly made to immediately adjacent regions—with a lesser fraction of long-distance connections—

in any size of brain.* The proportion of local connections compared to long-distance connections increases in larger brains, which is thought to be related to intractable problems of accommodating larger and larger volumes of interconnecting axons.

This is a note about CO₂ that we refer to in the reference index (page 139) to test HTML tags like <sub> in index tags and entries.

The researchers for this paper⁵ evaluated the network properties of cortical areas of the monkey versus the mouse cortex using the mathematics of graph theory, describing graphs of mouse and monkey cortical connections with respect to three essential properties. First, the graphs were “directed,” as the direction of connectivity from one “node”—in this case a tracer injection in a cortical area—to another was explicit in the method. Second, they were “weighted,” taking into account the relative numbers of projections from each area to any other. Finally, the nodes of the graphs, identical with tracer injection sites, were “spatially embedded”—that is, assigned relative locations in the cortex defined as the absolute distances between areas of cortex that axons would have to traverse. Prior graphical analyses of network structure, assessed from multiple neuroanatomical and imaging methods, have evaluated directed and weighted but not spatially embedded data corpora. In both the monkey and mouse, the distribution of axonal connection lengths as a function of distance exhibits an exponential decay, independently of area definition. The exponential decay rate along with distance distributions was found to determine the probability of a connection of any two cortical areas. The central question of the study is to explore the extent that the features of cortical connectional organization are predicted by this exponential falloff rule.

To get a better feel for the physical proportions of the two cortices, and the connection problem to be solved, the surface area of the cortex of the rhesus macaque is roughly 200 times greater than a mouse's. The outside range of axon extent from a single injection is about 50 mm in the monkey and 10 mm in a mouse.⁵ In the mouse, this range is adequate for any single cortical location to potentially extend to any other cortical location (with declining likelihood with distance), but the outside range for the monkey fails to reach the full cortical extent by about half. This shortfall in coverage is exacerbated by the fact that the falloff of the probability of making longer connections falls off more steeply in the monkey compared to the mouse

when the two brains are scaled to a common template. Interestingly, the falloff in connection probability for short-range, likely unmyelinated local connections (about 1.5 mm) is identical in the two species.

Even with this fundamental physical disparity, many features of connectivity are similar in the two species, some of which are predictable by the spatially explicit exponential distance falloff rule of connectivity. Both exhibit “cliques” of cortical areas, completely connected sets of areas that can be used to define “core” and “peripheral” areas. In the monkey, such highly interconnected cliques and core regions are found principally in the frontal and parietal cortex and exclude primary sensory and motor areas; in the mouse, the overall spatial distribution of clique regions is similar but includes some primary sensory and motor cortical regions. The distribution of “motifs”—the pattern of forward, backward, and reciprocal connections between areas—is well predicted by the exponential distance rule and is similar in some interesting respects in the two species. Despite great evolutionary time and massive volume changes, it appears that the cortex has evolved to retain an integrative structure and not devolve into a collection of computationally-isolated special function devices.

Notable deviations from an all-inclusive, distance-based rule of connectivity appear in both species as well—“association” areas are more likely to have reciprocal interconnections than “canonical” primary sensory and motor areas, independent of how closely they neighbor each other. This feature is less pronounced in the mouse but still present. Much variation remains to be explained, particularly that which might be produced by evolutionarily specified or experientially-stabilized connections. For example, if coactivation of particular modalities serves to preferentially stabilize the long-distance connections serving them (Hebbian “fire-together, wire together”), as in the case of auditory-vocal-visual coactivation in animals capable of vocal learning, the probability of connections between them would be greater than that given by a fundamental distance rule scaffolding alone.

The authors point out that the low probability of long-distance connections may make those connections particularly vulnerable in large brains, citing evidence of this kind in both autism and Alzheimer’s disease. This example, while well taken, brings up the caution that not every problem caused by connectivity (or any particular process) should be expected to be solved by connectivity, as multiple alterations in axon caliber and myelination varying with brain size have been described as partial solutions to this problem.^{12 13}

Finally, how a computational device scales may take advantage of how

it is made (or, be constrained by how it is made). The exponential distance rule points to a potential simple source of developmental scaffolding for cortical connections in the initial outgrowth of intracortical axons. Initial axon outgrowth, moreover, has some distinct features within and between species that should be investigated for their role in later network architecture. The frontal cortex is the first to send out axons as the cortex develops: * does this cause its position in “cliques” of cortical connectivity and its core status? Is the duration of development, though much longer in monkeys than mice, the effective limiting factor that determines how far intracortical axons have time to extend?¹⁴ Evolution has likely filtered from the original range of developmental rules organizing the cortex those rules generating computational architectures robust to common challenges. Variation in the sizes of animals and all their parts is certainly one of those repeated challenges.

Glossary

action potential: A brief electrical impulse by which a neuron sends a signal along its axon. Its duration and underlying biophysics are largely the same regardless of the size of the brain or body.

axon: The long, thread-like projection of a neuron that carries electrical signals away from the cell body towards other neurons. Because axons must span the whole brain, their length has to increase as brains and bodies get larger.

isocortex: The largest part of the cerebral cortex: a layered sheet of neurons. Sometimes just called the cortex, it is disproportionately large in primates and grows mainly in surface area as brain volume increases.

myelination: The process of wrapping an axon in a fatty insulating sheath (myelin), which speeds up the electrical signals travelling along it. Faster transmission comes at the cost of adding mass to the nervous system. *See also axon.*

neuron: A nerve cell, the basic signalling unit of the brain and nervous

* Finlay BL, Uchiyama R. Developmental mechanisms channeling cortical evolution. *Trends Neurosci.* 2015;38(2):69–76. doi: 10.1016/j.tins.2014.11.004. pmid:25497421

NOTES

system. Neurons receive, process, and pass on information through electrical and chemical signals, and connect to one another by way of axons.

Bibliography

- Bastos, A. M., W. M. Usrey, R. A. Adams, G. R. Mangun, P. Fries, and K. J. Friston. 2012. 'Canonical microcircuits for predictive coding'. *Neuron* 76 (4): 695–711.
- Cahalane, D. J., C. J. Charvet, and B. L. Finlay. 2014. 'Modeling local and cross-species neuron number variations in the cerebral cortex as arising from a common mechanism'. *Proceedings of the National Academy of Sciences* 111 (49): 17642–17647.
- Ercsey-Ravasz, M., N. T. Markov, C. Lamy, D. C. Van Essen, K. Knoblauch, Z. Toroczkai, et al. 2013. 'A predictive network model of cerebral cortical connectivity based on a distance rule'. *Neuron* 80: 184–197.
- Finlay, B. L., and P. B. Brodsky. 2006. 'Cortical evolution as the expression of a program for disproportionate growth and the proliferation of areas'. In *Evolution of Nervous Systems*, edited by J. H. Kaas, 73–96. Oxford: Elsevier/Oxford University Press.
- Finlay, B. L., and R. B. Darlington. 1995. 'Linked regularities in the development and evolution of mammalian brains'. *Science* 268 (5217): 1578–1584.
- Finlay, B. L., and R. Uchiyama. 2015. 'Developmental mechanisms channeling cortical evolution'. *Trends in Neurosciences* 38 (2): 69–76.
- Horvát, Sz., R. Gămănuț, M. Ercsey-Ravasz, L. Magrou, B. Gămănuț, D. C. Van Essen, et al. 2016. 'Spatial embedding and wiring cost constrain the functional layout of the cortical network of rodents and primates'. *PLoS Biology* 14 (7): e1002512.
- Innocenti, G. M., A. Vercelli, and R. Caminiti. 2014. 'The diameter of cortical axons depends both on the area of origin and target'. *Cerebral Cortex* 24 (8): 2178–2188.
- Kaas, J. 2000. 'Why is brain size so important: design problems and solutions as neocortex gets bigger or smaller'. *Brain and Mind* 1: 7–23.
- Krubitzer, L. 2009. 'In search of a unifying theory of complex brain evolution'. *Annals of the New York Academy of Sciences* 1156: 44–67.
- Markov, N. T., M. M. Ercsey-Ravasz, A. R. Ribeiro Gomes, C. Lamy, L. Magrou, J. Vezoli, et al. 2014. 'A weighted and directed interareal connectivity matrix for macaque cerebral cortex'. *Cerebral Cortex* 24 (1): 17–36.
- Miller, K. D. 2016. 'Canonical computations of cerebral cortex'. *Current*

Opinion in Neurobiology 37: 75–84.

- Oh, S. W., J. A. Harris, L. Ng, B. Winslow, N. Cain, S. Mihalas, et al. 2014. 'A mesoscale connectome of the mouse brain'. *Nature* 508 (7495): 207–214.
- O'Reilly, R. C., S. A. Herd, and W. M. Pauli. 2010. 'Computational models of cognitive control'. *Current Opinion in Neurobiology* 20 (2): 257–261.
- Wang, S. S.-H., J. R. Shultz, M. J. Burish, K. H. Harrison, P. R. Hof, L. C. Towns, et al. 2008. 'Functional trade-offs in white matter axonal scaling'. *Journal of Neuroscience* 28 (15): 4047–4056.
- Workman, A., C. J. Charvet, B. Clancy, R. B. Darlington, and B. L. Finlay. 2013. 'Modeling transformations of neurodevelopmental sequences across mammalian species'. *Journal of Neuroscience* 33 (17): 7368–7383.
- Zingg, B., H. Hintiryan, L. Gou, M. Y. Song, M. Bay, M. S. Bienkowski, et al. 2014. 'Neural networks of the mouse neocortex'. *Cell* 156: 1096–1111.

Endnotes

1. Finlay BL, Darlington RB. Linked regularities in the development and evolution of mammalian brains. *Science*. 1995; 268(5217): 1578–1584. pmid:7777856 doi: 10.1126/science.7777856
2. Krubitzer L. In search of a unifying theory of complex brain evolution. *Annals NY Acad Sci*. 2009; 1156: 44–67. doi: 10.1111/j.1749-6632.2009.04421.x
3. O'Reilly RC, Herd SA, Pauli WM. Computational models of cognitive control. *Current Opinion in Neurobiology*, 2010; 20(2), 257–261. doi: 10.1016/j.conb.2010.01.008. pmid:20185294
4. Kaas J. Why is brain size so important: design problems and solutions as neocortex gets bigger or smaller. *Brain and Mind*. 2000;1:7–23.
5. Sz Horvát, Gămănuț R, Ercsey-Ravasz M, Magrou L, Gămănuț B, Van Essen DC, et al. Spatial Embedding and Wiring Cost Constrain the Functional Layout of the Cortical Network of Rodents and Primates. *PLoS Biol* 2016 14(7): e1002512. doi: 10.1371/journal.pbio.1002512. pmid:27441598
6. Ercsey-Ravasz M, Markov NT, Lamy C, Van Essen DC, Knoblauch K, Toroczkai Z, et al. A predictive network model of cerebral cortical connectivity based on a distance rule. *Neuron* 2013 80, 184–197. doi: 10.1016/j.neuron.2013.07.036. pmid:24094111
7. Markov NT, Ercsey-Ravasz MM, Ribeiro Gomes AR, Lamy C, Magrou L, Vezoli J, et al. A weighted and directed interareal connectivity matrix for macaque cerebral cortex. *Cereb Cortex*. 2014;24(1):17–36. doi: 10.1093/cercor/bhs270. pmid:23010748
8. Oh SW, Harris JA, Ng L, Winslow B, Cain N, Mihalas S, et al. A mesoscale connectome of the mouse brain. *Nature*. 2014;508(7495):207–14. doi: 10.1038/nature13186. pmid:24695228
9. Zingg B, Hintiryan H, Gou L, Song MY, Bay M, Bienkowski MS, et al. Neural networks of the mouse neocortex. *Cell* 156, 1096–1111, doi: 10.1016/j.cell.2014.02.023 (2014). pmid:24581503
10. Cahalane DJ, Charvet CJ, Finlay BL. Modeling local and cross-species neuron number variations in the cerebral cortex as arising from a common mechanism. *Proc Natl Acad Sci U S A* 2014;111(49):17642–7. doi: 10.1073/pnas.1409271111
11. Finlay BL, Brodsky PB. Cortical evolution as the expression of a program for disproportionate growth and the proliferation of areas. In: *Evolution of Nervous Systems* 2006. Kaas J.H.,

NOTES

editor. Elsevier: Oxford University Press pp. 73–96

12. Wang SS-H, Shultz JR, Burish MJ, Harrison KH, Hof PR, Towns LC, et al. Functional trade-offs in white matter axonal scaling. 2008; *J Neurosci*, 28(15): 4047–4056. doi: 10.1523/JNEUROSCI.5559-05.2008. pmid:18400904
13. Innocenti GM, Vercelli A, Caminiti R. The diameter of cortical axons depends both on the area of origin and target. *Cerebral Cortex*, 2014; 24(8), 2178–2188. doi: 10.1093/cercor/bht070. pmid:23529006
14. Workman A, Charvet CJ, Clancy B, Darlington RB, Finlay BL. Modeling transformations of neurodevelopmental sequences across mammalian species. *J Neurosci* 2013; 17: 7368–7383 doi: 10.1523/jneurosci.5746-12.2013



Dividers

Some authors divide sections of work with little decorations, most commonly three asterisks. In our markdown, we use three asterisks, too, and this converts to an HTML horizontal rule or `<hr>` element. By default, we display that as empty space only. The space can be easily replaced with a symbol by setting the `$text-divider` value in each output's `.scss` styles file.

Here is a passage of text with a couple of these dividers in it. This is from Frederick W. Hamilton's *Division of Words: Rules for the Division of Words at the Ends of Lines, With Remarks on Spelling, Syllabication And Pronunciation*. We've added the dividers. Sorry, Fred.

Spelling

The idea that there is one right way to combine the letters representing a certain sound or group of sounds, that is a word, and that all other ways are wrong and little short of shameful is a comparatively new idea among us. The English speaking folk held down to a comparatively recent time that any group of letters which approximately represented the sound was amply sufficient as a symbol of the word. This sort of phonetic spelling was commonly followed, and followed with great freedom. No obligation was recognized to be consistent. In ordinary writing, such as letters and the like, it is not unusual to find the same word spelled in a variety of ways in the same document.

The last century has brought about an attempt to standardize spelling into conventional forms any departure from which is regarded as highly derogatory to the writer. In many cases these forms are fixed arbitrarily, and in some there is even now disagreement among the highest authorities. These difficulties and disagreements have two reasons: First, English is a composite language, drawn from many sources and at many periods; hence purely philological and etymological influences intervene, sometimes with marked results, while there is a difference of opinion as to how far these influences ought to prevail. Second, the English language uses an alphabet which fits it very badly. Many letters have to do duty for the expression of several sounds, and sometimes several of them have nearly or quite the same sound. For example, there are a number of distinct sounds of a, i, and o while g is sometimes indistinguishable from j and c from k. This is not always a matter of modification of sounds by the sounds of other letters combined with them. One has to learn how to pronounce

DIVIDERS

cough, dough, enough, and plough, the ough having four distinct sounds in these four words. Each one of these sounds, by the way, could be exactly as well represented by another combination of letters which would be unmistakable, viz., coff, doe, enuff, and plow. It is impossible to tell except by the context either the pronunciation or the meaning of bow. If the ow is pronounced as in low, it means a weapon. If the ow is pronounced as in cow it may mean either an obeisance or the front end of a boat.

This standardization of spelling is unfortunately not quite complete, although nearly so. Concerning the vast majority of the words in the English language there is no difference of opinion. A few words are differently spelled by different authorities. There are seven of these authorities of the first rank, three English, Stormonth, the Imperial Dictionary, and the Oxford Dictionary; and four American, Webster's International, Worcester, the Century Dictionary, and the Standard Dictionary. American printers may ordinarily disregard the English authorities.

Any one of the four American authorities may be safely followed. In cases where two spellings are given in the dictionary consulted, take the first one. Ordinarily a printing office adopts one of the great authorities as a standard and conforms the office style to it. All office copy will follow it and all errors in copy from outside will be corrected by it. Spellings differing from it will be regarded as errors, even though supported by other authorities.

This rule, however, is subject to one very important exception. The author has an unquestionable right to choose his own dictionary or to use any spelling for which there is any authority, English or American. If he has his own ideas on the subject of spelling he should be very careful that his manuscript is correctly spelled according to his ideas, and clearly written or typed. He should also indicate on the manuscript the authority he wishes used in correcting the spelling in case of mistakes or illegible passages. Every care should be taken to make the manuscript copy as correct as possible and as legible as possible. Such care may be very troublesome at first, but it will result in great saving of expense.

In addition to the authorities named there are the rules and "reformed" spellings adopted by the American Philological Association and published by the United States Government. These are followed fully in some offices, partly in others, and in many not at all. This is a question of the office style and the author's wish. If copy is clear and spelled according to any authority, it is the compositor's duty to follow it. If it is misspelled or illegible he is to correct it according to the office style unless otherwise directed by the author in writing. If furnished with such a direction he

is to follow it. This procedure will clear the compositor of all blame. Any questions which then arise lie between the author and the proofreader.

In the case of the reformed spellings, however, the departure from the ordinary appearance of the words is so great that the author cannot be allowed full freedom to set aside the office style. If he is paying for the printing he may insist on his spelling. If he is contributing to a periodical and the printing is done at the publisher's expense it is for the publisher to determine the style of printing to be used.

Any full consideration of the question of reformed spelling is hardly in place in this book. The author may perhaps be permitted one observation. Innovation in the use of the English language would appear to be primarily the work of scholars, and the adoption of such innovations would seem to belong to the book printer rather than to the commercial printer. The public mind as a whole is conservative. It is not hospitable to changes and does not soon become aware of them, much less familiar with them. The commercial printer makes his appeal to the mind of the general public. He will do well to use a vehicle familiar, intelligible, and acceptable to it.

Correct spelling is mainly a matter of habit and observation. To a certain extent it is a matter of careful pronunciation, but this is not always a safe or even a possible guide. The vowels preceding or following the one on which the primary accent falls, sometimes called obscure vowels, are so slurringly pronounced that even a pedantic precision will hardly make it possible to indicate clearly which vowel is used. The writer remembers seeing an examination paper written by a fourth year medical student in which the word fever was spelled fevor. A moment's thought will show that so far as pronunciation is concerned the word might be spelled fevar, fevir, fevor, fever, or fevur without any appreciable difference. The correct spelling is merely a matter of observation.

The author has on his desk at the moment of writing these lines half a dozen good books, each containing a set of rules for spelling. From these it would be easy to compile a set of fairly good rules. Each of these rules, however, has exceptions, in some cases quite numerous. To remember these rules with their exceptions would be a considerable mental task and to apply them would be cumbrous and time consuming. The effort would probably resolve itself into an actual learning of the words which present difficulties. The best way to become a good speller is to form the habit of careful reading, observing the form of every word as it passes before the eye and so unconsciously fixing it in the memory. The dictionary should be



DIVIDERS

consulted whenever there is any doubt.

If you are to write a word, call up a mental picture of it, and if the picture is not perfectly clear go to the dictionary and fix a correct image of it in your mind. Be careful to pronounce every word you use as correctly as possible and you will get all the aid pronunciation can give you. Careless speaking and careless reading are the two great sources of incorrect spelling.



Poetry

First, a bit of background about how we like to mark up poetry. This will help you understand what you're looking at when we get to the samples below.

Also see the official docs on poetry.

Encoding poetry can be tricky. Usually, poetry in HTML is structured by tagging each stanza as a paragraph, with line breaks after each line. You can do this by adding markdown line breaks (with double spaces or `\` at the end of each line) and tagging the paragraph with `{:.verse}`. However, this structure makes it impossible to have browsers, ereaders and PDF engines correctly indent runover lines (because there is no `nth-line` selector in CSS, unless you resort to a Javascript method that will bloat your code and won't run on many ereaders).

Tech note: Some text and code editors (e.g. Atom) strip out spaces at the ends of lines automatically. So use `\` for line breaks, not double-spaces.

We prefer another approach: the poem is an unordered list (`ul`) and each line (including each blank line between stanzas) is a list item (`li`). We just hide any list markers (bullets) with `list-style-type: none`. This way, we can control indents on runover lines. This is a non-semantic use of HTML, since a poem is technically not a list. But it's a healthy hack with universal browser and ereader support.

Our convention is to mark each line of a stanza with a hyphen -, and tag the list with `{:.verse}`:

```
- I wandered lonely as a cloud
- That floats on high o'er vales and hills,
- When all at once I saw a crowd,
- A host, of golden daffodils;
- Beside the lake, beneath the trees,
- Fluttering and dancing in the breeze.
{:.verse}
```

This gives us:

POETRY

I wandered lonely as a cloud
That floats on high o'er vales and hills,
When all at once I saw a crowd,
A host, of golden daffodils;
Beside the lake, beneath the trees,
Fluttering and dancing in the breeze.

We can also indent individual lines, where the poet wanted indents, by tagging individual list items.

- Alas for man! day after day may rise,
- {:.indent-3}Night may shade his thankless head,
- He sees no God in the bright, morning skies
- {:.indent-3}He sings no praises from his guarded bed.

{:.verse}

The 2 in {:.indent-2} refers to the number of em spaces to indent by. Our CSS allows for indents from 1 ({:.indent-1}) to 30 ({:.indent-30}).

This gives us:

Alas for man! day after day may rise,
 Night may shade his thankless head,
He sees no God in the bright, morning skies
 He sings no praises from his guarded bed.

Big gaps between words in a line must be created with spaces or space entities like ` ` in the poem text.

Centering poems

But wait, there's more! Best practice for poetry layout is that – in print – a poem should be centered on its longest line. That is *not* centering the lines of poetry, but placing the left-justified poem in the horizontal middle of the page. Put another way, the poem should be indented till its longest line is centered on the page.

To achieve this, put the entire poem, including its title, in a blockquote, by adding `>` to the start of each line. Tag the whole blockquote as {:.verse}, too. Finally, decide how wide you want the poem to as a percentage of the

page width. That is, if you reckon this poem's longest line reaches across 90 per cent of the page, use `.width-90`.

```
> - ### To One Who Has Been Long in City Pent
> - To one who has been long in city pent,
> - {:.indent-2}'Tis very sweet to look into the fair
> - {:.indent-2}And open face of heaven,—to breathe a
prayer
> - Full in the smile of the blue firmament.
> - Who is more happy, when, with heart's content,
> - {:.indent-2}Fatigued he sinks into some pleasant
lair
> - {:.indent-2}Of wavy grass, and reads a debonair
> - And gentle tale of love and languishment?
> -
> - Returning home at evening, with an ear
> - {:.indent-2}Catching the notes of Philomel,—an eye
> - Watching the sailing cloudlet's bright career,
> - {:.indent-2}He mourns that day so soon has glided
by:
> - E'en like the passage of an angel's tear
> - {:.indent-2}That falls through the clear ether
silently.
> {:.verse}
{:.verse .width-80}
```

In verse structured as a list like this, our CSS preserves white space. That is, if you type, say, three spaces you get three spaces. Normally, HTML collapses multiple spaces into one – which is great *except* when you want to deliberately use extra spaces, as some poets do. However, this doesn't work at the start of lines, where markdown strips leading spaces. There you must use HTML space entities (like ` `;) or our indent tags explained above.

However, the `whitespace: pre-wrap` CSS we use for this is not currently supported on Kindle. If that's important, it's best to stick to using HTML space entities like ` `;

Here's that poem rendered:

POETRY

To One Who Has Been Long in City Pent

To one who has been long in city pent,
 'Tis very sweet to look into the fair
 And open face of heaven,—to breathe a prayer
Full in the smile of the blue firmament.
Who is more happy, when, with heart's content,
 Fatigued he sinks into some pleasant lair
 Of wavy grass, and reads a debonair
And gentle tale of love and languishment?

Returning home at evening, with an ear
 Catching the notes of Philomel,—an eye
Watching the sailing cloudlet's bright career,
 He mourns that day so soon has glided by:
E'en like the passage of an angel's tear
 That falls through the clear ether silently.

Here's a long poetry example.

Gerontion

Thou hast nor youth nor age
But as it were an after dinner sleep
Dreaming of both.

Here I am, an old man in a dry month,
Being read to by a boy, waiting for rain.
I was neither at the hot gates
Nor fought in the warm rain
Nor knee deep in the salt marsh, heaving a cutlass,
Bitten by flies, fought.
My house is a decayed house,
And the jew squats on the window sill, the owner,

Spawned in some estaminet of Antwerp,
 Blistered in Brussels, patched and peeled in London.
 The goat coughs at night in the field overhead;
 Rocks, moss, stonecrop, iron, merds.
 The woman keeps the kitchen, makes tea,
 Sneezes at evening, poking the peevish gutter.

I an old man,
 A dull head among windy spaces.

Signs are taken for wonders. "We would see a sign":
 The word within a word, unable to speak a word,
 Swaddled with darkness. In the juvescence of the year
 Came Christ the tiger

In depraved May, dogwood and chestnut, flowering Judas,
 To be eaten, to be divided, to be drunk
 Among whispers; by Mr. Silvero
 With caressing hands, at Limoges
 Who walked all night in the next room;
 By Hakagawa, bowing among the Titians;
 By Madame de Tornquist, in the dark room
 Shifting the candles; Fraulein von Kulp
 Who turned in the hall, one hand on the door. Vacant shuttles
 Weave the wind. I have no ghosts,
 An old man in a draughty house
 Under a windy knob.

After such knowledge, what forgiveness? Think now
 History has many cunning passages, contrived corridors
 And issues, deceives with whispering ambitions,
 Guides us by vanities. Think now
 She gives when our attention is distracted
 And what she gives, gives with such supple confusions
 That the giving famishes the craving. Gives too late
 What's not believed in, or if still believed,

POETRY

In memory only, reconsidered passion. Gives too soon
Into weak hands, what's thought can be dispensed with
Till the refusal propagates a fear. Think
Neither fear nor courage saves us. Unnatural vices
Are fathered by our heroism. Virtues
Are forced upon us by our impudent crimes.
These tears are shaken from the wrath-bearing tree.

The tiger springs in the new year. Us he devours. Think at last
We have not reached conclusion, when I
Stiffen in a rented house. Think at last
I have not made this show purposelessly
And it is not by any concitation
Of the backward devils.
I would meet you upon this honestly.
I that was near your heart was removed therefrom
To lose beauty in terror, terror in inquisition.
I have lost my passion: why should I need to keep it
Since what is kept must be adulterated?
I have lost my sight, smell, hearing, taste and touch:
How should I use it for your closer contact?

These with a thousand small deliberations
Protract the profit of their chilled delirium,
Excite the membrane, when the sense has cooled,
With pungent sauces, multiply variety
In a wilderness of mirrors. What will the spider do,
Suspend its operations, will the weevil
Delay? De Bailhache, Fresca, Mrs. Cammel, whirled
Beyond the circuit of the shuddering Bear
In fractured atoms. Gull against the wind, in the windy straits
Of Belle Isle, or running on the Horn,
White feathers in the snow, the Gulf claims,
And an old man driven by the Trades
To a sleepy corner.

CENTERING POEMS

Tenants of the house,
Thoughts of a dry brain in a dry season.





Questions and quizzes

You can add questions to your book like this one.

A question title

Which of the following is true?

- ☐ You should not pick a fight with a Wookiee.
- ☐ Picking fights with Wookiees is safe.
- ☐ Wookiees love being laughed at.
- ☐ Wookiees make loyal companions.

See the docs for how to add quizzes and create exams.

You can also add _____ spaces (aka gap-fill or cloze questions), that give users feedback in web and app outputs.





Part 2

Images







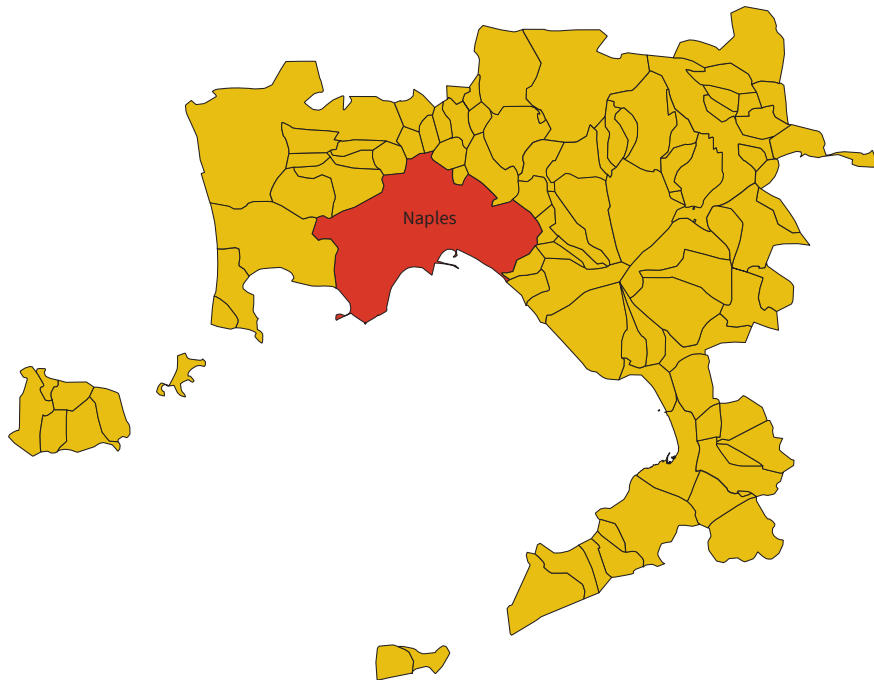
2.1 Plain images

For examples of plain images, here's the first chapter of *The Count of Monte Christo* by Alexandre Dumas. This and the next chapter also include a chapter number, to show how that is marked up and renders in the EBT.

First, here is a map of Naples, where the story is set.

Also see the docs on including images in content and adding them to your project.





A map of Naples. This is actually a paragraph that starts with an inline image, and has an `image-with-caption` class, which turns any text in this paragraph into a caption. The remaining images in this chapter are plain images, with no text or class attached.

Chapter 1. Marseilles—The Arrival

On the 24th of February, 1815, the look-out at Notre-Dame de la Garde signalled the three-master, the *Pharaon* from Smyrna, Trieste, and Naples.

As usual, a pilot put off immediately, and rounding the Château d’If, got on board the vessel between Cape Morgion and Rion island.

Immediately, and according to custom, the ramparts of Fort Saint-Jean were covered with spectators; it is always an event at Marseilles for a ship to come into port, especially when this ship, like the *Pharaon*, has been built, rigged, and laden at the old Phocée docks, and belongs to an owner of the city.

The ship drew on and had safely passed the strait, which some volcanic shock has made between the Calasareigne and Jaros islands; had doubled Pomègue, and approached the harbor under topsails, jib, and spanker, but so slowly and sedately that the idlers, with that instinct which is the forerunner of evil, asked one another what misfortune could have happened on board. However, those experienced in navigation saw plainly that if any accident had occurred, it was not to the vessel herself, for she

bore down with all the evidence of being skilfully handled, the anchor a-cockbill, the jib-boom guys already eased off, and standing by the side of the pilot, who was steering the *Pharaon* towards the narrow entrance of the inner port, was a young man, who, with activity and vigilant eye, watched every motion of the ship, and repeated each direction of the pilot.

The vague disquietude which prevailed among the spectators had so much affected one of the crowd that he did not await the arrival of the vessel in harbor, but jumping into a small skiff, desired to be pulled alongside the *Pharaon*, which he reached as she rounded into La Réserve basin.

When the young man on board saw this person approach, he left his station by the pilot, and, hat in hand, leaned over the ship's bulwarks.

He was a fine, tall, slim young fellow of eighteen or twenty, with black eyes, and hair as dark as a raven's wing; and his whole appearance bespoke that calmness and resolution peculiar to men accustomed from their cradle to contend with danger.

"Ah, is it you, Dantès?" cried the man in the skiff. "What's the matter? and why have you such an air of sadness aboard?"

"A great misfortune, M. Morrel," replied the young man,— "a great misfortune, for me especially! Off Civita Vecchia we lost our brave Captain Leclere."

"And the cargo?" inquired the owner, eagerly.

"Is all safe, M. Morrel; and I think you will be satisfied on that head. But poor Captain Leclere—"

"What happened to him?" asked the owner, with an air of considerable resignation. "What happened to the worthy captain?"

"He died."

"Fell into the sea?"

"No, sir, he died of brain-fever in dreadful agony." Then turning to the crew, he said, "Bear a hand there, to take in sail!"

All hands obeyed, and at once the eight or ten seamen who composed the crew, sprang to their respective stations at the spanker brails and outhaul, topsail sheets and halyards, the jib downhaul, and the topsail clewlines and buntlines. The young sailor gave a look to see that his orders were promptly and accurately obeyed, and then turned again to the owner.

"And how did this misfortune occur?" inquired the latter, resuming the interrupted conversation.





“Alas, sir, in the most unexpected manner. After a long talk with the harbor-master, Captain Leclere left Naples greatly disturbed in mind. In twenty-four hours he was attacked by a fever, and died three days afterwards. We performed the usual burial service, and he is at his rest, sewn up in his hammock with a thirty-six pound shot at his head and his heels, off El Giglio island. We bring to his widow his sword and cross of honor. It was worth while, truly,” added the young man with a melancholy smile, “to make war against the English for ten years, and to die in his bed at last, like everybody else.”

“Why, you see, Edmond,” replied the owner, who appeared more comforted at every moment, “we are all mortal, and the old must make way for the young. If not, why, there would be no promotion; and since you

assure me that the cargo—”

“Is all safe and sound, M. Morrel, take my word for it; and I advise you not to take 25,000 francs for the profits of the voyage.”

Then, as they were just passing the Round Tower, the young man shouted: “Stand by there to lower the topsails and jib; brail up the spanker!”

The order was executed as promptly as it would have been on board a man-of-war.

“Let go—and clue up!” At this last command all the sails were lowered, and the vessel moved almost imperceptibly onwards.

“Now, if you will come on board, M. Morrel,” said Dantès, observing the owner’s impatience, “here is your supercargo, M. Danglars, coming out of his cabin, who will furnish you with every particular. As for me, I must look after the anchoring, and dress the ship in mourning.”

The owner did not wait for a second invitation. He seized a rope which Dantès flung to him, and with an activity that would have done credit to a sailor, climbed up the side of the ship, while the young man, going to his task, left the conversation to Danglars, who now came towards the owner. He was a man of twenty-five or twenty-six years of age, of unprepossessing countenance, obsequious to his superiors, insolent to his subordinates; and this, in addition to his position as responsible agent on board, which is always obnoxious to the sailors, made him as much disliked by the crew as Edmond Dantès was beloved by them.

“Well, M. Morrel,” said Danglars, “you have heard of the misfortune that has befallen us?”

“Yes—yes: poor Captain Leclere! He was a brave and an honest man.”

“And a first-rate seaman, one who had seen long and honorable service, as became a man charged with the interests of a house so important as that of Morrel & Son,” replied Danglars.

“But,” replied the owner, glancing after Dantès, who was watching the anchoring of his vessel, “it seems to me that a sailor needs not be so old as you say, Danglars, to understand his business, for our friend Edmond seems to understand it thoroughly, and not to require instruction from any one.”

“Yes,” said Danglars, darting at Edmond a look gleaming with hate. “Yes, he is young, and youth is invariably self-confident. Scarcely was the captain’s breath out of his body when he assumed the command without consulting anyone, and he caused us to lose a day and a half at the Island of Elba, instead of making for Marseilles direct.”





“As to taking command of the vessel,” replied Morrel, “that was his duty as captain’s mate; as to losing a day and a half off the Island of Elba, he was wrong, unless the vessel needed repairs.”

“The vessel was in as good condition as I am, and as, I hope you are, M. Morrel, and this day and a half was lost from pure whim, for the pleasure of going ashore, and nothing else.”

“Dantès,” said the shipowner, turning towards the young man, “come this way!”

“In a moment, sir,” answered Dantès, “and I’m with you.” Then calling to the crew, he said—“Let go!”

The anchor was instantly dropped, and the chain ran rattling through the port-hole. Dantès continued at his post in spite of the presence of the pilot, until this manœuvre was completed, and then he added, “Half-mast

the colors, and square the yards!”

“You see,” said Danglars, “he fancies himself captain already, upon my word.”

“And so, in fact, he is,” said the owner.

“Except your signature and your partner’s, M. Morrel.”

“And why should he not have this?” asked the owner; “he is young, it is true, but he seems to me a thorough seaman, and of full experience.”

A cloud passed over Danglars’ brow. “Your pardon, M. Morrel,” said Dantès, approaching, “the vessel now rides at anchor, and I am at your service. You hailed me, I think?”

Danglars retreated a step or two. “I wished to inquire why you stopped at the Island of Elba?”

“I do not know, sir; it was to fulfil the last instructions of Captain Leclere, who, when dying, gave me a packet for Marshal Bertrand.”

“Then did you see him, Edmond?”

“Who?”

“The marshal.”

“Yes.”

Morrel looked around him, and then, drawing Dantès on one side, he said suddenly—“And how is the emperor?”

“Very well, as far as I could judge from the sight of him.”

“You saw the emperor, then?”

“He entered the marshal’s apartment while I was there.”

“And you spoke to him?”

“Why, it was he who spoke to me, sir,” said Dantès, with a smile.

“And what did he say to you?”

“Asked me questions about the vessel, the time she left Marseilles, the course she had taken, and what was her cargo. I believe, if she had not been laden, and I had been her master, he would have bought her. But I told him I was only mate, and that she belonged to the firm of Morrel & Son. ‘Ah, yes,’ he said, ‘I know them. The Morrels have been shipowners from father to son; and there was a Morrel who served in the same regiment with me when I was in garrison at Valence.’”

“*Pardieu!*, and that is true!” cried the owner, greatly delighted. “And that was Policar Morrel, my uncle, who was afterwards a captain. Dantès, you must tell my uncle that the emperor remembered him, and you will see it will bring tears into the old soldier’s eyes. Come, come,” continued he, patting Edmond’s shoulder kindly, “you did very right, Dantès, to follow Captain Leclere’s instructions, and touch at Elba, although if it were known that you had conveyed a packet to the marshal, and had conversed with the



emperor, it might bring you into trouble.”



“How could that bring me into trouble, sir?” asked Dantès; “for I did not even know of what I was the bearer; and the emperor merely made such inquiries as he would of the first comer. But, pardon me, here are the health officers and the customs inspectors coming alongside.” And the young man went to the gangway. As he departed, Danglars approached, and said,—

“Well, it appears that he has given you satisfactory reasons for his landing at Porto-Ferrajo?”

“Yes, most satisfactory, my dear Danglars.”

“Well, so much the better,” said the supercargo; “for it is not pleasant to think that a comrade has not done his duty.”

"Dantès has done his," replied the owner, "and that is not saying much. It was Captain Leclere who gave orders for this delay."

"Talking of Captain Leclere, has not Dantès given you a letter from him?"

"To me?—no—was there one?"

"I believe that, besides the packet, Captain Leclere confided a letter to his care."

"Of what packet are you speaking, Danglars?"

"Why, that which Dantès left at Porto-Ferraio."

"How do you know he had a packet to leave at Porto-Ferraio?"

Danglars turned very red.

"I was passing close to the door of the captain's cabin, which was half open, and I saw him give the packet and letter to Dantès."

"He did not speak to me of it," replied the shipowner; "but if there be any letter he will give it to me."

Danglars reflected for a moment. "Then, M. Morrel, I beg of you," said he, "not to say a word to Dantès on the subject. I may have been mistaken."

At this moment the young man returned; Danglars withdrew.

"Well, my dear Dantès, are you now free?" inquired the owner.

"Yes, sir."

"You have not been long detained."

"No. I gave the custom-house officers a copy of our bill of lading; and as to the other papers, they sent a man off with the pilot, to whom I gave them."

"Then you have nothing more to do here?"

"No—everything is all right now."

"Then you can come and dine with me?"

"I really must ask you to excuse me, M. Morrel. My first visit is due to my father, though I am not the less grateful for the honor you have done me."





“Right, Dantès, quite right. I always knew you were a good son.”

“And,” inquired Dantès, with some hesitation, “do you know how my father is?”

“Well, I believe, my dear Edmond, though I have not seen him lately.”

“Yes, he likes to keep himself shut up in his little room.”

“That proves, at least, that he has wanted for nothing during your absence.”

Dantès smiled. “My father is proud, sir, and if he had not a meal left, I doubt if he would have asked anything from anyone, except from Heaven.”

“Well, then, after this first visit has been made we shall count on you.”

“I must again excuse myself, M. Morrel, for after this first visit has been

paid I have another which I am most anxious to pay.”

“True, Dantès, I forgot that there was at the Catalans someone who expects you no less impatiently than your father—the lovely Mercédès.”

Dantès blushed.

“Ah, ha,” said the shipowner, “I am not in the least surprised, for she has been to me three times, inquiring if there were any news of the *Pharaon*. *Peste!* Edmond, you have a very handsome mistress!”

“She is not my mistress,” replied the young sailor, gravely; “she is my betrothed.”

“Sometimes one and the same thing,” said Morrel, with a smile.

“Not with us, sir,” replied Dantès.

“Well, well, my dear Edmond,” continued the owner, “don’t let me detain you. You have managed my affairs so well that I ought to allow you all the time you require for your own. Do you want any money?”

“No, sir; I have all my pay to take—nearly three months’ wages.”

“You are a careful fellow, Edmond.”

“Say I have a poor father, sir.”

“Yes, yes, I know how good a son you are, so now hasten away to see your father. I have a son too, and I should be very wroth with those who detained him from me after a three months’ voyage.”

“Then I have your leave, sir?”

“Yes, if you have nothing more to say to me.”

“Nothing.”

“Captain Leclere did not, before he died, give you a letter for me?”

“He was unable to write, sir. But that reminds me that I must ask your leave of absence for some days.”

“To get married?”

“Yes, first, and then to go to Paris.”

“Very good; have what time you require, Dantès. It will take quite six weeks to unload the cargo, and we cannot get you ready for sea until three months after that; only be back again in three months, for the *Pharaon*,” added the owner, patting the young sailor on the back, “cannot sail without her captain.”

“Without her captain!” cried Dantès, his eyes sparkling with animation; “pray mind what you say, for you are touching on the most secret wishes of my heart. Is it really your intention to make me captain of the *Pharaon*?”

“If I were sole owner we’d shake hands on it now, my dear Dantès, and call it settled; but I have a partner, and you know the Italian proverb—*Chi ha compagno ha padrone*—‘He who has a partner has a master.’ But the thing is at least half done, as you have one out of two votes. Rely on me to

procure you the other; I will do my best.”

“Ah, M. Morrel,” exclaimed the young seaman, with tears in his eyes, and grasping the owner’s hand, “M. Morrel, I thank you in the name of my father and of Mercédès.”

“That’s all right, Edmond. There’s a providence that watches over the deserving. Go to your father: go and see Mercédès, and afterwards come to me.”

“Shall I row you ashore?”

“No, thank you; I shall remain and look over the accounts with Danglars. Have you been satisfied with him this voyage?”

“That is according to the sense you attach to the question, sir. Do you mean is he a good comrade? No, for I think he never liked me since the day when I was silly enough, after a little quarrel we had, to propose to him to stop for ten minutes at the island of Monte Cristo to settle the dispute—a proposition which I was wrong to suggest, and he quite right to refuse. If you mean as responsible agent when you ask me the question, I believe there is nothing to say against him, and that you will be content with the way in which he has performed his duty.”

“But tell me, Dantès, if you had command of the *Pharaon* should you be glad to see Danglars remain?”

“Captain or mate, M. Morrel, I shall always have the greatest respect for those who possess the owners’ confidence.”

“That’s right, that’s right, Dantès! I see you are a thoroughly good fellow, and will detain you no longer. Go, for I see how impatient you are.”

“Then I have leave?”

“Go, I tell you.”

“May I have the use of your skiff?”

“Certainly.”

“Then, for the present, M. Morrel, farewell, and a thousand thanks!”

“I hope soon to see you again, my dear Edmond. Good luck to you.”

The young sailor jumped into the skiff, and sat down in the stern sheets, with the order that he be put ashore at La Canebière. The two oarsmen bent to their work, and the little boat glided away as rapidly as possible in the midst of the thousand vessels which choke up the narrow way which leads between the two rows of ships from the mouth of the harbor to the Quai d’Orléans.

The shipowner, smiling, followed him with his eyes until he saw him spring out on the quay and disappear in the midst of the throng, which from five o’clock in the morning until nine o’clock at night, swarms in the famous street of La Canebière,—a street of which the modern Phocéens

2.1 PLAIN IMAGES

are so proud that they say with all the gravity in the world, and with that accent which gives so much character to what is said, “If Paris had La Canebière, Paris would be a second Marseilles.” On turning round the owner saw Danglars behind him, apparently awaiting orders, but in reality also watching the young sailor,—but there was a great difference in the expression of the two men who thus followed the movements of Edmond Dantès.





2.2 Figures

Chris Fradkin

Figures, as we refer to them, are images, or similar visual artifacts (including tables) accompanied by a caption.

To ensure EPUB2 compatibility (which requires valid XHTML 1.1), and to give us more flexibility, we don't use the HTML `<figure>` tag, but rather use either:

- for very simple figures, a paragraph with an `image-with-caption` class that starts with an inline image followed by text; or
- a `div` with a `figure` class containing both an image and its caption, created with our `figure` include.

Old versions of the Electric Book template used a `blockquote` instead of a `div` for the figure, since a `blockquote` can be created with markdown `>` syntax.

To show figures in context, here is an article from *The Comics Grid: Journal of comics scholarship* by Chris Fradkin. The abstract, if you're curious:

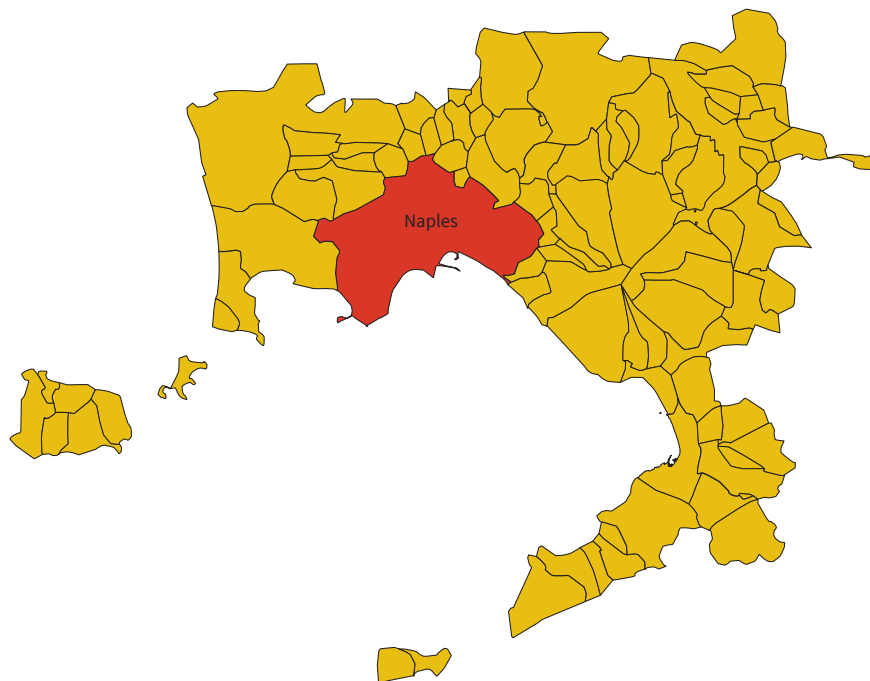


Figure 1 A map of Naples and surrounding areas

FIGURES

This note explores the notion of comic superheroes as tools for the empowerment of children. The author details interventions in Rwanda and Brazil, and their different usages of superheroes. With a focus on the superhero's pre-cloak stage—the stage prior to their employing superpowers—the author offers glimpses of current work in progress to help therapists empower orphaned children. While this area of research is at an early stage, its potential among health professionals is growing. Thus the comic superhero may be more than celluloid, as health professionals learn to use his superpowers.

Pre-Cloak Comic Superheroes: Tools for the Empowerment of Children

Fradkin, C., (2016). Pre-Cloak Comic Superheroes: Tools for the Empowerment of Children. *The Comics Grid: Journal of Comics Scholarship*. 6, p.13. DOI: <https://doi.org/10.16995/cg.85>. Creative Commons Attribution

The idea of using pre-cloak comic superheroes as tools for the empowerment of children has drawn notice from a multitude of corners: e.g., documentary filmmakers, a journalist from *Forbes*, the Joseph Campbell Foundation (2016).¹ It has predictably drawn interest from adoption networks and support groups, as well as agencies that deal with foster children. I say 'predictably' because these groups of at-risk children were the groups a recent study (Fradkin et al. 2016²) aimed to serve. As the findings in the 'pre-cloak' study noted: of the top-20 comic superheroes, 86% were orphaned or abandoned, in addition to enduring other woes. My co-authors and I stressed that this shared history has potential for empowerment: When these children hear of hardships overcome by superheroes, that may strengthen their resolve to carry on.

A program that employs the pre-cloak stage of superheroes has been active in Rwanda for some time. At the Rwandan Orphans Project (a center that serves former street children) (2016³), a life-skills group implemented by Lisa Meaney, MFT, has used superhero origin stories to inspire its orphaned boys (Figure 1). "For these boys," Meaney writes (2016, pers comm., 3 May), "childhood trauma can be a catalyst for positive change ... Just like it is for superheroes." For two years Meaney has incorporated superhero origin stories into her group-work with Rwandan boys. On Meaney's watch, superhero capes were made from old t-shirts, masks from cereal boxes. "Over and over again," she says, "the boys have made statements such as, 'If Batman can be an orphan and be a superhero then so can I'" (2016, pers comm., 22 April). When asked about the boys' openness to the disproportionately large percentage of Caucasian superheroes (~90%), Meaney reassured me that for her boys, the notion of a superhero role model is stronger than the color lines of race.



Figure 2 The Rwandan Orphans Project (near Kigali, Rwanda) uses comic superheroes to empower orphaned children. Capes from old t-shirts, masks from cereal boxes.

Published with permission from The Rwandan Orphans Project. Photo © Lisa Meaney, MFT, Rwandan Orphans Project, 2016 (Accessed 14 May 2016).

5,000 miles WSW of Rwanda is a program in São Paulo, Brazil. The ‘Superformula’ program at the A.C. Camargo Cancer Center (2016⁴), attempts to raise the spirits of its pediatric patients using special comics, videos, and superhero plastic covers for IV bags (Figure 2). The children are encouraged to do battle with their cancer, as the comic superheroes battle evil. Their chemo drip is ‘Superformula.’ While this approach may first seem novel, over time it may wear thin, as many patients on the ward will lose their battle. The Superformula program instills hope upon the ward, through the premise of ‘invincibility suggestion.’

So where to go? I pondered on the future of the project; an effort conceived in several stages. The Rwandans were standing in the background. Therapists were asking what to do.... The cart was put in motion; it was rolling down the hill; but still, the next step was elusive. Then, bang! the answer hit me: A compilation! A directory for the therapists; in particular, those who work with high-risk children. I envisioned a sleek database; a gold-foil embossed book. I celebrated, having settled on direction. My compass was following the course.



Historical Figure	Favourite Fruit	Ice-Cream Flavour	Favourite Game	Favourite Animal	Favourite Stage Play	Favourite Piece of Music
Cleopatra (69–30 BCE)	Figs	Rosewater sorbet	Senet	Cat	<i>The Book of the Dead</i> (recited)	Traditional Egyptian harp tunes
Leonardo da Vinci (1452–1519)	Pears	Honey-almond gelato	Chess	Horse	<i>La Mandragola</i> (Machiavelli)	Gregorian chant
William Shakespeare (1564–1616)	Apple	Mead-flavoured custard	Backgammon	Falcon	<i>Doctor Faustus</i> (Marlowe)	<i>Greensleeves</i> (folk melody)
Marie Antoinette (1755–1793)	Cherries	Pistachio	Jeu de paume	Pomeranian dog	<i>Le Mariage de Figaro</i>	<i>Queen of the Night</i> (Mozart)
Napoleon Bonaparte (1769–1821)	Grapes	Vanilla bean	Échecs (Chess)	Eagle	<i>Andromaque</i> (Racine)	<i>La Marseillaise</i>
Harriet Tubman (c. 1822–1913)	Peaches	Molasses ice cream	Checkers	Owl	<i>Uncle Tom's Cabin</i> (read aloud)	Spirituals (<i>Go Down Moses</i>)
Albert Einstein (1879–1955)	Plums	Chocolate	Violin duets	Dog	<i>Faust</i> (Goethe, adapted)	<i>Beethoven's Violin Concerto</i>
Genghis Khan (c. 1162–1227)	Wild berries	Fermented mare's milk (frozen)	Archery contests	Horse	Mongolian oral epics	Tuvan throat singing
Jane Austen (1775–1817)	Gooseberries	Clotted cream ice	Card games (Whist)	Dog (lapdog)	<i>She Stoops to Conquer</i>	<i>Rule, Britannia!</i>
Mansa Musa (c. 1280–c. 1337)	Dates	Date-honey sorbet	Mancala	Camel	Griot storytelling sessions	West African kora music

Figure 4 A wide table in a figure. In PDF, this is rotated to fit on the page, and also spans the sidebar.





FIGURES

Historical Figure	Favourite Social Media	Streaming Binge	Go-To Coffee Order	Weekend Hobby	Preferred App	Dream Travel Destination
Cleopatra	Instagram	<i>The Crown</i>	Iced hibiscus tea	Spa days + falconry	TikTok (beauty tips)	Maldives (luxury + water)
Leonardo da Vinci	YouTube (science side)	<i>Black Mirror</i>	Espresso doppio	Drone building	Procreate (sketching)	Tokyo (tech meets art)
William Shakespeare	Twitter (clever threads)	<i>Succession</i>	Flat white	Improv theatre	Goodreads	Edinburgh (Fringe Fest)
Marie Antoinette	Pinterest	<i>Bridgerton</i>	Lavender latte	Cake decorating	Instagram Reels	Dubai (opulent shopping)
Napoleon Bonaparte	LinkedIn	<i>House of Cards</i>	Strong black coffee	Strategy board games	Notion (planning wars)	Mongolia (Genghis Khan tour)
Harriet Tubman	Facebook (community)	<i>The Underground Railroad</i>	Herbal tea	Trail hiking	Signal (encrypted chat)	South Africa (Robben Island)
Albert Einstein	Reddit (r/askscience)	<i>Genius</i> (about him!)	Chai latte	Playing violin	Duolingo	Iceland (auroras + physics)

Figure 5 Another wide table in a figure. In PDF, this is rotated to fit on the page.

But in the morning, I felt weary after tallying the workload: a directory of comic superheroes' pre-empowered woes. Rebuking my resistance, I began. I sent word out to my students: from the past, the present, future: Would you like to gain some credit for your CV? When I mentioned comic superheroes, their eyes lit up like candles; one offered to enroll his aging mother. So, the routine was established: they would toil through DVDs, noting hardships and abuses that the superheroes suffered, in the pre-cloak stage, before empowerment. One student went to *Batman* (1989⁵); another to the *Hulk* (2003⁶); a third said he would work with *Spider-Man* (2002⁷).

The gears were grinding slowly. The students, as research assistants, buckled down. A directory was slowly in the works. At this stage we built the format in an Excel spreadsheet, time-coded from main title to the end. The time-code ran from top to bottom on the left-hand column, with adversities and comments to the right. Every hardship from each film was logged: pre-cloak and post. They were coded for the different categories: loss of parents, bullying, poverty, abuse; romance problems, hardships while at school. In the end, we finalized two separate versions for each film: the first in chronological from downbeat to the end; the second chronological by hardship (e.g., bullying: from main title through the scroll; then hardships while at school in the same format). We opted for this doubling to allow the therapist quick access to focus on one specific hardship. The students worked in shifts, scene-to-scene they chugged along. Assistant Chua logged bullying (00:03:55–00:04:26), in *Spider-Man 3* (2007⁸): 'During lecture, Parker's classmates are shooting spitballs at him ... [then they're] shining light into his eyes.' Assistant Sullivan logged loss of parents (01:30:15–01:32:10), in a flashback from *Batman* (1989⁹): He notes Bruce Wayne and his parents walking from a movie theatre ... He then describes the details of their murder.

As progress inches forward, challenges arise. In terms of presentation ... Should the pre-cloak stage be separate from the post-? Should categories meld into each other? Should the series of the films (e.g., *Spider-Man* [2002¹⁰], *Spider-Man 2* [2004¹¹], *Spider-Man 3* [2007¹²]) be listed one-after-the-other, or stand-alone—independently? Decisions will be made; options will be weighed. And opinions gathered from disparate sources. The directory, when whole, will be reviewed by focus groups: by critics, colleagues, friends and foes alike.

So here I sit. Contemplating the potential of pre-cloak comic superheroes. While *Deadpool* (2016¹³) slaughters the box office; the *Force Awakens* (2015¹⁴) breaks the roof; *X-Men: Apocalypse* (2016¹⁵) has come and gone; likewise *Independence Day: Resurgence* (2016¹⁶). *Doctor Strange* (2016¹⁷) is

FIGURES

coming soon; while next year: *Wonder Woman* (2017¹⁸); *Spider-Man* (*Spider-Man: Homecoming* 2017¹⁹). On this planet of consumers, we can never get our fill: our addiction is to heroes on the screen. Make them larger than we are; give me guidance like a star; and in the process show the path my feet should go.

Endnotes

1. Joseph Campbell Foundation (2016). Available at: <https://www.jcf.org> (Accessed 26 March 2016).
2. Fradkin, C, Weschenfelder, G V and Yunes, M A M (2016). Shared adversities of children and comic superheroes as resources for promoting resilience: Comic superheroes are an untapped resource for empowering vulnerable children. *Child abuse & neglect* 51: 407–415, DOI: <https://dx.doi.org/10.1016/j.chiabu.2015.10.010>
3. Rwandan Orphans Project (2016). Available at: <https://www.rwandanorphansproject.org/> (Accessed 14 May 2016).
4. A. C. Camargo Cancer Center (2016). Available at: <https://www.accamargo.org.br> (Accessed 14 May 2016).
5. *Batman* (1989). Burton, Tim USA: Warner Bros. [film].
6. *Hulk* (2003). Lee, Ang USA: Universal Pictures. [film].
7. *Spider-Man* (2002). Raimi, Sam USA: Columbia Pictures. [film].
8. *Spider-Man 3* (2007). Raimi, Sam USA: Columbia Pictures. [film].
9. *Batman* (1989). Burton, Tim USA: Warner Bros. [film].
10. *Spider-Man* (2002). Raimi, Sam USA: Columbia Pictures. [film].
11. *Spider-Man 2* (2004). Raimi, Sam USA: Columbia Pictures. [film].
12. *Spider-Man 3* (2007). Raimi, Sam USA: Columbia Pictures. [film].
13. *Deadpool* (2016). Miller, Tim USA: 20th Century Fox. [film].
14. *Star Wars: The Force Awakens* (2015). Abrams, J. J. USA: Walt Disney Studios Motion Pictures. [film].
15. *X-Men: Apocalypse* (2016). Singer, Bryan USA: 20th Century Fox. [film].
16. *Independence Day: Resurgence* (2016). Emmerich, Roland USA: 20th Century Fox. [film].
17. *Doctor Strange* (2016). Derrickson, Scott USA: Walt Disney Studios Motion Pictures. [film].
18. *Wonder Woman* (2017). Jenkins, Patty USA: Warner Bros. Pictures. [film].
19. *Spider-Man: Homecoming* (2017). Watts, Jon USA: Columbia Pictures. [film].

Part 3

Code and mathematics





Code

See the docs on writing code in markdown.

Trust the geeks, they've got this sorted. Here's a code block from xkcd.

```
int random()
{
    return 5; // Chosen by dice roll, guaranteed to be
    random
}
```

What about two consecutive code blocks? Here's some YAML book metadata from this project, followed by some funny code from StackOverflow.

```
project:
  # The name of the organisation or entity that owns
  this project
  organisation: "Electric Book Works"
  # A live web address for the organisation
  url: "https://electricbookworks.com/"
  # The name of the project of books in this folder.
  name: "The Electric Book workflow"

// Replaces with spaces the braces in cases where
braces in places cause stasis
$str = str_replace(array("\{","\}")," ",$str);
```





Mathematics

Jekyll uses MathJax, which the Electric Book workflow supports.

LaTeX-style maths via MathJax can be inline ($E = mc^2$) or displayed:

$$P_{t+1} - P_t = \frac{b+d}{s}(P_t - \overline{P})$$

The maths above and below is borrowed and adapted from *The Economy* by CORE.

$$P_{t+2} - P_{t+1} = \frac{b+d}{s}(P_{t+1} - \overline{P})$$

It follows from $P_t = \overline{P} - \frac{s}{b+d}E_t$, $E_t = P_{t+1} - P_t$ and the equation above that

$$\begin{aligned} \frac{P_{t+2} - P_{t+1}}{P_{t+1} - P_t} &= \\ \frac{P_{t+1} - \overline{P}}{P_t - \overline{P}} &= R \end{aligned}$$





Part 4

Interactivity





Links and buttons

Links are pretty straightforward. For instance, you might link to the Electric Book template (<https://ebw.co/template>).

Buttons should be defined by adding a `.button` class to any element, such as a link. The template allows for inline buttons (<https://example.com>) and block-level buttons:

Example (<https://example.com>)

Again, the print version should have a way to deal with these. Alternatively, the book author can hide buttons in the print version. For instance, designs may support adding a `.non-printing` class to the element. Or the author can use Liquid control-flow tags, like `{% if site.output == "print-pdf" %}`, to control whether and how buttons appear in print.





Video

The Electric Book template includes a tag for easily inserting videos from YouTube and Vimeo. See the docs for more.

In this example, we have a YouTube video, without a description, set to start 14 seconds in. There's an option to watch it elsewhere, too, which won't show on PDF by default.



And a Vimeo one, set to start 5 seconds in.



The Hey Duggee series trailer

Neither have thumbnail images specified. On the web, if an image hasn't been specified here, when you're online the page will fetch a thumbnail for

VIDEO

YouTube videos automatically. For Vimeo and in other formats (PDF, epub, app), you'll get a placeholder.

Clicking the description will take you to the video online (except in print-PDF, where the print-ready PDF's profile doesn't allow clickable links).



Audio

We don't yet provide an easy include tag for audio. Instead, you can manually embed audio using HTML.





iFrames

You can embed an iFrame in a book, for instance an embeddable quiz like Betterquiz.

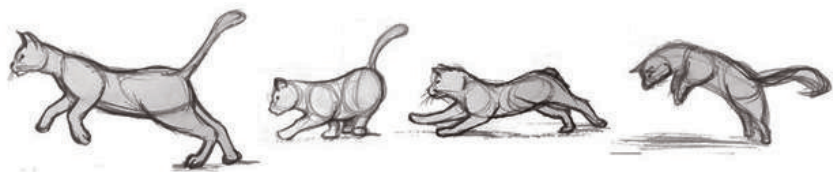
At this stage, it's best to use Liquid control flow tags to show text-only alternatives in PDF and epub, and embedded iFrames only in web output.





Slides

You can gather a number of figures into a set of slides. On the web and in apps, you can click through the slides. In print, you'll see one summary slide, and step-by-step text below the image describing each stage in the slide sequence. In epub, you'll see the summary slide, then each slide in turn, with its slide caption.



Slide 1 Three stages of an artistic way to draw animal shapes.
'Medialness and the Perception of Visual Art', Frederic Fol Leymarie and Prashant Aparajeya, in Art & Perception 5:2.

Ovals of varying sizes positioned at important junctions and capturing some of the main body parts. The particular orientations and combinations of these primitives indicate different body poses and leads to an animation of natural-looking movements.	Some of the initial primitive outlines are erased, and further details and shadows are added.
--	---

Details are added and some shading is shown.





Part 5

Indexes





Traditional index

Also see the docs on indexing.

A traditional index includes hard-coded page numbers. So the print edition must be finalised, and its pages fixed, before it is created.

We recommend creating book indexes with an unbroken, unordered list, with a `reference-index` class.

This is the index from *The Future of Ideas* by Lawrence Lessig (CC BY NC).

A

access:

- competition in, 173
- granting of, 13–14, 20

access fees, 20, 138

adaptations, creative, 8

Adobe Corporation, 208

advertising, 7

“ah-ha” technologies, 130

AIM, in instant messaging, 27

AirPort technology, 81, 172, 224–25

ALGOL, 51

algorithms:

- patents on, 207–8
- peer-to-peer querying, 137

Allman, Eric, 56

altruism, 13

Amateur Radio Service (ARS), 225, 244

Amazon.com, 132–33, 167, 169, 211

Amnesty International, 184

Andreessen, Marc, 265–66

Antheil, George, 79

anticircumvention program, 187–90

anticommons, 214–15

antitrust laws, 67, 110

AOL (America Online), 162–68

- access vs. control in, 163–66
- and competition, 167, 267
- and copyright law, 183
- early days of, 147–48, 162–63
- and Internet, 7, 163, 165

Time Warner merger with, 7, 164–66, 267

wires of, 163–64

Apache server, 55–56, 60, 69–70, 71

Apple Computer:

AirPort technology of, 81, 172, 224–225

and DVD movies, 189

GUI interface of, 63

iMac computer of, 124

Macintosh computers of, 27–28, 42, 63, 66

Mac OS, 60–61, 63, 90–91

and PC design, 159

“rip, mix, burn” ad of, 9, 10, 11

and unknown applications, 90–91

AppleScript language, 58

architecture:

changes in, 140

as choice, 37

code of, 35, 113

of communication systems, 112–14, 232–33

control via, 34, 151, 176

in cyberspace, 120–21, 140

for distribution, 119

freedom of, 35

importance of, 35

and innovation, 35–36, 92, 139, 140, 162

of Internet, 7, 15, 16, 23, 35–37, 40, 44, 119, 135, 140, 210, 238, 264

as physical constraint, 120

politics of, 35

of the Web, 169, 170

Armstrong, C. Michael, 154



TRADITIONAL INDEX

ARS (Amateur Radio Service), 225, 244

artwork:

code layer in, 111–12

content layer in, 105–07

control in, 107–10

copyright law applied to, 3

creativity in, 104–12, 200

digital, 8–9

physical layer in, 110–11

asynchronous transfer mode (ATM) systems, 149

AT&T:

as Bell system, 27

breakup of, 45, 53, 149

cable of, 153–54, 164–65, 241

in Canada, 164

compensation system of, 30

and competition, 28–29, 32–33, 37–38, 62, 158, 226

in computing business, 53

consent decree with government (1956), 51–52

and FCC, 27, 30, 81, 221

and foreign attachments rule, 30

and Internet, 154

Kingsbury Commitment of, 28–29

and monopoly power, 28, 29, 30, 32, 148

in open access movement, 164, 165

permissible businesses of, 45

shareholder obligations of, 32

smart network of, 38

and telephone system, 24, 26–34, 44, 62

and Unix operating system, 51–52, 53

and unknown applications, 90

@Home, 154, 158

Atkinson, Bill, 42

ATM (asynchronous transfer mode) systems, 149

auction sites, 169

audio changes, MP3, 123–24

Audio Home Recording Act, 181

B

Baby Bells, 149

backbone providers, 157

Bagdikian, Ben, 117

Baldwin, Carliss, 92

band managers, 221

bandwidth:

broadening of, 229

infinite, 47

overuse of, 138

pricing system for allocation of, 47

Bank, David, 66, 67

Bar, François 158–59

Baran, Paul, 26–27, 29–32, 33–34, 267

Batman Forever (film), 4

Beam-it, 129, 192

Behlendorf, Brian, 56

Bell companies:

as Baby Bells, 149

DSL used by, 155

lines leased by, 45

patents held by, 27

regulated to be open, 155, 248

regulation model for, 149

services unbundled by, 149, 248

see also AT&T

Bell Labs:

innovations channeled through, 30

innovative spirit of, 176

inventions of, 29, 38

and Unix operating system, 51–52

Bender, James, 182–83

Benkler, Yochai, layer concept of, 23–25, 103, 104, 113, 167,

238, 240

Berkeley Internet Name Domain (BIND), 56

Berlin, Irving, 107

Berners-Lee, Tim, 37, 40, 41–44, 134, 166, 213

BestBookBuys, 169

Bezos, Jeff, 211

Bidder's Edge, 169, 170–71

BIND (Berkeley Internet Name Domain), 56

Bingaman, Anne, 64

“Bluetooth” protocols, 81
 Boies, David, 196
 books:
 banning of, 184
 copyright protection of, 215
 HTML, 122–23, 196–99
 sales of, 211
 Boston Common, 19, 21
 bots:
 in copyright, 180–83
 spiders and, 169–71
 Bradner, Scott, 89
 Brief History of the Future, A (Naughton), 32
 broadband:
 AOL and, 163–64
 cable, *see also* cable broadband
 in Canada, 249
 competition in, 173, 248
 control of ISPs in, 246
 emerging technology of, 159, 161, 175
 faster downstream than upstream, 159
 FCC and, 159, 164
 markets in, 159
 open access for, 164, 165, 168
 broadcasting, *see* radio spectrum; television
 Brotz, Douglas, 208
 Brown, John Seely, 235–36
 browsers, 41, 43, 134
 BSD Unix, 53, 67
 Buchanan, James, 214, 215
 bugs, in code, 61
 bundling behavior, 66
 bureaucratization, 140–41
 Burk, Dan, 170
 Bush, Vannevar, 42
 business methods:
 patents on, 207, 208–9, 210
 software-implemented, 208–9

C

cable broadband, 151–54, 166–68
 building control into, 156–58, 168
 and Communications Act, 155
 and competition, 161–62, 167
 and DOCSIS standard, 153–54
 and FCC, 151–52
 incentives to build, 174
 and Internet, 153–54, 155–58, 167
 and streaming video, 156–57, 158
 and TCP/IP, 248
 two-way communication on, 153–54
 Cable Labs, 153
 cable television:
 advertising on, 7
 broadcasts on, 111
 CATV, 151
 concentrated ownership of, 117, 159, 173
 creative rights in, 109
 development of, 151
 distribution of, 119
 as end-to-end system, 151
 incentives for, 151–52
 layers of, 24
 licensing of, 201
 monopoly control of, 152, 153
 pricing of, 153
 satellite TV and, 153
 caches, 136–37
 Canada:
 free TV in, 190–91
 open access in, 164, 192, 249
 Carlton, Jim, 90–91
 Carterfone decision, 148
 CATV, 151
 CBS, and spectrum allocation, 74
 CDDDB (culture databases), 124–26
 censorware, 184, 186–87
 CERN labs, 41, 42–44
 Charmed Technologies, 81–82



TRADITIONAL INDEX

- children, protection of, 178–79, 184, 200
- China Online, 191
- chip fabricators, 70
- Christensen, Clay, 89, 91, 92, 139–40, 146, 210
- Cisco technologies, 156, 158
- Clark, David, 34, 37, 38
- Clark, Kim, 92
- Coase, Ronald, 12, 75, 114, 126, 221
- COBOL (programming language), 51
- code:
 - anticircumvention program and, 187–90
 - bugs in, 61
 - changes in, 140–41, 179
 - commons of, 49–50, 53, 55, 57, 138
 - compiled, 50, 253
 - as content, 50
 - content production in, 11, 53, 58–60, 67
 - derived from open code, 55, 59, 60
 - encryption cracked by, 190
 - fair use of, 188, 190
 - forking of, 52, 67–68
 - hard-wired, 50–51
 - hidden, 58
 - incentives to build, 68–69, 71
 - as law of cyberspace, 35
 - licensing of, 59–60
 - limits on, 256–57
 - as logical layer, *see also* code layer
 - nature of, 35–36, 50
 - object, 50
 - openness of, 52–54, 55–57, 59–60, 61, 67–69, 71–72, 210, 247, 253
 - performative, 57–58
 - source, 50, 60, 67, 72
- code layer, 23–25
 - in the arts, 111–12
 - changes to, 246–49
 - closing of, 217, 238
 - in commerce, 112–14
 - end-to-end at, 58
 - innovation commons at, 85, 103, 138, 167, 175
 - of Internet, 48, 49–50, 56, 138, 161
 - neutral platforms of, 72, 246–49
 - of the Web, 57, 85
- Code and Other Laws of Cyberspace* (Lessig), 35, 140, 179, 238
- Cohen, Julie, 257
- Cohen Theorem, 257
- Cold War, 12, 13
- command line interface, 62, 63
- commerce:
 - censorship in, 186–87
 - changes in policy of, 165
 - code layer in, 112–14
 - competition in, 114–15
 - creativity in, 112–19
 - of the different, 112
 - economics and, 115
 - innovation in, 6, 10
 - markets in, 114–15
 - vertical integration in, 165–66
- commercial radio, 74
- commons, 19–23
 - access fees for use of, 20, 138
 - anticommons vs., 214–15
 - built on control layer, 176
 - of code, 49–50, 53, 55, 57, 138
 - coexistence of open and closed, 93–94
 - contributing back to, 97
 - as default, 228–29
 - determinants of, 21
 - freedom of, 15
 - as free resource, 19–21, 44, 69, 86–99
 - idea of, 15
 - innovation of, *see also* innovation commons
 - inverse, 68
 - of knowledge, 49, 55, 57–58
 - market linked to, 138, 228
 - norms for control of, 96
 - peer-to-peer innovations in, 137–38
 - rebuilding of, 255–56

- as resource for decentralized innovation, 85
- state intervention in, 228
- tragedy of, 22, 23, 48, 168
- undermining technology added to, 48
- value of, 87–88, 93, 173
- Communications Act (1934), 74
 - on “cable services” (Title VI), 155
 - on telecommunications services (Title II), 155
- Communications Decency Act (1996), 178–79
- communications systems:
 - architecture of, 112–14, 232–33
 - on computers, 112–14
 - layers of, 23–25
 - organizational models for, 167
 - security of, 26–34
 - spectrum, *see also* spectrum
 - two-way, 153–54
 - wireless, 74–75
 - wires of, 88
- community:
 - as built by noise, 133
 - CATV, 151
 - connectivity within, 80
 - monopoly power over, 87, 216
 - networks built by, 162
 - on-line, 162–63
 - overconsumption regulated by, 22
 - relationship of resource to, 21
 - spectrum of, 223–24
 - town square in, 87
- compensation without control, 201–2
- compression technology, 123
- compulsory licensing right, 109, 201, 255
- CompuServe, 147–48
- computer-communication systems, architecture of, 112–14
- computers:
 - ad hoc supercomputers, 136
 - applications run on, 36–37, 88
 - competition in field of, 27–28
 - costs of, 113
 - denial of service attack on, 169
 - dominant companies in, 27
 - early coding of, 50–51
 - at edge vs. within networks, 34, 36–37
 - IBM-compatible, 13, 62, 65, 159, 189
 - icons of, 60–61
 - incompatibility between, 41–42, 51
 - Intel chip architecture in, 62, 63, 66
 - Internet and, *see also* Internet
 - Mac design, *see also* Apple Computer
 - object code of, 50
 - as physical layer, 23
 - as property, 241
 - software of, *see also* software
 - wearable, 81–82
 - Web and, *see* World Wide Web
- Computer Telephony*, 38
- Congress, U.S.:
 - and cable television, 151–53, 201
 - and copyright law, 98, 196–99, 200, 201, 254
 - First Amendment restrictions on, 187
 - and indecent content, 178–79, 184
 - influence of lobbyists on, 267
 - and intellectual property, 187–88
 - new technologies confronted by, 108–10, 190
 - and patent law, 207, 259
 - and phone company, 148
 - rebuilding the creative commons, 255–56
 - and recording industry, 201–2, 254
 - and spectrum, 224, 255
- connectivity, as Internet goal, 36
- consent decree, 64
- Constitution, U.S.:
 - Copyright and Patent Clause in, 177, 197–98, 237
 - Dormant Commerce Clause of, 114
 - Exclusive Rights Clause of, 116
- constraints:
 - of architecture, 120
 - on creativity, 7, 8, 11, 104, 202, 216

TRADITIONAL INDEX

- on markets, 114–15
- in scarcity-based economy, 103
- consumption vs. production, 13
- content:
 - code as, 50
 - control of, 7, 107–8, 110, 156–58, 176, 179–80, 183, 248
 - CSS, 189
 - distribution of, 238
 - dynamic, 137
 - encryption of, 188–90
 - exclusivity of, 128
 - firewalls for blocking of, 172–73
 - free, 20, 50, 106, 108, 110, 123, 177, 196, 256
 - indecent, 178–79, 184, 199
 - legal protection of, 10–11, 59, 183, 185, 250–59
 - policy-based routing of, 156
 - range of, 131
 - as separate from medium, 166
 - space-shifting of, 194
- content layer, 23–25
 - in the arts, 105–7
 - changes in, 249–61
 - closing of, 217, 238
 - copyrights in, 250–59
 - digital technology in, 23, 120–21
 - free code at, 58, 72, 85
 - freedom at, 50
 - innovation at, 103
 - patent law and, 259–61
 - tension between freedom and control in, 177, 249
- content scramble system (CSS), 189
- contract law, 58–60, 97, 186, 257
- control:
 - architecture of, 34, 151, 176
 - balance of freedom vs., 14–15, 97, 139
 - compensation without, 201–2
 - constitutional limits on, 105–6
 - of content, 7, 107–8, 110, 156–58, 176, 179–80, 183, 248
 - in creative process, 107–10
 - in cyberspace, 237–38
 - filtering as, 157
 - as good, 13, 15, 86
 - imposed via copyright, 180–217
 - imposed via norms, 96, 146, 151
 - imposed via technology, 96–97, 140
 - increasing, 99
 - by lawyers, 4, 11, 58–60
 - local, of worldwide technologies, 192
 - of networks, 150–51, 156–58
 - perfect, 116, 217, 250
 - pressure toward, 168
 - private incentives for, 168, 174
 - of property, 22, 170–71
 - of resources, *see also* resources
 - scarcity and, 47
 - by state vs. market, 12, 14
 - techniques of, 96
 - trends toward, 249
 - in vertical integration, 165–66
 - via firewalls, 172–73
 - of wires, 26, 147–76
- Cook, Gordon, 267
- copy protection system, 253
- copyright law:
 - as agent for control, 180–217
 - aim of, 98
 - anticircumvention program in, 187–90
 - application of, 58
 - in art industry, 3
 - bots applied to, 180–83
 - broad application of, 4–5
 - censorship via, 184, 186–87
 - commercial exploitation limited in, 258–59
 - in Constitution, 177, 197–98, 237
 - constraints of, 8
 - content protected in, 10–11, 183, 185, 250–59
 - contract law and, 58–60, 186
 - contributory infringement under, 196
 - control uncontrolled in, 183, 200
 - for derivative works, 106–7, 123, 198–99

DMCA, 187, 88, 190

exceptions built into, 181

expansion of, 98, 106–7, 110, 196–97, 216, 264

fair use of, 105, 108, 181, 185, 188, 190, 195, 254

false claims under, 256

in film industry, 3–4, 11, 200

and First Amendment, 198, 264

foreigners excluded from, 106, 185, 190–91

incentives provided by, 59, 98, 107, 197, 201

intent of, 11, 187

on the Internet, 98, 178–79, 190–92, 200, 216

in music industry, 3, 192–94, 254–55

narrow scope of, 8, 106–7

noninfringing use under, 195–96

old vs. new in, 202–5

potential use under, 195–96, 200

and site shutdowns, 183

as state-backed monopoly, 250

“thick” vs. “thin” protection in, 202–3

time limits in, 106, 107, 188, 197, 251–52

unenforceable violations of, 181, 182

unlicensed uses in, 180–81

Corel Corporation, 262

Cox, Alan, 57

CPHack, 184–87, 188, 190

creation vs. discovery, 106

creativity:

- in adaptations, 8
- in the arts, 104–12, 200
- before the Internet, 8, 104–19
- centralization of, 44
- in commerce, 112–19
- constraints on, 7, 8, 11, 104, 202, 216
- control in process of, 107–10
- with digital tools, 9
- in filmmaking, 235
- free resources as basis of, 12–15, 44
- incentives for, 4, 21, 59, 109
- and innovation, 10
- intellectual property in, 203

- in the Internet, 6, 9, 14, 23, 76
- legal control of, 4, 58–60
- licensing and, 216
- new built on old in, 9, 13, 202, 204, 214
- potential for, 9–11
- rebuilding commons of, 255–56
- and subsidiarity, 121
- tangible forms of, 58

Crocker, Steve, 36

CSS (content scramble system), 189

culture, production of, 263–64

culture databases (CDDb), 124–26

CyberPatrol, 184–86

cyberspace:

- architecture in, 120–21, 140
- character of, 121
- code as law of, 35
- control in, 237–38
- as nonrivalrous, 116
- vs. real space, 103–4, 121, 181, 182
- world of ideas in, 116

D

dark fiber, 245

data:

- balance of power changed by, 128, 133
- and code layer, 138
- different classes of, 46–47
- real space costs of, 138

Davies, Donald, 31

DeCSS, 187–90

Defense Advanced Research Projects Agency (DARPA), 242

Defense Communication Agency (DCA), 33–34

Defense Department, U.S., 27, 30, 32, 44, 79

democracy:

- blocked transactions in, 93
- centralization as opposed to, 112
- free resources in, 12–13, 92–93

derivative works, 8, 106–7, 123, 198–99

DES (digital encryption standard)-

TRADITIONAL INDEX

encrypted message, 136
design evolution, 92
Devil's Advocate, The (film), 4
Dickinson, Q. Todd, 210, 211
digital age, 14
Digital Millennium Copyright Act of 1998 (DMCA), 187, 188, 190
Digital Research, 62
digital subscriber line (DSL), 154–55, 161–62, 248
digital technology:
 competition in, 201
 compulsory licensing rights to, 109
 in content layer, 23, 120–21
 for copying, 98
 falling costs of, 8, 235
 in film industry, 9, 235
 music processor, 8–9, 96
 potential of, 7, 9
 for telecommunications, 31
DirecTV, 153
discovery vs. creation, 106
discrimination, technologies for, 173
disk drive industry, 90
Disney Corporation, 234
distributed.net, 136
distributed processing, 136
distributed technology, 136
distribution:
 architecture for, 119
 of content, 238
 costs of, 7, 126, 138
 of media, 119, 234
 models of, 7
 of music, 119, 123–24, 127–29, 131, 200–202
 new means of, 126–29
DMCA (Digital Millennium Copyright Act, 1998), 187, 188, 190
DOCSIS standard, 153–54, 161–62
Dormant Commerce Clause, 114
DOS (disk operating system), 62–66

DR-DOS, 62, 63, 64, 65
DSL (digital subscriber line), 154–55, 161–62, 248
Dunkin' Donuts, 182
DVD disks, 188–90, 256

E

Easterbrook, Frank, 186
eBay, 169–71
economics, and scarcity, 103, 115
Economist, The, 206
Edelman, Ben, 191
education, and filmmaking, 235–36
efficiency, 92 802.11b protocol, 81
Einstein, Albert, 12, 13, 14, 20, 21
Eldred, Eric, 122–23, 196–99, 202, 264, 265
Eldritch Press, 123
electricity grid, as end-to-end, 39
Electronic Frontier Foundation, 196
Emacs, 54
e-mail, 10
EMI Publishing, 183
encryption systems, 188–90
end-to-end (e2e) argument, 34–39, 88–89
 central control disabled in, 40, 167–68
 compromises to, 172–73
 consequences of, 41–44, 48, 58
 costs of, 46
 design choice of, 121
 on different layer, 45–46
 discrimination in, 173
 importance of, 91
 and innovation commons, 36–37, 40–41, 48, 58, 147, 156, 210, 238–39
 intelligence in, 149
 neutral network of, 58, 61, 161
Enlightenment, 94, 95
Epstein, Richard, 170
essential volunteerism, 56
Ethernet, 77–78, 231
European Union, 64

exclusivity, 116, 128

F

fair use:

- reverse engineering and, 185
- rights of, 105, 108, 181, 188, 190, 254
- VCR and, 195

Fanning, Shawn, 130

Fano, Robert, 42, 112–14

fat pipes, 151–53

Faulhaber, Gerald, 176

Federal Communications Commission (FCC):

- and AT&T, 27, 30, 81, 221
- and broadband, 159, 164
- and cable, 151–52
- and future of technology, 263
- and spectrum, 74, 75, 80–81, 218–21, 223–27, 244, 246

Federal Radio Commission (FRC), 73

Federal Trade Commission (FTC), 166

Feely, Pat, 234

Ferguson, Warren, 195, 196

fiber optic cable, 245

fiber optic technology, 29, 47

File Synchronization, 60–61, 66

film industry:

- concentrated ownership in, 117
- copyright law applied to, 3–4, 11, 200
- digital technology in, 9, 235
- distribution in, 119, 234
- DVD disks and, 188–90, 256
- education and, 235–36
- lawyers in control of, 4–5, 11
- production costs in, 8, 124
- and VCR, 195

filtering, 157, 184

Firefly, 133

firewalls, 172–73

First Amendment:

- and code layer, 112, 190
- Congress restricted by, 187

copyrights as immune from, 198, 264

free speech in, 116, 197–98, 224

protest under, 93

First Year, The (film), 3

foreign attachments rule, 30

forking, 52, 67–68

Fourneau, Henri, 108

Fox Studios, 181, 182

Frankfurter, Felix, 74–75

Franklin Benjamin, 206

FRC (Federal Radio Commission), 73

free, meanings of the word, 11, 12, 15, 20

free culture, 9–10

freedom:

- in the architecture, 35
- vs. communism, 12, 13
- in computer world, 52
- as constitutional question, 11
- at content layer, 50
- vs. control, 14–15, 97, 139
- as enhancing the social value of the controlled, 48
- of the press, 197
- of speech, 116, 140, 178–79, 184, 197–98, 224
- to tinker, 61, 68

Free Software Foundation, 12, 53–54, 59

Frost, Robert, 197

FTC (Federal Trade Commission), 166

G

Gates, Bill, 62–67, 206, 214

Gaynor, Mark, 89

General Public License (GPL), 44, 54, 59–60, 97, 185–86

geography, constraint of, 114

Gilder, George, 47–48, 83–84, 227–28, 232

Gilmore, John, 265

Gingrich, Newt, 118

GNU C Compiler (GCC), 54

GNU/Linux OS, 54–55, 56, 57, 60, 62, 69

GNU OS, 53–54

Gnutella, 137, 138

TRADITIONAL INDEX

Goldsmith, Jack, 192
Gone with the Wind (Mitchell), 198–99
Gopher, 43
government:
 avoiding control by, 80–81
 commons created by, 45, 155
 control of, 14, 92, 93
 exclusive rights granted by, 7
 in open IP service, 176
 phone companies regulated by, 27, 30, 45, 81, 148, 161, 221
 spectrum controlled by, 73–76, 79–81, 83, 232, 246, 255
 spectrum hoarded by, 242
 see also Federal Communications Commission
GPL (General Public License), 44, 54, 59–60, 97, 185–86
graphical user interface (GUI) operating system, 62, 63
Guggenheim, Davis, 3–4
GUI-OS, 62, 63

H

Hamilton, Alexander, 265
Hand, Learned, 106
Hardin, Garrett, 22, 168, 175
hardware, code of, 35
Harry Fox Agency, 183
Hart, Michael, 123
Hartford Courant, 118
Hatch, Orrin G., 262–64, 267
Hayek, Friedrich von, 212
Hazlett, Thomas, 74, 75–76, 84, 218–19, 225, 229–30, 241
Heller, Michael, 214, 215
Hemingway, Ernest, 252–53
Hendricks, Dewayne, 80–81, 96
highways:
 access to, 14
 as commons, 20, 76–77, 87, 174–75, 244–46
 as end-to-end systems, 39
 as free resources, 12–13, 20, 228
 privatization of, 244
 production costs of, 13
 value of, 87, 173

history, access to, 13–14
Hollywood, *see also* film industry
home networks, 157–58
Hoover, Herbert, 73, 74
HTML (hypertext markup language), 41, 43, 57–58
HTML books, 122–23, 196–99
HTTP (hypertext transfer protocol), 41, 43
Huber, Peter, 227–28
Hughes, David, 79–81, 96, 223, 226
“Hush-a-Phone” device, 30
HyperCard, 42
hypertext, 42, 57

I

IBM:
 and Apache server, 69–70, 71
 and Linux, 69
 modular design in, 92
 and open code, 69–71
 OS/2 computers, 28, 65
IBM-compatible PCs, 13, 62, 65, 159, 189
icons, use of term, 60–61
iCraveTV, 190–92
ideas:
 copyright of, 98, 187
 expression of, 188
 free exchange of, 49, 94, 95
 nonrivalrous character of, 94
 taken for granted, 5, 13, 15
 as thinking power, 94
 world of, 116
iMac computers, 124
impact statement, need for, 211
infringement, contributory, 196
innovation:
 advantages to, 229
 and architecture, 35–36, 92, 139, 140, 162
 centralized control of, 32
 changes in the environment for, 236–37
 in commerce, 6, 10

- commons of, *see also* innovation commons
- cost of, 57, 162, 209–10
- and creativity, 10
- decentralization of, 85, 139–40
- defense of, 6
- in end-to-end networks, 36–37, 156
- entrepreneurial, 6
- in environment of minimal control, 140
- from experiments, 10
- freedom for, 11, 39, 139
- free resources as basis of, 12–15, 71, 76, 91, 156
- incentives for, 71, 203
- legal control of, 4, 32, 139
- licensing as deterrent to, 76, 216
- new and unimagined, 37
- old vs. new, 6, 199–201, 202–5, 236–39
- protection of, 253–54
- relocation of, 140–42
- in stupid networks, 38
- in technology, 5, 9–11, 48, 200
- in telecommunications, 45, 176
- tragedy of the commons in, 175–76, 177
- innovation commons, 49
 - at code layer, 58, 85, 103, 138, 167, 175
 - e2e and, 36–37, 40–41, 48, 58, 147, 156, 210, 238–39
 - Internet as, 23, 26, 40–41, 48, 85, 167, 175, 238–39, 266
- Innovator's Dilemma, The* (Christensen), 89–91, 139, 210
- Intel chip architecture, 62, 63, 66
- Intel Corporation, 27
- intellectual property:
 - in the creative process, 203
 - as property, 237
 - protection of, 204, 217
- intellectual property law, 7, 57, 97, 177, 187–88; *see also*
 - copyright law; patent law
- Internet:
 - advertising on, 7
 - and AOL Time Warner, 7, 163, 165
 - architecture of, 7, 15, 16, 23, 35–37, 40, 44, 119, 135, 140, 210, 238, 264
 - and AT&T, 154
 - backbone providers to, 157
 - birth of, 44–46, 148–49
 - blocking access to, 184
 - building control into, 156–58
 - and cable broadband, 153–54, 155–58, 167
 - capacity of, 46, 47, 229
 - changes in, 7–8, 15, 16, 23, 25, 41, 46, 99, 135, 139, 140, 146, 156, 175, 176
 - code layer of, 48, 49–50, 56, 138, 161
 - as commons built on control layer, 176
 - as communication system, 25
 - connections made by, 7, 36, 41–42, 44, 48, 82, 130, 147–49
 - constraints on, 175, 216–17
 - copyright on, 98, 179–80, 190–92, 200, 216
 - creativity in, 6, 9, 14, 23, 76
 - dynamic content of, 137
 - early promise of, 7, 14, 121
 - as end-to-end network, 34–39, 48, 149, 156, 210
 - entrepreneurs of, 6
 - file sharing on, 157
 - filtering on, 157, 184
 - free resources of, 14–15, 40, 44, 48, 50, 56–57, 86–99, 254
 - government-funded research for, 44, 45–46
 - growth as open in, 37
 - HTML books on, 122–23
 - as innovation commons, 23, 26, 40–41, 48, 85, 167, 175, 238–39, 266
 - layers of, 23, 25
 - and mass media, 178–79, 200
 - and Microsoft competition, 65–66
 - name space for, 172
 - network of, 26, 34, 147, 157–58
 - neutral platform of, 10, 37, 91, 161, 167, 174, 175, 266
 - new applications of, 40, 122
 - new demand created by, 132–34
 - old regime threatened by, 6, 16, 199–200
 - on-line services of, 163
 - P2P networks on, 134–37
 - pornography on, 178–79, 184, 199

TRADITIONAL INDEX

protocols (IP) of, 35, 36, 37, 41, 42, 48, 122, 176
public domain and, 56
search engines on, 137
service providers (ISPs) for, 34, 82, 148, 157, 176, 246
TCP/IP, 41, 56, 149, 248
television content on, 162, 190–92
threats to future of, 5–6
trespass law and, 170
unpredictability of, 7, 39–40, 88, 91
vertical integration and, 165–66
video limits on, 156–57, 158, 166
weaknesses of, 46
and wearable computing systems, 81–82
Web as separate from, 41
wires of, 26, 34–37, 44–46, 79, 148–49, 151–53, 167
worldwide audience of, 181–82, 191
Internet Explorer, 41
inventions, *see also* patent law; patents
inverse commons, 68
IP (Internet protocols), 35, 36, 37, 41, 42, 48, 122, 176
IPSec technology, 173
IPv4 of IPv6 name space, 172
Isenberg, David, 38, 89–90
ISPs (Internet service providers), 34, 82, 148, 157, 176, 246

J

Jaffe, Adam, 206, 207
Jansson, Eddy, 184
Jaszi, Peter, 198
Java, 65, 264–65
jazz, 9, 13
Jefferson, Thomas, 94–95, 104, 206–7
jukeboxes, 109
jurisdiction, problems of, 185

K

Kahin, Brian, 213
Kapor, Mitch, 13, 35
Kennard, William, 80, 223
Kepler, Johannes, 13

Kingsbury, Nicholas C., 28
Kingsbury Commitment, 28–29
Kleinrock, Leonard, 31
knowledge, commons of, 49, 55, 57–58
Kozinski, Alex, 203–4
Kreig, Andrew, 118

L

Lamarr, Hedy, 79
Lange, David, 256
language, as nonrivalrous commons, 21
Law and Order in Cyberspace (Huber), 227–28
layers, 23–25
 content, 23
 logical (code), 23
 physical (wires), 23
 see also Benkler, Yochai; specific layers
liability rule, 110
licensing:
 compulsory right of, 109, 201, 255
 free software, 60
 GPL, 44, 54, 59–60, 97, 185–86
 innovation chilled by, 76, 216
 nature of, 59
 open source, 60, 67, 72, 97
 and patent law, 205, 210
 and spectrum, 73, 223–25
Lightman, Alex, 81–82, 222
Linux OS, 54–55, 56, 57, 60, 62, 69
Linux PCs, 189
Litman, Jessica, 4, 258–59
local area networks, 77
logical layer, *see also* code layer
Lotus 1-2-3, 13
lyric servers, 124–26

M

Machiavelli, Niccolò, 6
Macintosh, *see also* Apple Computer
McChesney, Robert, 117

- McPherson, Isaac, 94
- Madison Square Garden, New York, 24, 25
- Madonna, 128
- management theory, 89–91, 92
- Maritz, Paul, 67
- markets:
- as agent for control, 96, 151, 162, 165, 166
 - competition in, 232
 - constraints on, 114–15
 - free, 71
 - vs. government regulation, 12, 14, 75–76, 263
 - incentives in, 71, 162
 - linked to commons, 138, 228
 - networks and, 46–47, 162
 - power of, 159
 - probable returns in, 90
 - for spectrum, 75, 228, 231, 232, 242
 - unidentified or undeveloped, 90
- Marx, Groucho, 13
- Mattel Corporation, 184–86, 187, 188
- mechanical reproduction right, 108–9
- media:
- bias in, 118–19
 - competition in, 119
 - constraints on, 216
 - content as separate in, 166
 - distribution of, 119, 234
 - homogeneity of output in, 118, 132
 - Internet and, 178–79, 200
 - ownership of, 116–19, 159, 173
 - real-space, 132
 - technology and, 116, 118
- mercantilism, 71
- Microsoft:
- code controlled by, 67, 263
 - and competition, 28, 56, 63–66, 167, 267
 - consent decree signed by, 64–65
 - as dominant in computer industry, 27
 - and DOS, 62–66
 - government lawsuit against, 61–68, 246, 262, 264
 - innovation killed by, 62
 - Internet Explorer of, 41
 - and Mac OS, 90–91
 - monopoly position of, 64, 65, 66–67
 - MSN of, 183
 - on open code values, 57
 - and software patents, 206, 214
 - and Windows, 28, 62–66
- Minix, 54
- Mitchell, Margaret, 198–99
- mobile devices, connections of PCs and, 81
- modularity, 92
- monopoly:
- antitrust laws and, 67
 - and AT&T, 28, 29, 30, 32, 148
 - benefits of, 29, 30, 175
 - and cable television, 152, 153
 - centralized control of innovation in, 32
 - in closed societies, 250
 - control built into, 32, 175
 - as exclusive right, 58
 - of a malevolent giant, 91–92
 - and Microsoft, 64, 65, 66–67
 - power over the community in, 87, 216
 - state-backed, 71, 205–7, 211, 212, 215–16, 242, 250
- Moody, Glyn, 56
- Motion Picture Association, 185
- Mouse, Mickey, 107
- MP3, 123–24, 125, 192–94, 201
- my.mp3.com, 127–29, 133, 192
- Napster and, 130–32, 194–95
- MS-DOS, 62, 63, 64, 65
- MSN, 183
- MUDs (multi-user domains), 163
- Mueller, Milton, 28
- Murphy, Frank, 75
- music:
- access to, 128, 131
 - album sales, 200
 - availability of, 7, 13, 111



TRADITIONAL INDEX

breaking into, 134
capital markets in, 201
compulsory licensing rights to, 109
concentrated ownership of, 117
copies of, 130, 181
copyright law applied to, 3, 192–94, 254–55
digital composition of, 8–9, 96
distribution of, 119, 123–24, 127–29, 131, 200–202
growth of, 199–200
on jukeboxes, 109
lyric servers in, 124–26
MP3, 123–24, 125, 192–94, 201
my.mp3.com, 127–29, 133, 192
Napster and, *see also* Napster
new forms of, 9
in noncommercial broadcasting, 109
OLGA, 182–83
ownership of the artist in, 128
piano rolls, 108–9, 254
production of, 127
unregulated, 8
my.mp3.com, 127–29, 133, 192

N

Napster, 130–32, 194–96, 201, 254
and album sales, 200
bandwidth used by, 138
court shutdown of, 264, 265
firewalls against use of, 172–73
Hatch's views on, 262–63
as P2P technology, 135, 194, 258
National Center for Supercomputing Applications (NCSA),
55
National Parks Service, 231
National Research Council (NRC), 34–35, 40, 165–66, 176
Native American tribes, as sovereign nations, 81
NATs (network access technologies), 171–72
Naughton, John, 32, 33
navigable waterways, 87
NBC, and spectrum allocation, 74

NCSA Server, 55
Negroponte switch, 243
Net, *see also* Internet
Netscape Corporation, 55, 65
Netscape Navigator, 41
network access technologies (NATs), 171–72
network design, principles of, 34–35
networks:
access to, 34, 45, 155, 171–72, 244–46
ATM design of, 149
built by communities, 162
closed, 162
collision on, 77
control of, 150–51, 156–58
data transport provided by, 35, 36, 40
end-to-end argument for, 34–39, 156
Ethernet, 77–78, 231
government monopoly of, 242
home, 157–58
incentives to build, 174
intelligence located in, 34
linking of, 147–49
local area, 77
market influences on, 46–47, 162
neutral platforms of, 37, 45, 61, 161
as open resource, 45, 61, 88
organization models for, 167
P2P, 134–137, 159, 194, 258
real options theory applied to, 89
smart, 38
speed of, 47
stupid, 38
vertical integration in, 165–66
wires of, 26
neutral platforms:
increasing controls vs., 175
of Internet, 10, 37, 91, 161, 167, 174, 175, 266
of networks, 37, 45, 61, 161
of open code, 72, 246–49
public benefits of, 168

Newton, Harry, 38
 Newton, Sir Isaac, 13
 Noam, Eli, 219, 226, 231
 noise, communities built by, 133
 nonexclusive rights, 86
 nonrivalrous resources:
 nature of, 21, 58, 94–95, 115–16
 need for creation of, 95, 96
 system of control for, 95–96
 norms:
 control via, 96, 146, 151
 protection by, 58, 161, 167, 170
 NRC (National Research Council), 34–35, 40, 165–66, 176

O

object code, 50
 old vs. new, 6, 16, 139, 145–46, 199–201, 202–5, 212, 236–39, 265
 OLGA, 182–83
 1-Click technology, 211
 on-line bulletin boards, 79–80
 on-line communities, 162–63
 on-line services, 147–48, 163
 open access movement, 164, 165
 open source licenses, 60, 67, 72, 97
 operating systems (OS):
 built on open code, 68, 247
 common platform of, 51
 DOS, 62–66
 free, 52
 GNU, 53–54
 GNU/Linux, 54–55, 56, 57, 60, 62, 69
 GUI, 62, 63
 IBM OS/2, 28, 65
 incompatibility of, 51
 licensing of, 60
 Mac OS, 60–61, 63, 90–91
 as nonrivalrous resource, 96
 Unix, 51–52
 Oracle, 208

Oram, Andy, 137
 O'Reilly, Tim, 210–11
 Osterman, Jack, 32
 Ostrom, Elinor, 95
 overconsumption, prevention of, 22

P

P2P networks, 134–37, 159, 194, 258
 packet-switching technology, 31–34, 40, 46, 79, 157
 Parker, Sean, 130
 parks:
 as commons, 19, 20, 21, 87
 as free resource, 13, 20
 patches, 55–56
 Patel, Marilyn Hall, 196
 patent bar, 211–12
 patent law, 205–15
 as barrier to idea theft, 211
 and content layer, 259–61
 damages under, 260
 duration and scope of, 207–8, 212–13, 214, 216, 259
 “holdup problem” of, 214
 innovation affected by, 205, 206, 209–11, 214
 and licensing, 205, 210
 and litigation, 213
 progress as goal of, 205
 reform of, 260–61
 as state-backed monopoly, 205–7, 211, 212
 technological competition and, 206
 Patent Office, U.S., 207, 208, 213, 259–61
 patents:
 on algorithms, 207–8
 as anticommons, 214–15
 of AT&T, 27
 on business procedures, 207, 208–9, 210
 Congress and, 207, 259
 costs of, 213
 Franklin on, 206
 Jefferson on, 94–95, 104, 206–7
 limited rights of, 71



TRADITIONAL INDEX

- nature of, 212
- power of, 94–95
- on software, 206, 207–8, 213, 214
- Supreme Court and, 207
- transaction costs of, 210
- Patterson, L. Ray, 258
- PC-DOS, 62
- Peacefire.org, 184
- peer-to-peer (P2P) networks, 134–37, 159, 194, 258
- Perl, 56
- permission:
 - absence of need for, 20
 - granting of, 11
 - necessity of, 12
 - neutral granting of, 12, 20
- physical layer, 23–25
 - access to, 138
 - in the arts, 110–11
 - changes in, 240–46
 - of computer-communications architecture, 113
 - control at, 110–11, 238
 - creativity at, 110–11
 - free spectrum at, 86, 241–44; *see also* spectrum
 - innovation at, 103
 - wires of, *see also* wires
- physics, different, 104
- piano rolls, 108–9, 254
- Pitofsky, Robert, 166
- plasticity, 39
- Platt, Charles, 176
- player pianos, 108–9
- poetry:
 - new forms of, 9
 - new markets for, 126
- Poetry Daily, 126
- policy-based routing, 156
- pornography, 178–79, 184, 199
- Posner, Richard, 202–3
- preference matching engines, 133–34
- press, freedom of, 197
- Prince, The (Machiavelli), 6
- privacy, loss of, 133, 140, 213
- Prodigy, 147–48
- production:
 - consumption vs., 13
 - costs of, 7, 8, 13–14, 124, 203
 - incentives for, 14, 21, 71, 95, 97, 203, 241
 - of music, 127
- Project Gutenberg, 123
- property:
 - control of, 22, 170–71
 - as exhaustible resource, 230
 - incentives to produce, 97, 203, 241
 - intellectual, 7, 57, 97, 177, 187–88, 203, 204, 217, 237
 - open to the public, 86, 228
 - and prosperity, 5
 - protection of, 20, 187, 236–37
 - with public interest, 87
 - resources as, 13
 - scope of, 6, 11, 94
 - trespass on, 170
- property right, exercise of, 20
- property rule, 110
- protocols:
 - “Bluetooth,” 81
 - 802.11b, 81
 - HTML, 41, 43, 57
 - HTTP, 41, 43
 - open and free, 57
 - for sharing resources, 97
 - TCP/IP, 41, 56, 149, 248
 - see also* IP
- public domain:
 - free content in, 20, 50, 106, 108, 110, 123, 177, 196, 256
 - Internet protocols in, 56
 - scope of, 8
- publicity, right of, 203
- publishing industry, *see also* books

Q

Quality of Service (QoS) solutions, 46–47

R

radio:

- commercials on, 74
- early programming on, 74
- noncommercial broadcasting, 109
- smart, 78
- software-defined, 242

Radio Act (1912), 73

Radio Act (1927), 73

radio spectrum, 218–26

allocation of, 74, 218–21, 227, 229, 233

amateur use of, 225

AM stations, 242

community-owned, 223–24

concentrated ownership of, 117, 159

FCC and, 74, 218–21, 223–27, 244, 246

FM stations, 223–25, 232

and freedom of speech, 224

nature of, 73–75

as property, 225–26, 229

social value in, 227

technical interference with, 225

transmission on, 79

see also spectrum

radio waves, 73

Randall, Alice, 198–99, 202

Rand Corporation, 26

rap music, 9

Raymond, Eric, 68

RCA, 234

Real Audio, 127

real options theory, 89

real space, 15

barriers to innovation in, 120, 138

vs. cyberspace, 103–4, 121, 181, 182

friction of, 181, 183

law of, 181, 199

media in, 132

Recording Industry Association of America (RIAA), 130,

192–96, 200, 215

recordings, *see also* music

Red Hat, 213

Reed David P., 34, 35, 37, 38, 39–40, 77, 90

regulator, and target for influence, 74

relativity, Einstein's theory of, 12, 14, 20, 21

Reno v. ACLU, 179, 180, 265

resources:

access to, 13–14

clear use of, 89

community relationships to, 21

depletion of, 21–22, 95, 96–97

as foundation to participation in society, 93

free vs. controlled, 11–15, 22, 44, 47, 48, 69, 71, 86–99, 156, 228

incentive for production of, 14, 71, 95

“in common,” 19–21, 86–99

privatization of, 227, 244

production costs of, 13

protocols for sharing of, 97

real cost of, 120

recreational, 20

rivalrous vs. nonrivalrous, 21, 87, 95–97, 115–16

scarcity of, 75

state vs. market control of, 12, 75–76, 228

unfair capture of, 87

unknown applications of, 88–89

unused, 230

value of, 87–88

RFC 1958, 36–37, 40

Rheingold, Howard, 136

RIAA (Recording Industry Association of America), 130,

192–96, 200, 215

Ritchie, Dennis, 51–52, 55

TRADITIONAL INDEX

rivalrous resources:

depletion of, 21–22

nature of, 21, 95, 115–16

need for control of, 95, 115

Road Runner, 154

roads, *see also* highways

Robertson, Michael, 127–28, 133, 201

Rose, Carol, 13, 20, 86, 87–88, 96

Rosen, Hilary, 200, 215

RSA Labs, 136

Rubinfeld, Daniel, 166

rules, control via, 151

S

Saltzer, Jerome, 34, 37, 38, 156

satellite TV, 153

scarcity:

and control, 47

economics of, 103, 115

search engines, 137

“sendmail,” 56

servers:

centralized, 131

competition among, 82

open code projects, 56

patches to, 55–56

restrictions on, 157

use of term, 55

Web, 41, 56, 134

see also ISPs

SETI (Search for Extraterrestrial Intelligence), 135

Shapiro, Carl, 214

Shavell, Stephen, 205

Simpsons, The, 180, 181, 182

Singer, Hal, 166

Skala, Matthew, 184

Smith, Adam, 71

software:

on business methods, 208–9

code of, 35

commons of, 49, 69, 85

content protection in, 11, 59, 252–53

copyright of, 252–53

development of, 70

Free Software Foundation, 12, 53–54, 59

host “IMP,” 36

licensing of, 59–60

open code of, 51, 59, 68–69

patents on, 206, 207–8, 213, 214

restrictions on copying of, 59

shrink-wrapped licenses with, 185–86

source carried with, 53, 55

Somers, Daniel, 158

Sony, 195, 196

source code, 50, 60, 67, 72

Soviet Union, collapse of, 145–46

Speakers’ Corner, London, 24, 25

spectrum, 73–84, 218–33

access to, 80–83, 161–62, 232–33, 241, 244

allocation of, 76–77

auction of, 75, 225–27, 231–32

bottom-up sharing of, 78–79

as commons, 83, 84, 86, 221–22, 228–30, 241–44

competition in use of, 222, 232, 245

congestion in, 241

cost of, 231

digital processors on, 79

DOCSIS standard for, 153–54, 161–62

dual strategy for, 242

government control of, 73–76, 79–81, 83, 232, 246, 255

government hoarding of, 242

innovation in use of, 84

licensing of, 73, 223–25

managing policies of, 232

market for, 75, 228, 231, 232, 242

nature of, 73–75

as nonrivalrous resource, 230, 231

optimal use of, 231

overlay of, 76

pollution of, 242–43

as property, 221, 242
 radio, *see also* radio spectrum
 as rivalrous resource, 230–31
 shared resources of, 76, 78–79, 96, 219–20, 230–31
 spread, 76, 80
 sunk cost in, 232
 tragedy of the commons in, 229
 unlicensed, 82–83, 226, 244
 unused, 230
 wireless, 82, 154, 161–62
 speech, freedom of, 116, 140, 178–79, 184, 197–98, 224
 spiders, 168–71
 spreadsheet, 13
 spread spectrum, 76, 80
 Sprint, 38
 Stallman, Richard, 12, 52–54, 59, 69, 208, 213, 267
 Starbucks coffee shops, 70
 Story, Joseph, 198
 storytelling, creativity in, 8, 9
 streaming technologies, 127, 137
 streaming video, 156–57, 158, 159, 190–91
 subsidiarity, 121
 Supreme Court, U.S.:
 on copyright law, 105–6, 108, 109, 195–96, 197
 on patent law, 207
 on sovereignty of Native American tribes, 81
 on spectrum regulation, 74–75, 255
 Sykes, Alan, 192
 synchronicity, 10

T

Tannenbaum, Andrew, 54
 TCP/IP, 41, 56, 149, 248
 technology:
 “ah-ha,” 130
 balance of control vs. freedom in, 99, 201–2
 changes in, 176
 competition in, 206, 214
 compression, 123
 control of, 232, 234

controls imposed via, 96–97, 140
 costs of, 113
 digital, 7–9
 disruptive change of, 89–90, 91–92, 146, 175
 distributed, 136
 experimentation in, 10
 falling costs of, 7, 8
 innovation in, 5, 9–11, 48, 200
 for local control, 192
 and markets, 114–15
 and media outlets, 116, 118
 new, congressional dealings with, 108–10, 190
 possible but not profitable, 48
 potential for, 10, 263
 probable returns of, 90
 progress in, 5
 protection by, 256
 retrofitting by, 139
 scarcity of, 47
 streaming, 127, 137
 unpredictable future uses of, 39
 wireless, 82
see also specific technology
 telecommunications:
 alternative systems of, 31–34
 and Communications Act, 155
 end-to-end network in, 149–51
 government intervention in, 27, 45, 81, 148, 161, 221
 innovation in, 45, 176
 research in, 29–30
 wires of, 26–34, 45–48, 88, 147, 148–49, 152, 154–55, 163–64
 telephone system:
 AT&T, 24, 26–34, 44, 62
 Bell system of, 27
 and birth of the Internet, 44–46, 148–49
 circuit-switching technology for, 33
 competition in, 27, 28–29, 32–33, 158
 DSL in, 154–55, 248
 independents in, 28



TRADITIONAL INDEX

innovation in, 30
Internet wired to, 44–48
layers of, 24, 25
modems connected to, 45
packet-switching technology for, 31–34, 46
patents of, 27
security of, 26–27, 31–34
selective interconnection in, 28, 29
universal service in, 28
wires of, 147, 148–49, 152, 154–55

television:
and AOL Time Warner, 7
cable, *see also* cable television
competition and, 132
freedoms in, 111, 112
and iCraveTV, 190–92
and Internet content, 162, 190–92
pay-per-view, 153
satellite TV, 153
as spectrum guzzler, 243
and VCR, 195
wires of, 152

Thompson, Ken, 51–52, 55
Time Warner, AOL merger with, 7, 164–66, 267
Titanic, 73
Tonga, high-speed Internet access in, 80–81
Torvalds, Linus, 54–55, 57, 62
town square, as commons, 87
trademark law, 204–5
tragedy of the commons, 22, 23, 48, 168
in innovation, 175–76, 177
in spectrum, 229
transaction costs, 210
transistor, invention of, 29
transparency, nature of, 71
trespass:
by bots, 169–70
classic definition of, 170
Turner, Ted, 152
Twelve Monkeys (film), 4 2001, 196

U

uncertainty:
and free vs. controlled resources, 88–89
plasticity and, 39
Uniform Computer Information Transactions Act (UCITA),
257
United States v. Microsoft, 61–68, 246, 262, 264
Unix operating system, 51–52
BSD Unix, 53, 67
command line interface of, 62
as commons, 52, 54
GNU and, 53–54
licensing of, 53
USENET, 96

V

Vaidhyanathan, Siva, 202
Valenti, Jack, 195
value, dimensions of, 93
VCR, 195
Verser, Locke, 136
vertical integration, 165–66, 176
video:
development of, 234
digital forms of, 96
limits on, 156–57, 158, 166
streaming, 156–57, 158, 159, 190–91
VCR, 195
VisiCalc, 13
Vixie, Paul, 56
volunteerism, essential, 56

W

Walker, Jay, 212, 267
Wall, Larry, 56
Walzer, Michael, 93
Wayner, Peter, 60, 68, 213
Web, *see also* World Wide Web
WebSphere, 70
Western Union, 29

- Wheel of Fortune, 203
 - White, Vanna, 203–4
 - wideband technologies, 76, 78
 - Wind Done Gone, The (Randall), 198–99
 - Windows, 28, 62–66, 189
 - wired, use of term, 49, 178
 - Wired magazine, 49
 - wireless, *see also* spectrum
 - wires, 26–48
 - of AOL, 163–64
 - constraints from, 47
 - control of, 26, 147–76
 - end-to-end design, 34–44, 46, 88, 156
 - fiber optic cable, 245
 - government-created commons on, 45, 155
 - of Internet, 26, 34–37, 44–46, 79, 148–49, 151–53, 167
 - monopoly control of, 32
 - network connections on, 26
 - neutrality of, 149
 - as physical layer, 23–25
 - as property, 241
 - Quality of Service solutions, 46–47
 - as rivalrous resource, 96
 - sharing of, 33
 - of telecommunications system, 26–34, 45–48, 88, 147, 148–49, 152, 154–55, 163–64
 - and voice call vs. Internet use, 148
 - woods, cathal, 183
 - WordPerfect, 262
 - World Wide Web:
 - auction sites on, 169
 - audience of, 181–82, 191
 - bots of, 169–71, 182
 - browsers on, 41, 43, 134
 - code layer of, 57, 85
 - core protocols for, 56
 - denial of service attack on, 169
 - emergence of, 41–44, 134
 - as fantasy, 7
 - HTML as mark-up language of, 57–58
 - HTML books on, 122–23
 - indexing of, 168–69
 - interconnectivity in, 43
 - Internet as separate from, 41
 - many-to-many publishing of, 88
 - open architecture of, 169, 170
 - as “out of control,” 37
 - patent law and, 213
 - pornography on, 184
 - potential for, 182
 - privacy on, 213
 - programming language for, 56
 - search engines of, 169–71, 182
 - spiders in, 168–71
 - as universal resource, 37, 44
 - Web servers on, 41, 56, 134
 - writers:
 - as authors, 111, 204
 - royalty expenses of, 203
 - Wu, Tim, 47–48
 - WWW, *see also* World Wide Web
- X–Z**
- Xerox PARC, 235–36
 - Yahoo!, 28, 167, 169, 183
 - Young, Bob, 213
 - Zittrain, Jonathan, 129, 130





Dynamic index

Also see the docs on indexing.

A dynamic index uses hyperlinks to the points in the text that it indexes, so that in print the page numbers are always accurate even if the text reflows, and online links to text are clickable. Here is a very short sample.

A

Adler, Irene 3, 18, 21–2
Adventures of Sherlock Holmes 3

B

Briony Lodge 11, 12, 13, 21

C

cases
 Atkinson brothers 4
 Holland royal family 4
 Scandal in Bohemia 3–23
 letter 6–8, 21
 Study in Scarlet 4
 Trepoff murder 4
Charing Cross 21
CO₂ 53

F–L

Fradkin, Chris 89
Holmes, Sherlock 3–23
King of Bohemia 20–21, 22
luxuries
 cigars 4
 spirits 4
 gasogene 4

M–Z

marriage 4, 9
 unhappy 4
 secret 9
Norton, Irene 21
occupations
 clergy 16
 fake 21
 Nonconformist 16
temple 22

